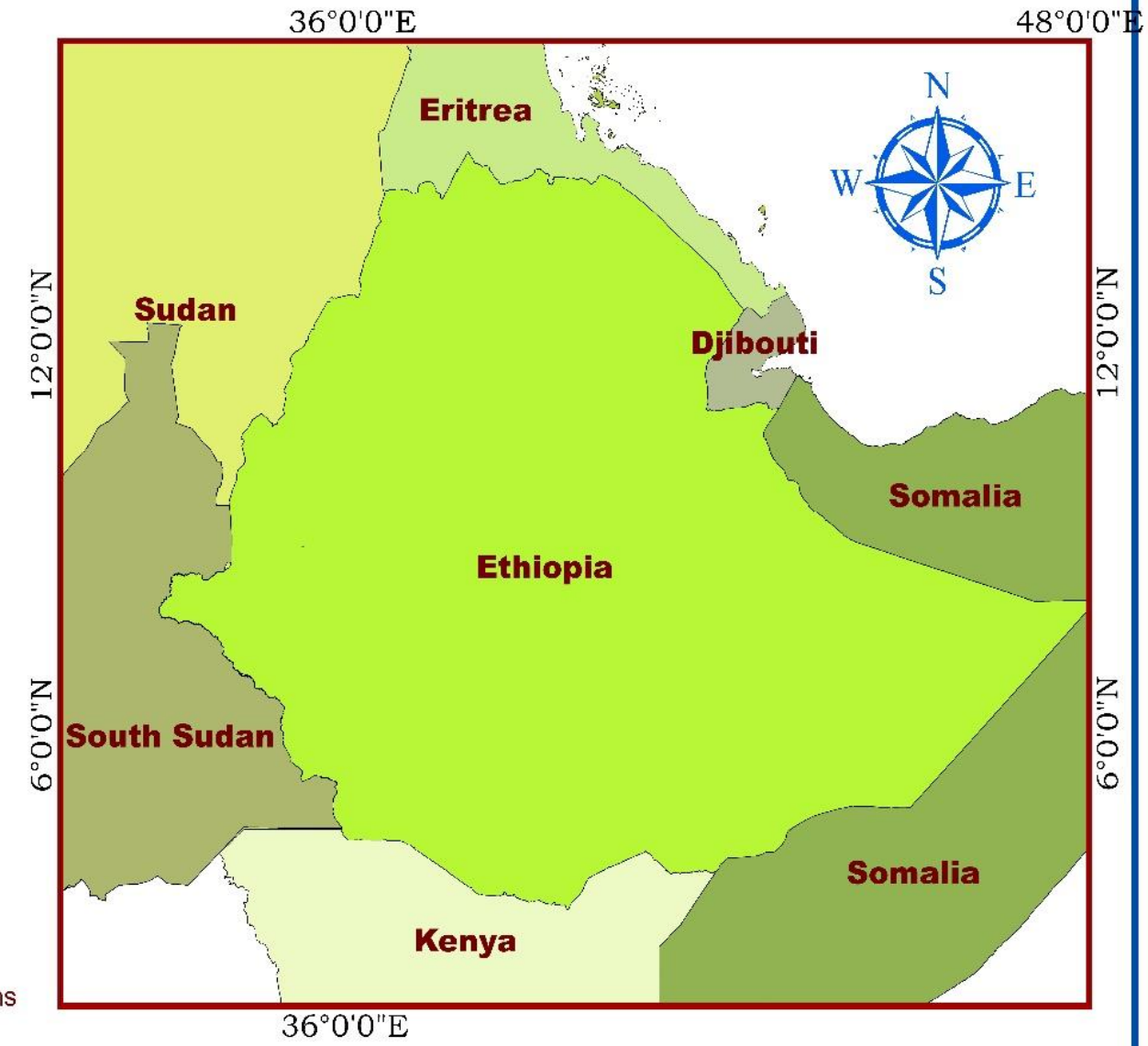
GEOGRAPHY OF ETHIOPIA AND THE HORN (CC0 171)

Geography of Ethiopia and The Horn





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1.1. Geography: Definition, Scope , Approach and Themes

1.1.1 Meaning of Geography

- It is difficult to forward a definition acceptable to all geographers at all times due to the **dynamic nature the subject and its scope in its long history**.
- The following may be accepted as a working definition.
  - **Geography is the scientific study of the Earth that describes and analyses spatio-temporal distribution, variations, interactions and relationships of phenomena** (physical, biological and human) **over the surface of the Earth.**

1.1.2. The Scope, Approaches and Themes of Geography

i) Scope

- Geography is **a holistic and interdisciplinary field of study** contributing to the understanding of the changing spatial structures from the past to the future
- The **scope of geography** is **the interface of the atmosphere, lithosphere, hydrosphere biosphere and anthropospher**, which provides the habitable zone in which humans are able to live.

ii)Approaches

- Geography **can be approached by considering a topical - regional continuum**
- The **topical (systematic) approach** - takes up particular categories of physical or human phenomena as distributed over the Earth
- **The Regional approach-** focuses on concerns with the associations within regions of all or some of the elements and their interrelationships

iii) Themes of Geography

- Geography has **five basic themes** namely
    1. ***Location***
    2. ***Place***
    3. ***Human-environment interaction***
    4. ***Movement and***
    5. ***Region***

1. **Location** is defined as **a particular place or position**(Absolute and relative)

2. **Place** refers to **the physical and human aspects of a location**.

- This theme is associated with **toponym** (**the name of a place**), **site** (the **description of the features of the place**), and **situation** (the **environmental conditions of the place**).
- **Each place in the world has its unique characteristics** expressed in terms physical (climate topography, etc) and human elements (population, culture)
- The concept of "**place" aids geographers to compare and contrast two places on Earth**.

3. Human-Environment Interaction

- humans are an active elements in the ecology of the earth
- human-environment interaction involves three distinct aspects:
  - **Dependency**- Dependency refers to the ways in which humans are dependent on nature for a living/ **environmental determinism**
  - **Adaptation**- Adaptation relates to **how humans modify themselves**, their lifestyles and their behavior to live in a new environment with new challenges
  - **Modification** - Modification allowed humans to "conquer" the world for their comfortable living and desirable end / **environmental possiblism/**

4. Movement

- entails to the **translocation of human beings, their goods, and their ideas** within different spatial and temporal scale.

5. Region

- A region is a **geographic area having distinctive characteristics** that distinguishes itself from adjacent unit(s) of space
- It could be a **formal region** that is characterized by homogeneity in terms of a certain phenomenon (soil, temperature, rainfall, or other cultural elements like language, religion, and economy).
- It can also be a **functional or nodal region** characterized by functional interrelationships in a spatial system defined by the linkages binding particular phenomena.

1.2. Location, Shape and Size of Ethiopia and the Horn

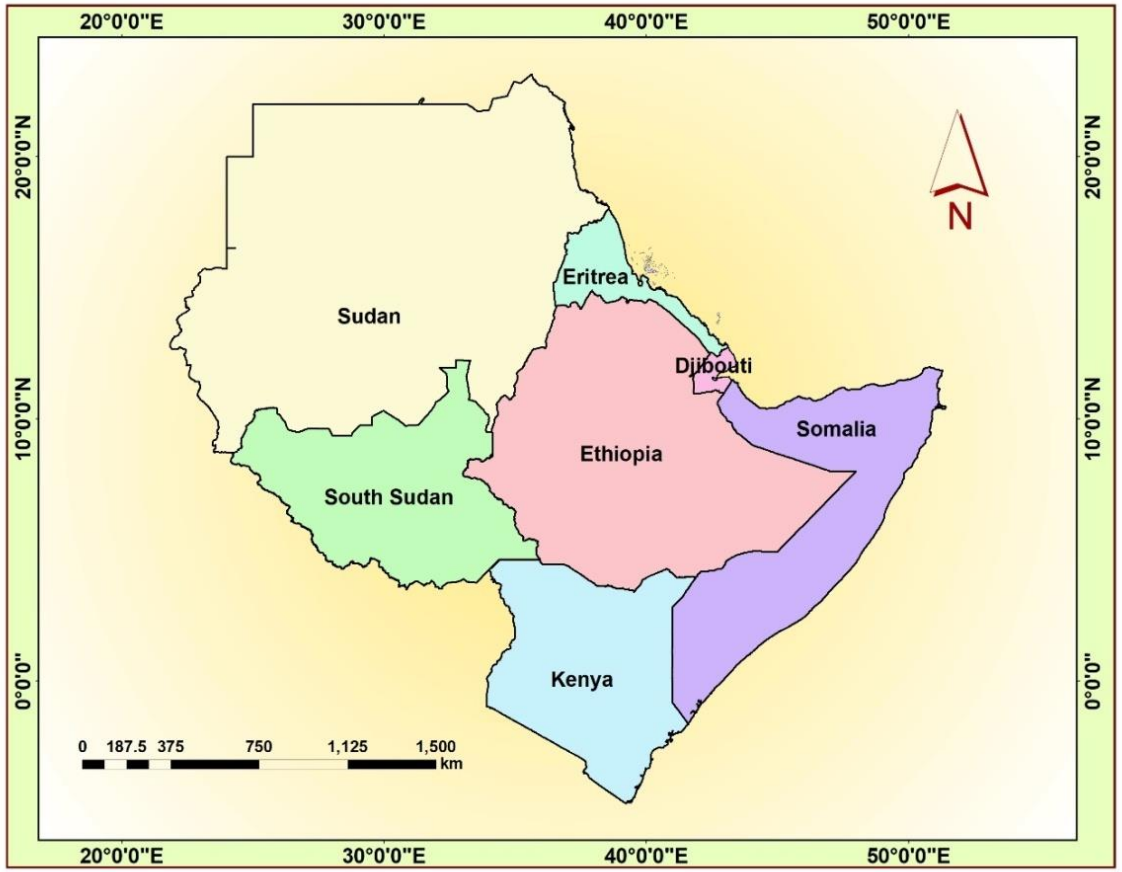
- The **Horn of Africa**, a region of eastern Africa, is a narrow tip that **protrudes into the northern Indian Ocean, separating it from the Gulf of Aden**.
- It includes the countries of **Djibouti, Eritrea, Ethiopia, and Somalia,** whose **cultures have been linked throughout their long history**.
- In terms of size, **Ethiopia is the largest of all the Horn of African countries**, while Djibouti is the smallest.
- The Horn contains **diverse physical elements** like the **highlands of the Ethiopian plateau**, **the Ogaden desert**, and **the Eritrean and Somali coasts** as well as the **human aspects.**

1.2.1. Location of Ethiopia

- The location of a country or a place on a map or a globe is expressed in two different ways. These are **astronomical** and **relative locations**

A) Astronomical location, also known as absolute or mathematical location, states location of places using the lines of latitudes and longitudes.

- Astronomically, Ethiopia is located located between
  - **3°N (Moyale) and 15°N (Bademe** - the northernmost tip of Tigray) **latitudes** and
  - **33°E (Akobo) to 48°E (the tip of Ogaden** in the east) **longitudes**.



- The latitudinal and longitudinal extensions are important in two ways.
  - First, as a result of its **latitudinal extension** the country experiences **tropical climate**
  - secondly due to its **longitudinal extension** there is **a difference of one hour between the most easterly and most westerly points of the country**

B) Relative location expresses the location of countries or places with reference to the location of other countries (vicinal), landmasses or water bodies

- In relative terms, **Ethiopia is a landlocked** country.
- The relative location of Ethiopia is indicated in the following Table 1.1.

Vicinal location	In relation to water bodies & land masses
➲ Sudan to the west and northwest ➲ South Sudan to southwest ➲ Djibouti to the east ➲ Somalia to the east and southeast ➲ Eritrea to the north and northeast ➲ Kenya to the south	➲ In the Horn of Africa ➲ Southwest of the Arabian Peninsula ➲ South of Europe ➲ Northwest of the Indian Ocean ➲ In the Nile Basin

- The relative location of Ethiopia has implications for its:

- **Climate:** The location of Ethiopia relative to the Indian Ocean, the Atlantic Ocean and the African and Asian landmass has also various bearings on the climate of Ethiopia.
- **Socio-cultural**: Ethiopia is one of the earliest recipients of the major world religions namely **Christianity, Islam and Judaism** due to its proximity to the Middle East, which was the origin of these religions. The linguistic and other cultural relationships, which Ethiopia shares with its neighbours, reflect the influence of location.
- **Geopolitics:** the political history of Ethiopia has been considerably influenced by Geopolitical considerations of superpowers, Adjacency to the Red Sea (a major global trade route) and The Middle East geopolitical paradigms. As a result, Ethiopia has been exposed for external invasions in a number of times; though the country resisted foreign intervention and remains free of external domination.
- **Hydropolitics:** Ethiopiacontrols the head water of the blue nile which contributes more than 86%of water of the Nile. The down stream countries have great interest in this river. The country also controls may of the rivers that flow to the neighbouring countries, e.g Wabishebele, Baro-Akobo, Tekeze, Ghibe, etc

1.2.2 Size of Ethiopia

- Ethiopia with a total area of approximately **1,106,000 square kilometers** is the **8th largest country in Africa and 25th in the World**.
- The size of Ethiopia also affects both the natural and human environment of the country. The advantages and disadvantages of the size of Ethiopia are listed below

Advantages	Disadvantages
• Possess diverse agro ecological zones • Variety of natural resources • Own extensive arable land • Has larger population size • Home for diverse cultures • Greater depth in defense external invasion	• Demands greater capital to construct infrastructural facilities • Requires large army to protect its territory • Difficult for effective administration • Difficult for socio-economic integration

1.2.3. The shape of Ethiopia and its Implication

- Countries of the World have different kinds of shape that can be divided into five main categories: **compact, fragmented, elongated, perforated, and protruded**.

- These shapes have **implications on defense, administration and economic integration within a country**.

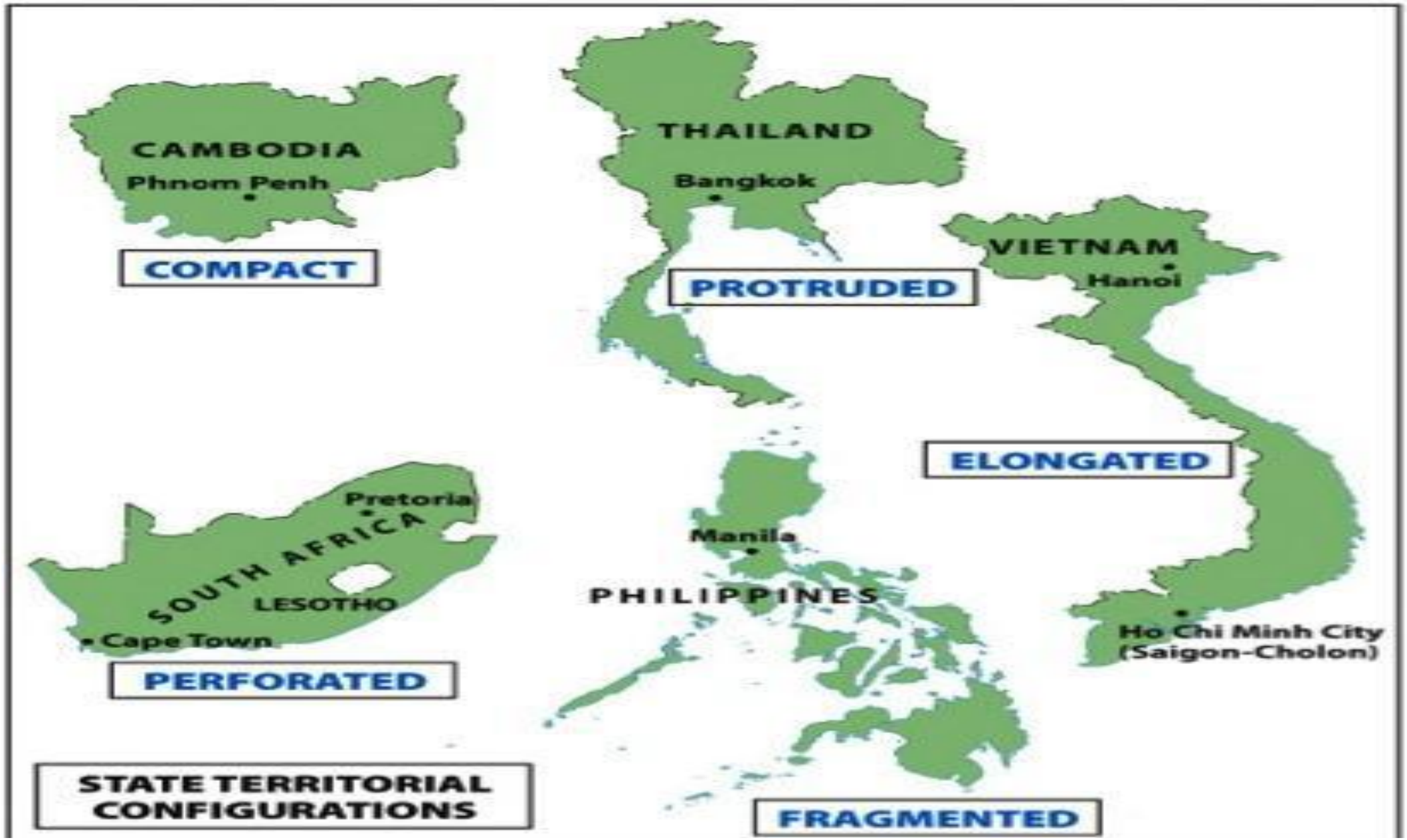
- **Compact shape countries:** The distance from the geographic centre of the state to any of the borders does not vary greatly. It is easier for **defence, socioeconomic integration.**
- **Fragmented shape countries***:* They are divided from their other parts by either water, land or other countries.
- **Elongated shape countries***:* are geographically long and relatively narrow like Chile.

- **Perforated shape countries**: A country that completely surrounds another country like the Republic of **South Africa**.
- **Protrude shape countries**: Countries that have one portion that is much more elongated than the rest of the country like **Myanmar and Eritrea.**

Indices of compactness

- There are various ways of measuring shape of countries. These measures are known as the **indices of compactness**.
- These indices **measure the deviation of the shape of a country from a circular shape**, which is the most compact shape.
- Since there is no country with absolutely circular shape, those approximating a circular shape are said to be more compact

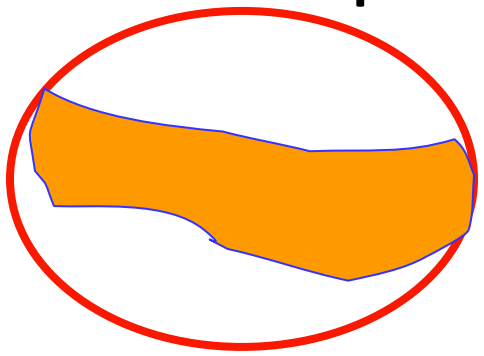
The shape of countries…..



- There are **four most commonly used measures of compactness**. These are:

1. **Area-Boundary (A/B) ratio**: The ratio of area of country to its boundary length. **The higher the A/B ratio, the greater the degree of compactness**.
2. **Boundary - Circumference(B/C) ratio**: The ratio of **boundary length of a country** to the circumference of a circle having the same area as the country itself.
   - It measures how far the boundary of a country approximates the circumference of a circle of its own size.
   - Therefore, **the nearer the ratio to 1 the more compact the country is.**

3. **Area-Circumference(A/C) ratio**. The ratio of the area of the country to the circumference of the smallest inscribing circle.
   - It compares the area of the country with the circumference of a circle that passes touching the extreme points on the boundary of the country.
   - **The higher the A/C ratio, the greater the degree of compactness.**
4. **Area-Area (A/A') ratio**: The ratio of the actual area of a country to the smallest possible inscribing circle
   - The area of the inscribing circle is the area of the smallest possible circle whose circumference passes through the extreme points on the boundary.
   - Half-length of the longest distance between two extreme points gives radius of the inscribing circle.
   - **The nearer the ratio to 1, the more compact the country is**.



Ethiopia's shape compared to some other countries

Country	Area(km^2)	Boundary (km)	A/B ratio	B/C ratio	A/C ratio
Ethiopia	1,106,000	5,260	**210.27**	**1.41**	**296.61**
Djibouti	22,000	820	26.83	1.56	41.83
Eritrea	117,400	2,420	48.51	1.99	96.83
Kenya	582,644	3,600	161.85	**1.33**	215.28
Somalia	637,657	5,100	125.03	1.80	225.22

1.3. Basic Skills of Map Reading

✓A map is a two-dimensional, diminished, simplified representation of part or whole of the Earth surface and its elements as viewed vertically from the above.

✓Although many disciplines use maps, they have a special significance for Geographers as **primary tools for displaying and analyzing spatial distributions, patterns and relations.**

Importance of maps

- Provide the basis for **making geographical details of regions**
- are powerful tools for **making spatial analysis of geographical facts** of areas represented.
- are useful for **giving location of geographical features** by varied methods of grid reference, place naming etc.
- are used on various disciplines like **land use planning, military science, epidemiology, geology, economics, history, archaeology, agriculture etc**.
- make **storage of the geographical data of areas represented**.
- are potentially **used to asses' reliable measurements of the geographical features**. The measurements can be of area size, distance etc.

Types of Map

1. Topographical maps: Topographic maps **depict one or more natural and cultural features of an area.**

- They could be **small, medium or large scale** depending on the size of the area represented.

2. Special purpose/statistical maps: These are maps, which show distribution of different aspects such as temperature, rainfall, settlement, vegetation etc.

Marginal Information on Maps

– These are explanatory notes put on the margin of the maps for the purpose of clear communication between the map and the map user.

- These are **Title, scale, Legend, North arrow, date of compilation** etc…

Basic Principles of Map Reading

- map readers need to have

- ideas about the symbol and also the real World (landscapes)
- knowledge of directions is an important principle in reading maps
- locating *places using latitudes and longitudes or grid references*
  - ✓ left to right,- easting (heading eastward), and
  - ✓northing's (heading in a northward direction).

- 1. What are the implications of location, shape and size of Ethiopia and the Horn on the physical environment, socioeconomic and political aspects?
- 2. Explain the themes of Geography.
- 3. Uses of map and What are the basic principles of map reading?

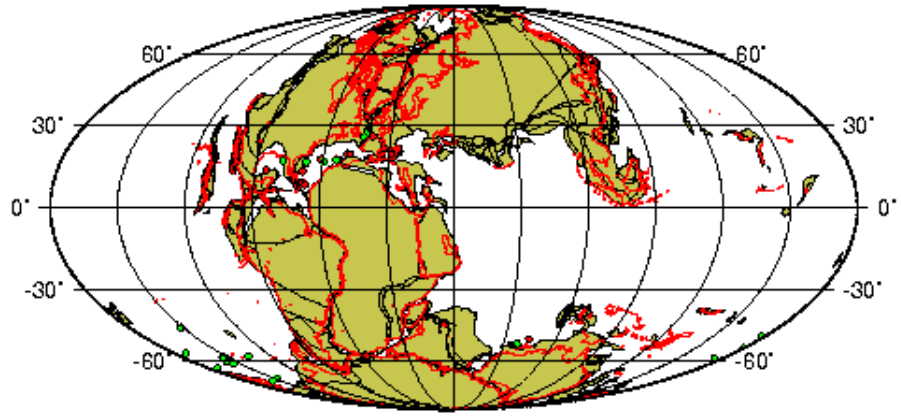
CHAPTER TWO
THE GEOLOGY OF ETHIOPIA AND THE HORN

2.1 Introduction

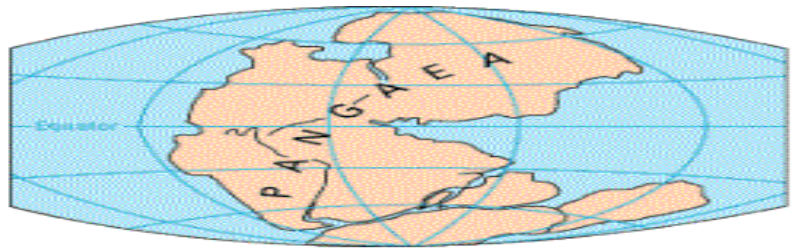
- ✓ **Geology** is an **Earth science** that **studies the evolution of the earth, the materials of which it is made of, and the processes acting upon them**.
- ✓ Much of Geology is concerned with **events that took place in the remote past**
- ✓ Geological understanding must, therefore, be obtained by **inferences and clues**
- ✓ These clues are based on
  - **Rocks and landforms** - Direct Method
  - **Characteristics of seismic waves** – Geo-Physics
  - **Geochemistry** (analysis of the detailed composition of rocks which can give clues as to their origin)
  - **Geochronology** (methods for finding the ages of rocks, usually from the radioactive elements they contain).

ALFRED WEGENER
THEORY OF CONTINENTAL DRIFT

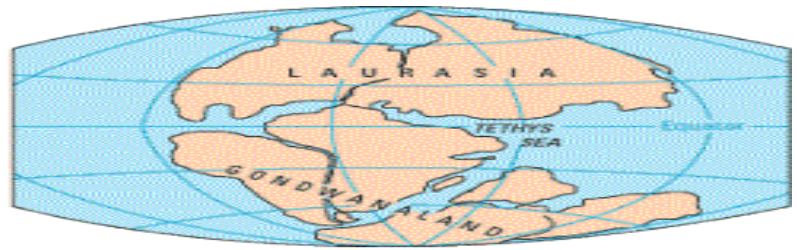
- Alfred Wegener proposed the hypothesis of continental drift in 1911
- The **continental drift hypothesis** proposes that the earth's continents were once assembled to form the super single continent **Pangaea**.
- The large super continent was then split into **Gondwanaland** where Africa is a part and **Laurasia**; and later into smaller fragments over the last million years.
- These then drifted apart to form the present arrangement of continents.



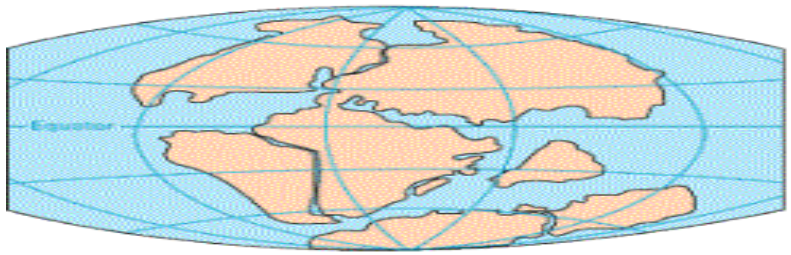
150 My Reconstruction



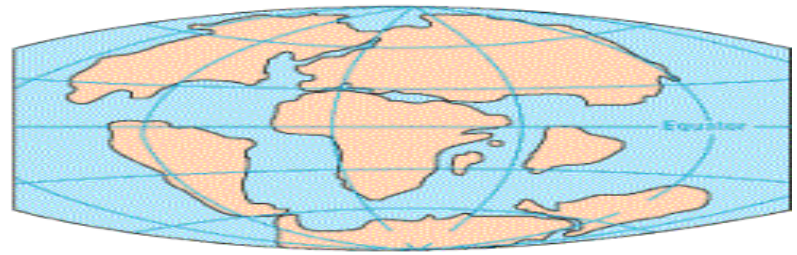
PERMIAN
225 million years ago



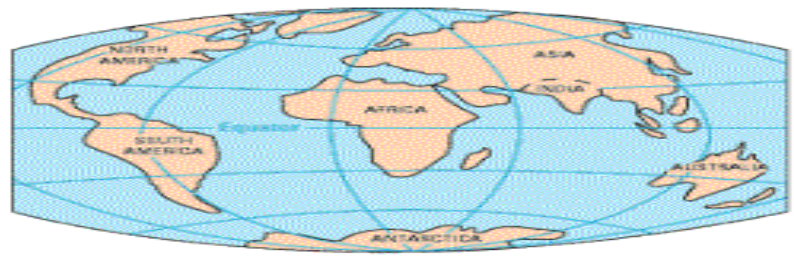
TRIASSIC
200 million years ago



JURASSIC
135 million years ago



CRETACEOUS
65 million years ago



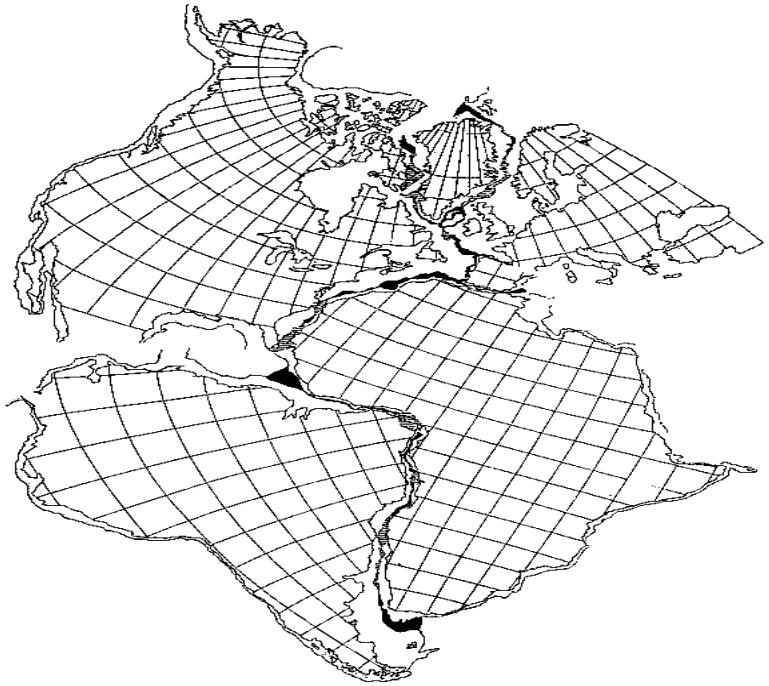
PRESENT DAY

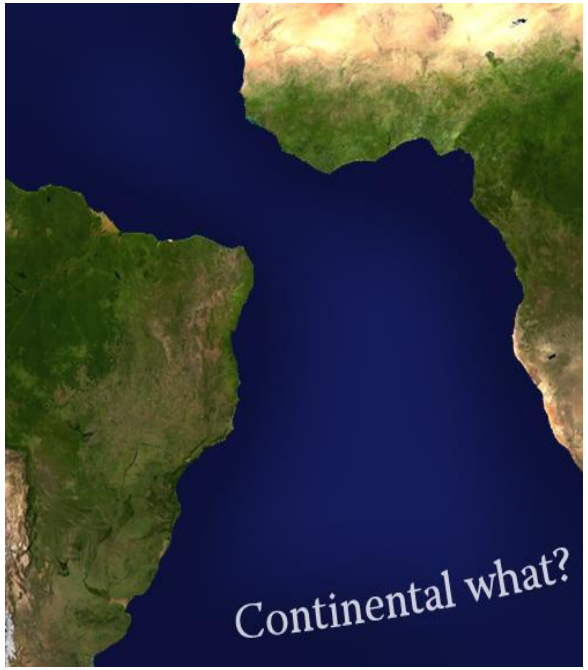
WEGENER'S EVIDENCE

- He gathered information from many different sources and used it as evidence for his hypothesis

1. Structural Evidence (*Fit of the continents):* Continents "fit together" like puzzle pieces

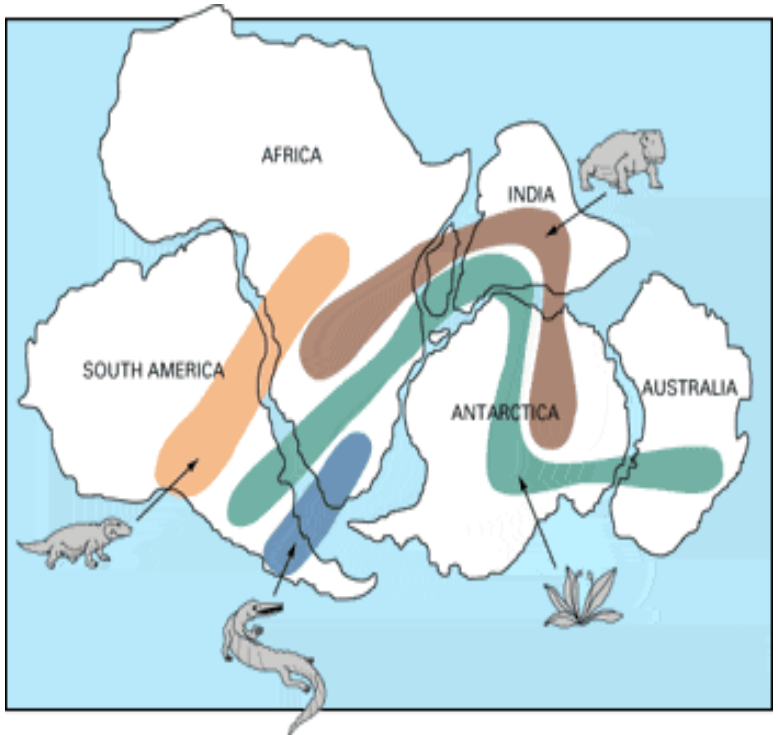
- The opposing coastlines of continents often fit together.





2. Palaeontological - Distribution of fossils:

- The distribution of plants and animal fossils (remains of living things) on separate continents forms definite linked patterns if the continents are reassembled.
- similar fossils have been discovered in matching coastlines on different continents.



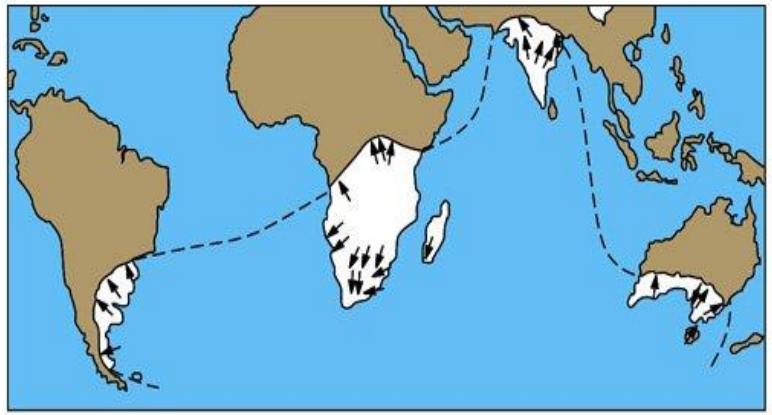
3. Geological - Match of mountain belts, rock types:

- If the continents are reassembled as Pangaea, mountains in West Africa, North America, Greenland, and Western Europe match up.
  - Some mountain ranges on different continents seem to match.
    - Ex: ranges in Canada match Norway and Sweden
    - Ex: Appalachian Mtn. match UK mtn

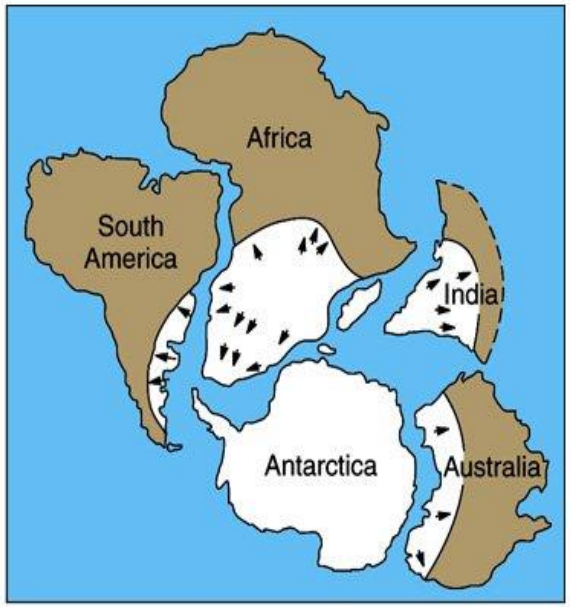




4. Paleoclimatic- : rocks formed 200 million years ago in India, Australia, South America, and southern Africa all exhibited evidence of continental glaciations.



A



B

2.2 The Geologic Processes: Endogenic and Exogenic Forces

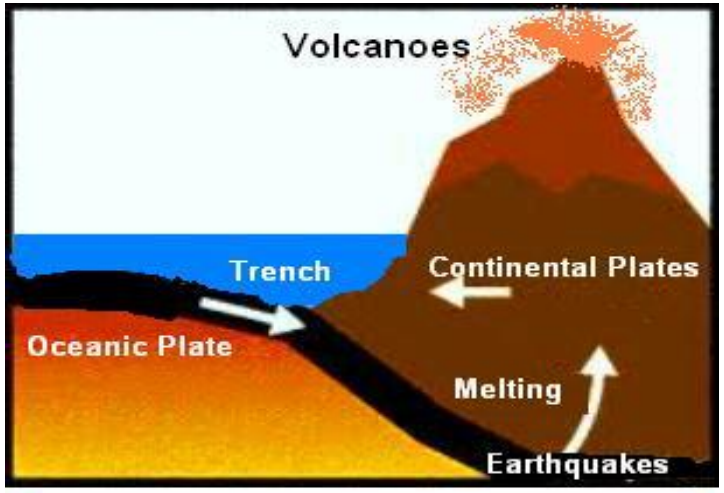
- ✓ Geology studies how Earth's materials, structures, processes and organisms have changed over time.
- ✓ These processes are divided into **two major groups: internal and external processes.**
- ✓The landmass of Ethiopia, as elsewhere, is the result of the combined effect of endogenic and exogenic processes

1. The internal processes (endogenic)

- include **volcanic activity** and all the **tectonic processes** (folding, faulting, **orogenesis** (mountain building), and **epeirogenesis** (slow rising and sinking of the landmass).
- These processes result in **building of structural and volcanic features like plateaus, rift valleys, Block Mountains, volcanic mountains, etc**.

2. The external (exogenic) processes are

- **geomorphic processes**.
- They include **weathering, mass transfer, erosion and deposition**.
- They act upon the volcanic and structural landforms by modifying, roughening and lowering them down.



The internal processes



The external processes

2.3. The Geological Time Scale and Age Dating Techniques

2.3.1 Geological Time Scale

- The Earth is believed to have been formed approximately **4.6 billion** years ago and the earliest forms of **life were thought to have originated** approximately **3.5 billion** years ago.
- **Geological time is difficult to measure precisely.** However, The geological time scale measures time on a scale involving **four main units:**

epoch, period, era, eon

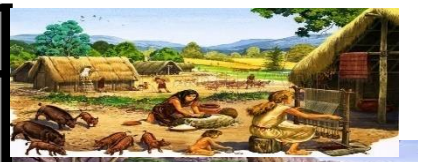
- **An epoch** is the smallest unit of time on the scale and encompasses a period of millions of years.

- Chronologically, epochs are clumped together into larger units **called periods.**
- Periods are combined to make subdivisions called **Eras.**
- An **eon** is the largest period of geological time.
- Two or more geological eras form an Eon, which is the largest division of geologic time, lasting many hundreds of millions of years
- The division of time units in the geological time scale is usually **based on the occurrence of significant geological events (e.g. mass extinctions).**
- The geological time scale, illustrated in Table 2.1, is built largely on the basis **of life and evolution**.

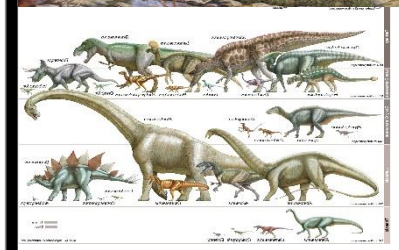
R. Steinberg
DMC 1-12

Geologic Time Scale

Eon	Era	Period	Epoch	Boundary Dates (Ma)
Phanerozoic	Cenozoic	Quaternary	Holocene	0.012
			Pleistocene	2.6
		Neogene / Tertiary	Pliocene	5.3
			Miocene	23.0
		Paleogene / Tertiary	Oligocene	33.9
			Eocene	55.8
			Paleocene	66
	Mesozoic	Cretaceous		146
		Jurassic		200
		Triassic		251
	Paleozoic	Permian		299
		Carboniferous: Pennsylvanian		318
		Carboniferous: Mississippian		359
		Devonian		416
		Silurian		444
		Ordovician		488
		Cambrian		542
PRECAMBRIAN: Proterozoic	Neo-, Meso-, Paleo-	Ediacaran		~635
				2500
Archean				4000
Hadean		*No Rock Record on Earth*		~4600
		ORIGIN OF EARTH		









Note #1: Vertical timeline of boundary dates *is not* drawn with a uniform scale.
Note #2: Boundary dates from the International Commission on Stratigraphy 2010 Geologic Time Scale
Note #3: Carboniferous, Paleogene, and Neogene are more commonly used outside of the U.S.
Note #4: Epochs for the Mesozoic and Paleozoic are too numerous to be shown.
Note #5: The Hadean Eon is not formally recognized.

Table 2.1: The Geological time scale

Era	Period	Began (in Million Years)	End (my)	Major Events (million years ago)
Cenozoic	Quaternary	1.6	Present	Major glaciers in North America and Europe (1.5)
	Tertiary	70	1.6	Rocky Mountains (65), individual continents take shape.
Mesozoic	Cretaceous	146	70	Dinosaurs extinct (65), western interior seaway and marine reptiles (144 – 65)
	Jurassic	208	146	Pangaea (one land mass) begins to break up (200)
	Triassic	225	208	First mammals and dinosaurs
Paleozoic	Permian	290	225	Greatest extinction on Earth (245)
	Pennsylvanian	322	290	First reptiles
	Mississippian	362	322	Coal-forming forests
	Devonian	408	362	First land animals and first forests (408)
	Silurian	439	408	Life invades land
	Ordovician	510	439	First fish appeared
	Cambria	600	510	Great diversity of marine invertebrates
Precambrian	Proterozoic	2,500	600	Marine fossil invertebrates (600)
	Archean	4,500	2,500	Earliest fossils recorded (3,500), earliest rock formation (4,000)

2.3.2 Age Dating Techniques

- There are two techniques of knowing the age of rocks: **Relative and absolute age dating.**
- **Relative dating**
  - geological evidence to assign comparative **ages of fossils**.
  - **use the layers of rock formation and sequences of depositions** and associate to
  - **"What is on top of the older rocks?"**
  - This only works if the area has been undisturbed.



- **Absolute Dating -** Also known as **Radiometric** techniques.
  - **Uses radioactive elements** such as **uranium (U) and thorium (Th)** decay naturally to form different elements or isotopes of the same element.
  - Every radioactive element has its own **half-life**.
  - **Half-life** is the time needed for half of a sample of a radioactive element to undergo radioactive decay and form daughter isotopes.
- Two of the major techniques include:

1. ***Carbon-14 Technique****:* Upon the organism's death, carbon-14 begins to disintegrate at a known rate, and no further replacement of carbon from atmospheric carbon dioxide can take place.
   - Carbon-14 has half-life of 5730 years.

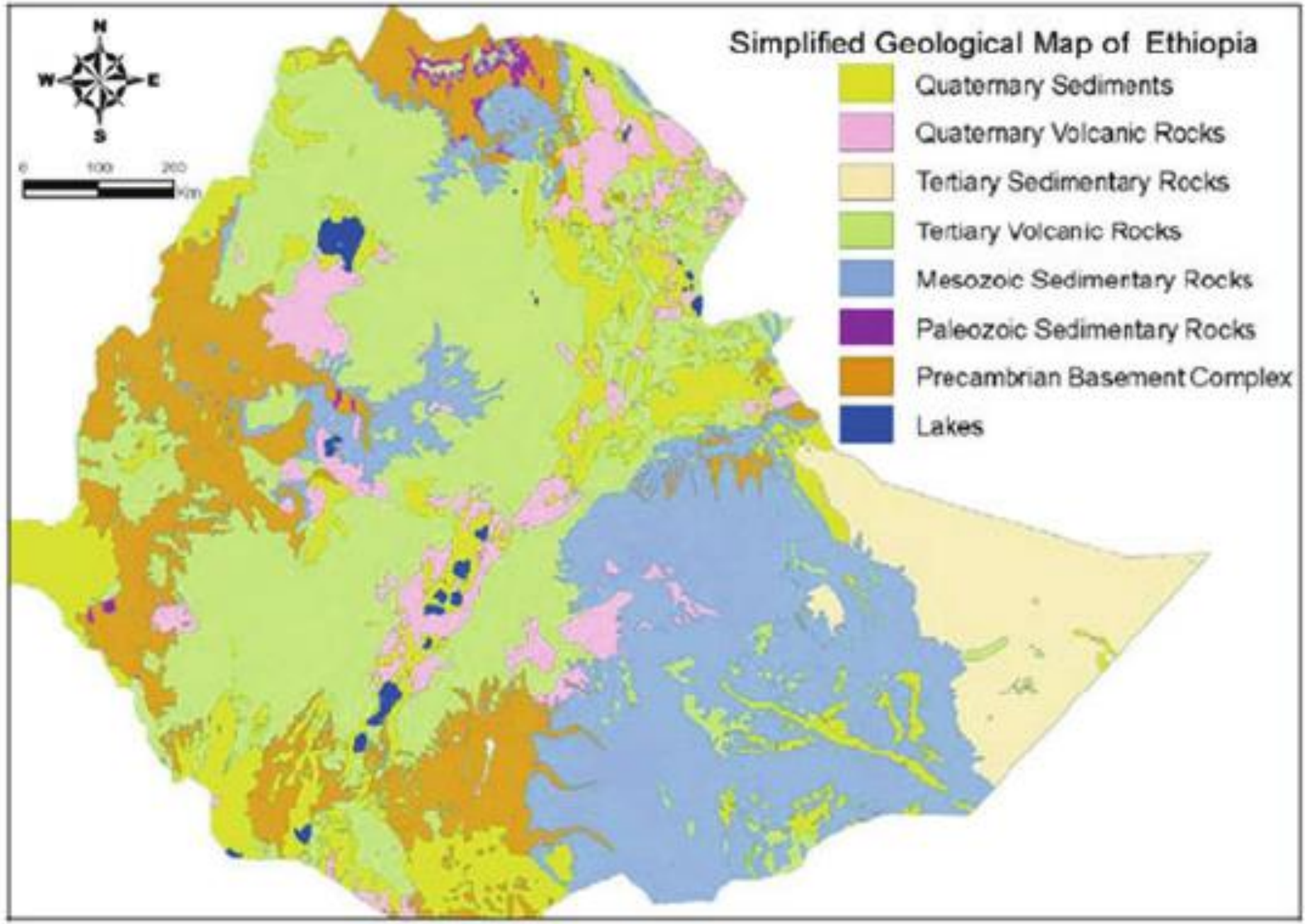
2. ***Potassium-Argon Technique:*** **(K-Ar) Dating**
   - The decay is widely used for dating rocks.
   - Geologists are able to date entire rock samples in this way, because potassium-40 is abundant in micas, feldspars, and hornblendes.
   - Potassium-40 has half-life of 1.3 billion years.

2.4. Geological Processes and the Resulting Landforms of Ethiopia and the Horn

2.4.1. The Precambrian Era Geologic Processes (4.5 billion – 600 million years ago)

- It covers **5/6th** of the Earth's history.
- Due to its **remoteness in time** and the **absence of well-preserved fossils**, our **knowledge of the events is limited**.
- The major geologic event of this Era was **Orogenesis** - that resulted in **intense folding**
- In addition, the Precambrian era was known for extensive **denudation** responsible for **peneplanation**.
- The penepnained land was covered with younger rock formation later in the Mesozoic and Cenozoic eras.

- As such, the Precambrian rocks are overlaid by later rock formations of different ages.
- Despite the Precambrian rocks are overlaid by sequences of rock formation, they are **fond exposed** in the following areas:
  - **In the northern part**: Western lowlands, parts of northern and central Tigray.
  - **In the western Part:** Gambella, Benishangul-Gumuz (Metekel and Asossa), western Gojjam, western Wellega, Illuababora, and Abay gorge.
  - **In the southern Part:** Guji, southern Omo, and parts of southern Bale and Borena.
  - **In the eastern part:** Eastern Hararghe.


Simplified Geological Map of Ethiopia
Quaternary Sediments
Quaternary Volcanic Rocks
Tertiary Sedimentary Rocks
Tertiary Volcanic Rocks
Mesozoic Sedimentary Rocks
Paleozoic Sedimentary Rocks
Precambrian Basement Complex
Lakes
N
W
E
S
0
100
200
Km

2.4.2. The Paleozoic Era Geologic Processes (600million - 225 million years ago)

- The major geological process of this Era was **denudation.**
- The gigantic mountains that were formed by the Precambrian orogeny were subjected to **intense and prolonged denudation**.
- At the end, the once gigantic mountain ranges were **reduced to a "peneplained" surface.**
- Undulating plain with some interspersed residual features (**inselbergs**) here and there were formed.
- Because of the limited deposition within Ethiopia, **rocks belonging to this Era are rare in the country**

2.4.3. The Mesozoic Era Geologic Processes (225-70 million yrs ago)

- This Mesozoic Era lasted for about 155 million years.
- It was an Era of **slow sinking and rising** (**epeirogenesis**) of the landmass.
- This process affected the **whole present-day Horn of Africa and Arabian landmass.**
- As a result the land was **tilted eastward** and therefore **lower in the southeast and higher in the northwest.**
- The three periods of the Mesozoic era experienced epeirogenesis in one form or another.

1. Triassic Period

- During this period, about 225 million years ago, land subsided around the horn of Africa beginning from southeast to northwest of Ethiopia.
- Sinking was followed by sea invasion (**transgression) from the Indian Ocean** starting from **Somalia and Ogaden** and slowly spreading **northwestward**, in the late **Triassic period and continued up to Jurassic period.**

- As the shallow sea spread towards the land, **sands were deposited** over the peneplained Precambrian rock surface.
- These are known as the **Adigrat sand stone.** They are named after place names in Tigray where they might have been first identified.
- As such, the Adigrat sandstone is ***older and thicker in the southeast*** and progressively decreases in age and thickness north-westward

2. Jurassic Period

- As the water stayed longer deposits were also made of remains of plants and animals such as **gypsum and lime** .
- As a result sedimentary rocks known as **Hintalo limestone were formed**.
- Mesozoic rocks are considered to have the greatest potential for **oil and gas** deposits.

3. Cretaceous period

- During this period, the Horn of Africa and Ethiopia, experienced **slow uplifting** and consequently the regression of the sea began in the late Jurassic **and continue throughout the cretaceous period**.
- As the result of the **regression of the sea**, the uppermost layer is known as the **Upper sandstone** was formed.
- The **Upper sandstone**, on the other hand, is thicker and younger (Upper Cretaceous) in **the Southeast**, while in the **Northwest** it is **older and thinner**.

- By the end of the Mesozoic Era, when the land emerged out of the sea, **three major sedimentary** formations were laid and formed upon the Precambrian rock surface.
- These were the **Adigrat or lower sandstone, Hintalo limestone** and **Upper Sandstone.**
- Due to the tilting of the landmass during the transgression and regression of the sea, and due to the direction of the invading and retreating sea, the age and thickness of the Sandstone layers vary in a **Southeast - Northwest direction**.

- The transgressing sea and Mesozoic sediments nearly covered the whole of Ethiopia.
- In most parts of Ethiopia, the Mesozoic rocks are overlaid by the Cenozoic rocks and recent deposits.
- As surface rocks, these old marine sediments are extensively found in the **Southeast lowlands.**
- Other exposures include **central Tigray, and along the gorges of Abay and Wabishebelle rivers**

Major geological events around the Horn of Africa and Ethiopia

era	period	epoch	Geological event	Rock formation
Precambrian			Orogenesis	Basement rock
Paleozoic	**carboniferous**		Denudation and peneplanation	
Meseozoic	**Triassic**		- Subsidence of the Horn of Africa (Epeirogenesis) -Transgression of the Indian Ocean to the Horn of Africa	**Adigrat sandstone**
	Jurassic		-Depth of the sea increased and life formed	**Hintalo limestone, shale, gypsum**
	Cretaceous		- Uplifting of the Horn area (Epeirogenesis) - Regression of water to the Indian Ocean	Upper sandstone mainly along the shore of the Indian Ocean
Ceneozoic	**Tertiary**	**Paleocene**	Uplifting of the horn of Africa	
		Eocene	Uplifting of the horn of Africa followed by fracturing and out pouring of magma	**Trappean lava**
		Oligocene	Uplifting of the horn of Africa followed by fracturing (rift valley formation) and out pouring of magma (formation of lava plateau)	
		Miocene	Uplifting of the horn of Africa followed by fracturing (rift valley formation) and out pouring of magma (formation of lava plateau)	
		Pliocene	Tectonicand Volcanic processes	**Aden volcanic series**/ scoriaceous basalt
	Quaternary	**Pleistocene**	Tectonic and Volcanic processes - Pluvial rain	Aden volcanic series/ scoriaceous basalt - lacustrine deposits, Fluvial deposits, Glacio-fluvial deposition, coastal and marine deposition
		Holocene	volcanic activities,	

2.4.4. The Cenozoic Era Geologic Processes (70million years ago - Present)

- The Cenozoic Era is the **most recent of the geologic Eras**.
- The **tectonic and volcanic activities** that took place in this Era have an important effect in the making of the present-day landmass of Ethiopia and the Horn of Africa.
- These geologic activities are:

1. **Uplifting of the Arabo-Ethiopian landmass and outpouring of huge quantity of lava.**
2. **Formation of the Rift Valley.**
3. **Quaternary volcanism and deposition.**

1. Uplifting of the Arabo-Ethiopian landmass and outpouring of lava flood

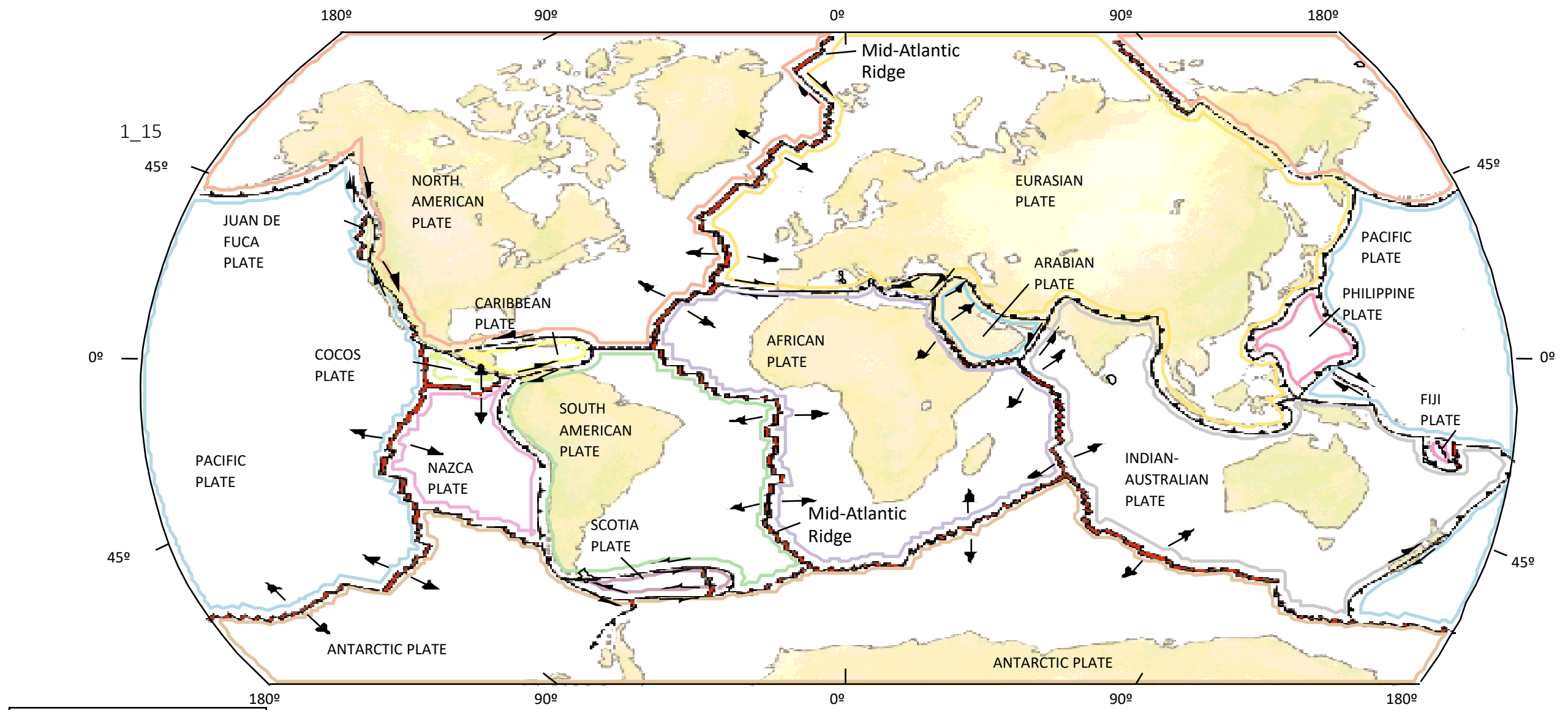
- The uplifting of the Arabo-Ethiopian landmass is **a continuation of the slow rise that began in the Upper Jurassic and Cretaceous periods**.
- This huge uplift continued to the **Paleocene and Oligocene epoch** of the Tertiary period.
- Where the uplifting was of greater magnitude, the land was pushed up to a maximum height of **2,000 meters** above sea level particularly during the **Eocene epoch**.
- This uplifting was of an **epeirogenic** character.
- The greatest uplift was in **central Ethiopia**.
- This **immense tectonic force** also fractured the crust at many places and **huge quantity of lava** came out through these fractures.

- The out pouring of this flood of basalt spread widely and extensively and covered a large part of the Mesozoic sedimentary layer **to form the Ethiopian plateau surface** and also the **floor of the present-day Rift Valley.**
- The mass of lava formed a thick layer of **volcanic rocks on the plateau**, which mounted to more than **1,000 meters above sea level in the north Central Highlands**.
- The lava comes out through vents, huge volcanic rocks were piled up to form many and huge **Volcanic Mountains**' towering the flat basaltic plateau.
- This volcanic material is known as **Trappean lava or Trap Series lava.**
- Where it is not dissected by erosion, the Ethiopian plateau had flat and nearly horizontal surface.
- This is an expression of the peneplaned Precambrian surfaces, the smothering effect of the Mesozoic deposition, and the spreading of the Trap series lava over the Mesozoic sediments.

2. The Formation of the Rift Valley

- The formation of the Rift Valley is said to be related with **the theory of plate tectonics**.
- According to the theory, lateral movement of the crust in opposite directions producing **tensional forces**(in the **divergent plate boundaries**) caused parallel fractures or faults on the sides of the up-arched swell.
- As the tension widened the fractures, the central part of the landmass collapsed to form an extensive structural depression known as the **Rift Valley.**
- The major faulting movement probably began in the late **Oligocene and Miocene Epochs of the Cenozoic Era**.
- But the major rifting, affecting the whole African Rift System, including that of Ethiopia and the Gulf of Aden took place in the **Miocene Epoch**.

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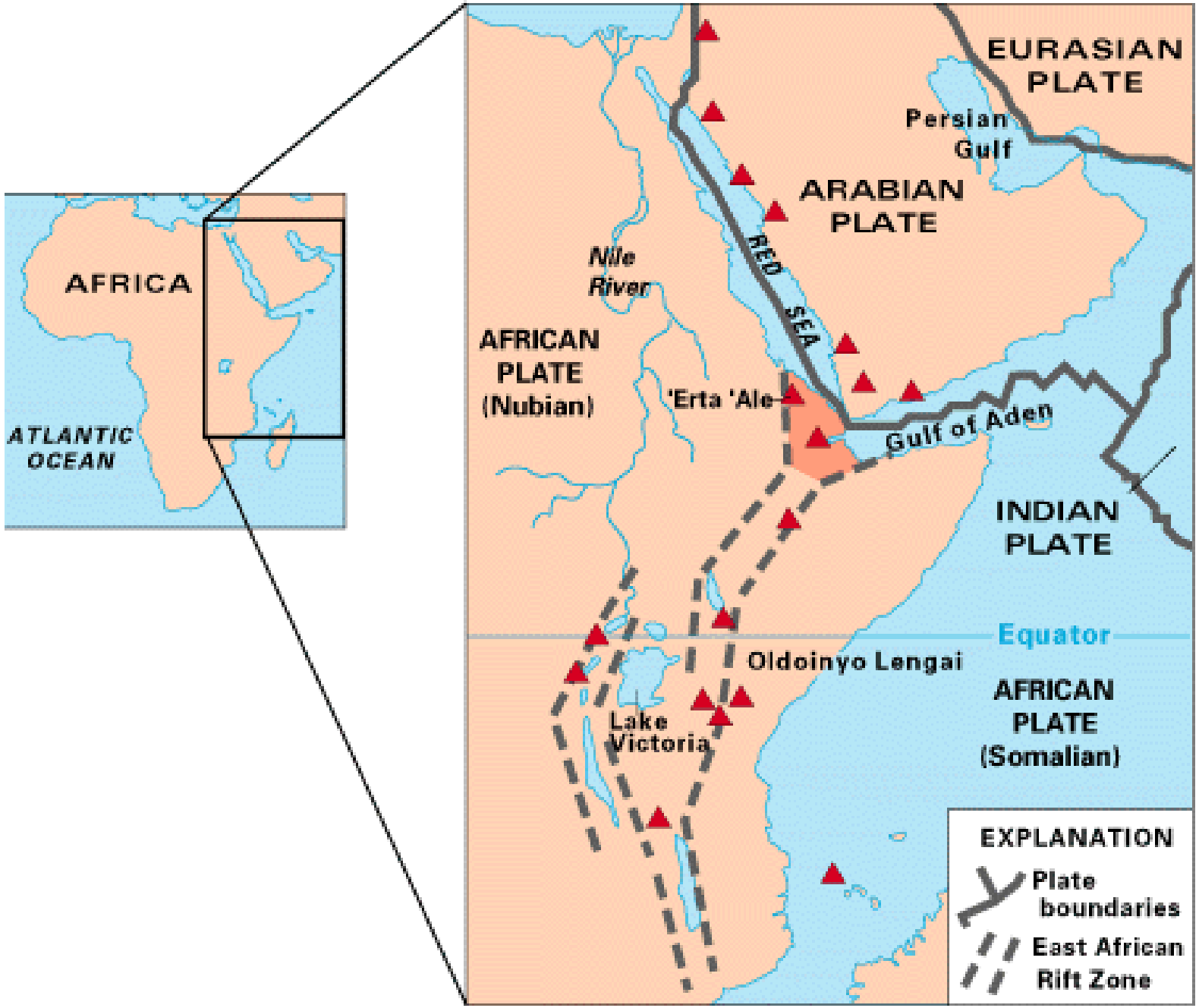


Convergent plate boundary

Divergent plate boundary

Transform plate boundary

Seven or so major plates, about an equal number of small plates


AFRICA
ATLANTIC OCEAN
EURASIAN PLATE
Persian Gulf
ARABIAN PLATE
Nile River
RED SEA
AFRICAN PLATE (Nubian)
'Erta 'Ale
Gulf of Aden
INDIAN PLATE
Equator
Oldoinyo Lengai
Lake Victoria
AFRICAN PLATE (Somalian)
EXPLANATION
Plate boundaries
East African Rift Zone

- Rifting and faulting, however, continued all the time throughout the **Pliocene and even the Pleistocene Epochs**.
- The Red Sea and the Gulf of Aden were connected as a result of the rifting and faulting of the land bridge that separated them.
- **This rifted the Red Sea trough, which began to be flooded from the north.**
- During the Pliocene, the Afar depression (including the Gulf of Zula) was down-faulted allowing the Red Sea water to penetrate far inside.
- **Reversed tilting and volcanic activity**, during the Pleistocene epoch blocked the connection and isolated the extension of the sea, allowing much of the water to evaporate. **As a result, thick saline materials accumulated.**
- During the same period, the area between the Danakil Depression and the Red Sea was uplifted to form the Afar Block Mountains.

- The Ethiopian Rift Valley is part of the Great East African Rift system that **extends from Palestine-Jordan in the north to Malawi-Mozambique** in the south, for a distance of about **7,200 kilometers**. Of these, **5,600 kilometers** is in Africa and **1,700 kilometers in Eritrea and Ethiopia**.
- On land, **the widest part of the Rift Valley is the Afar Triangle (200-300 km).**
- **The Red Sea, meet and form the triangular depression of the Afar, Gulf of Aden, and the East African System** where the **Kobar Sink lies about 125 meters below sea level**.
- The formation of the Gulf of Aden and the separation of the Arabian Peninsula from the Horn of Africa also took place during the **Tertiary period**.
- The **Rift Valley region of Ethiopian is the most *unstable*** part of the country.
- There are numerous **hot springs, fumorales, active volcanoes, geysers, and frequent earthquakes.**

- The formation of the Rift Valley has the following structural (physiographic) effects:
    - **It divides the Ethiopian Plateau into two.**
    - **It separates the Arabian landmass from African landmass.**
    - **It causes the formation of the Dead Sea, Red Sea and the Gulf of Aden troughs.**
    - **It creates basins and fault depressions on which the Rift Valley lakes are formed**.
- Faulting and graben formation are not only limited to the Rift Valley. For example, similar tectonics activities have occurred in the **Lake Tana Basin**.
- However, the formation of **Lake Tana** had been accentuated by volcanic activity so that lava flow in the southeast had dammed part of the rim to deepen the basin.
- Faulting in other places had a structural control along some part of the river courses, e.g Awash River

3. Quaternary Volcanic Eruptions and Depositions

Quaternary Volcanic Eruptions

- They are **recent volcanic activities known as Aden volcanics** that took place **after the formation of the Rift Valley.**
- This occurred in the **Pliocene-Pleistocene Epochs**.
- This is a **continuation of the tectonic and volcanic processes** that earlier affected the Rift Valley formation.
- This activity was generally **limited to the floor of the Rift Valley and the region south of Lake Tana,** where the lava covers an area of more than 3,000km^2.
- **Aden volcanic** and recent faulting are more extensively developed in the **Afar region**.
- The area is shattered by numerous faults and subjected to extensive **scoriaceous basalt eruptions**.
- Because of their recent occurrence, the Aden volcanic have relatively **well-preserved and visible morphological features**.

- The basic volcanic features of the Aden series include the following:
  - Numerous and freshly preserved **volcanic cones**, many of which have **explosive craters**. Some of these are active **Dubi, Erta Ale, Afrera** etc. Of these, **Erta Ale is the most active volcano** in Ethiopia.
  - **Volcanic hills and mountains**, some of which are semi-dormant (Fantale, Boseti-Gouda near Adama, Aletu north of Lake Ziway, Chebbi north of Lake Hawassa etc.).
  - **Extensive lava fields and lava sheets** some of which are very recent.
  - **Lava ridges**.
  - **Thermal springs, fumaroles** etc.

Quaternary deposits

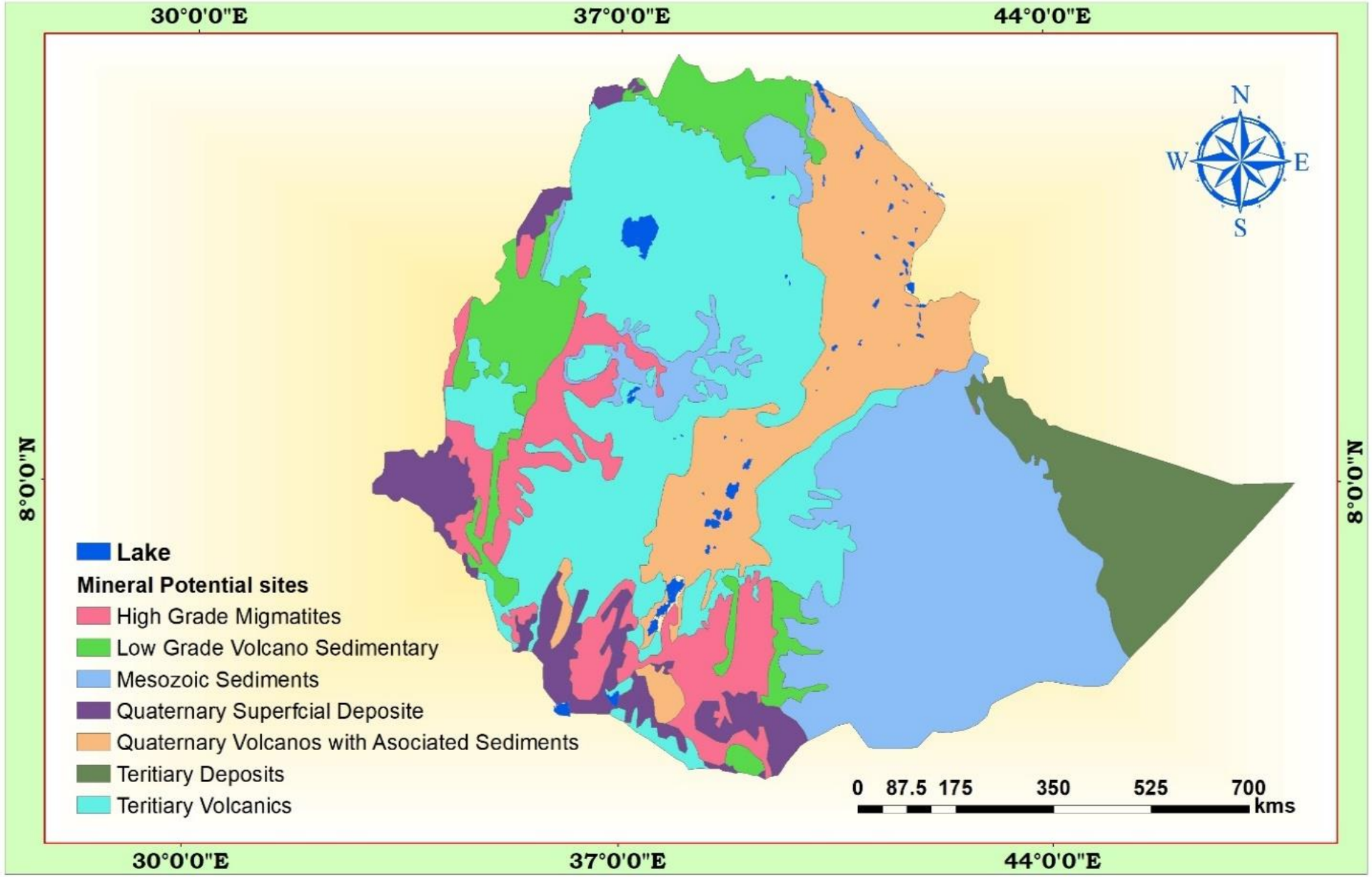
- During the Quaternary period of the Cenozoic Era, the Earth experienced a **marked climatic change**, **where warmer and dry periods were alternating with cooler and wet periods**.
- This was the time of the last "**Ice Age**" in the middle and high latitude areas and the time of the "**Pluvial Rains**" in Africa.

- The heavy Pluvial Rains **eroded the Ethiopian plateau** and the eroded materials were **deposited in the Rift Valley lakes**.
- Lake and marshy areas became **numerous and deep**. Many were enlarged and covered much area and even merged together.
- For example, **Ziway-Langano-Shalla**; **Hawasa-Shallo; Chamo-Abaya; and Lake Abe and the nearby smaller lakes and marsh basins** formed huge lakes.
- After the "Pluvial Rains", the Earth's climate became **warmer and drier.**
- Thus, it **increased the rate of evaporation that diminished the sizes of the lakes.**
- Today, there are **lacustrine deposits of continental origin around many of the Ethiopian lakes, river valleys and lowlands.**

- According to the **place** and **manner of deposition** and **depositing agents** these deposits are divided as follow.
    - ***Lacustrine deposits***: Deposits on former lakebeds, and swampy depressions.
    - ***Fluvial deposits***: Deposits on the banks of rivers, flood plains both in plateau, foothills etc.
    - ***Glacio-fluvial deposits and erosional features***: These are occurred on high mountains, such as Bale and Kaka Mountains.
    - ***Aeolian deposits***: Are windblown deposits.
    - ***Coastal and marine deposits***: Deposits on sea invaded and sea-covered places.
    - The quaternary deposits are mainly found in the **Rift Valley (Afar and Lakes Region), Baro lowlands, southern Borena, and parts of northwestern low lands.**
    - Generally, **the Cenozoic rocks cover 50% of the land mass of the country.** These include **Highland Tertiary volcanic (basalts), Tertiary as well as Quaternary volcanic, and sediments of the rift valley.**

2.5. Rock and Mineral Resources of Ethiopia

- **The occurrence of metallic** minerals in Ethiopia is associated with the **Precambrian rocks**.
- A great variety of such minerals occur in the **basement rocks**.
- These rocks contain most of the **metallic deposits** known at present.
- The exploitation and search for mineral deposits in Ethiopia has been taking place **for the past 2,000 years or so**, and its early cultures were based partially on the mineral wealth of the day.
- Such has been the case of **gold production** and utilization, which has become part of Ethiopia's history, tradition and folklore (legend).
- The mining and working of **iron** for the manufacture of tools, utensils and weapons, and the use of **salt and salt-bar** all these indicate to a fairly **long mining tradition**.
- However, presently mineral production from Ethiopia has been **negligible by World standards**.


30°0'0"E
37°0'0"E
44°0'0"E
8°0'0"N
N
W
E
S
Lake
Mineral Potential sites
High Grade Migmatites
Low Grade Volcano Sedimentary
Mesozoic Sediments
Quaternary Superfcial Deposite
Quaternary Volcanos with Asociated Sediments
Teritiary Deposits
Teritiary Volcanics
0 87.5 175 350 525 700
kms

2.5.1. Brief Facts and Current State of Main Minerals in Ethiopia

- Geological surveys proved that **Ethiopia has abundant mineral resources of metals and precious metals, coal, and industrial minerals**.

Gold

- has been mined in Ethiopia for quite long time, mainly from **Benishangul-Gumuz (Metekel) and Adola.**

- Operating mines produce gold from **primary sources** in such localities as **Dermi-dama, Sakoro and Lega-dembi.**

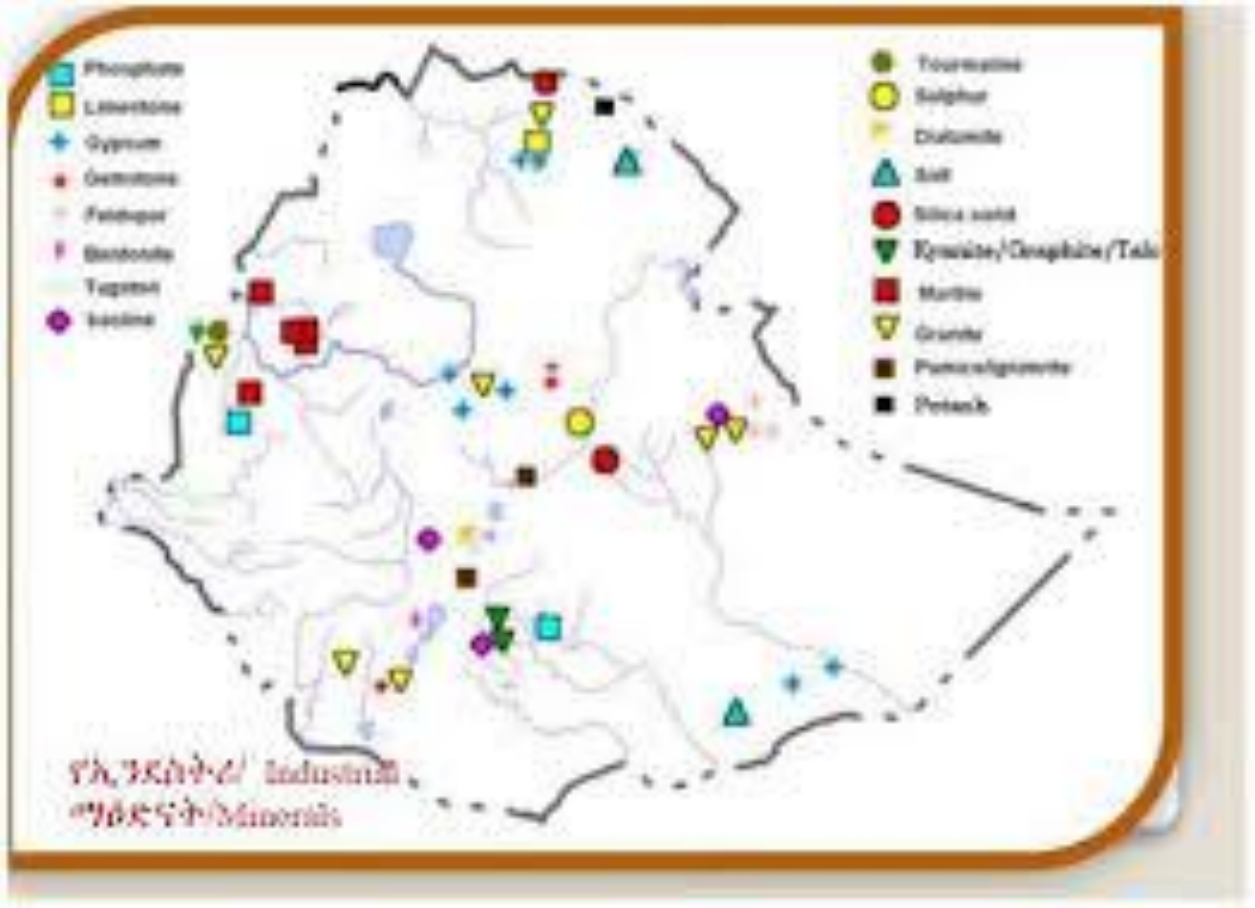






- **Mechanised alluvial working** is confined to the state operated gold field of **Adola.**

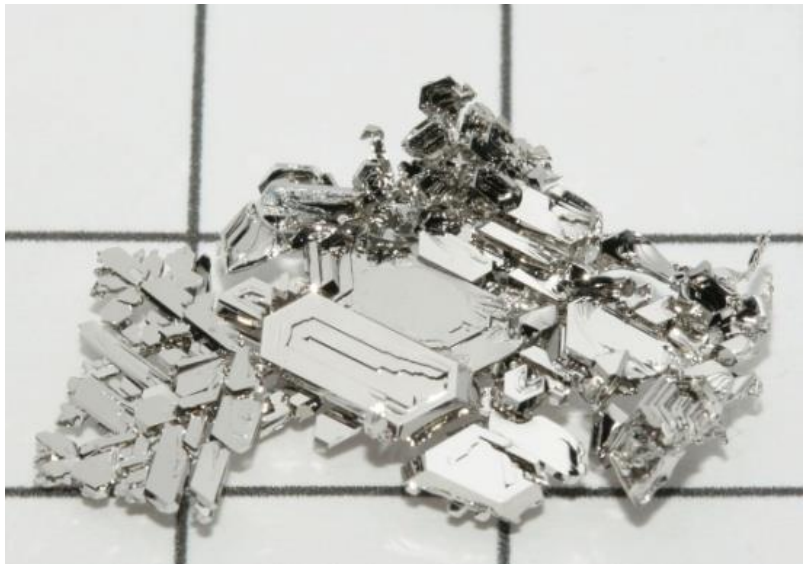
- **Secondary gold deposits** are common in the following localities: **Adola**, Murmur Basin, **Shakiso**, Awata Basin, **Dawa Basin**, **Ghenale Basin**, Ujama Basin, Makanisa (Guba and Wombera), **Kaffa In Gambella and Illuababora** (Akobo River), in **Sidama** (Wondo), **Borena** (Negele-Yabelo area) and in **Benishangul-Gumuz** (Sherkole), **West Wellega**, Mengi-Tumat-Shangul areas to the Sudanese border, and the drainage of the Didessa and Birbir.



Platinum

- used extensively for **jewellery**. Its main use, however, is in **catalytic converters for cars, trucks and buses.**
- The **Yubdo area in Wellega**, is the only active Ethiopian Platinum mine. Platinum occurrences have been reported from **Delatti in Wellega, and the valley of Demi-Denissa and Bone Rivers as well as Tullu Mountain area in Sidama**.





Tantalum:

- used in **the electronics industry for capacitors and high power resistors**
- Significant deposit of tantalum and niobium is found in southern Ethiopia mainly in **Adola area where Kenticha Tantalum mine** with resources of more than 17,000 metric tons of **world class** ore reserve is found.





Lignite/ low quality coal/

- **Extensive lignite deposits** in Ethiopia are found in **Nedjo (Wellega), and in small amounts in Chilga (Gonder)** are found in the sedimentary formations laid in between Trapean lava.
- However, **important Lignite,** one of the **lowest ranked coal**, is known to occur in many localities such as in the Beressa Valley and **Ankober** (North Shewa), Sululta (near Addis Ababa), Muger Valley (West Shewa), Aletu valley (near Nedjo), Kariso and Selmi Valleys (Debrelibanos), Zega wodem gorge (near Fiche), **Didessa Valley** (southwest of Nekemte), Kindo and Challe Valley (Omo confluence), Adola,Wuchalle (north of Dessie), Chukga area (on Gonder-Metema road), Dessie area (near Borkena River).
- These areas are promise to be a good prospect to meet some of the local industrial and domestic needs.



Gemstones

- a piece of mineral crystal which, in cut and polished form, is used to make jewelry or other adornments.
- Gemstones, including amethyst, aquamarine, emerald, garnet, opal, peridot, sapphire, and tourmaline occur in many parts of Ethiopia, mainly in Amhara and Oromia Regional States.
- Quality **Opal** was first discovered by local people in Wadla and Dalanta woredas, North Wello in Amhara Regional State.





Potash

- is primarily **used in fertilizers**
- The potash reserve in the **Danakil (Dallol Depression**) of the Afar region is believed to be significant.





Gypsum

- **used as a fluxing agent, fertilizer, filler in paper and textiles, and retarder in portland cement.**
- A limited amount of gypsum is produced for domestic consumption in Ethiopia, mainly for the **cement industry**, but very large deposits are known to occur in sedimentary formations of **the Red Sea coastal area, Danakil Depression, Ogaden, Shewa, Gojjam, Tigray, and Hararghe.**
- Total reserves are probably enormous because the thickness of the gypsum deposits is many hundreds of meters and the formation are known to extend laterally for hundreds of kilometers.







Clay

- Ethiopia is endowed with industrial clay material.
- Alluvial clay deposits for **bricks and tile, pottery and pipe industry** occur in Adola, Abay gorge, and the Rift Valley lakes region.
- **Ceramic clay** for the production of glasses, plates, bricks is found at **Ambo and Adola**.
- **Tabor ceramic** industry in Hawassa gets most of its raw materials from local sources.





Marble

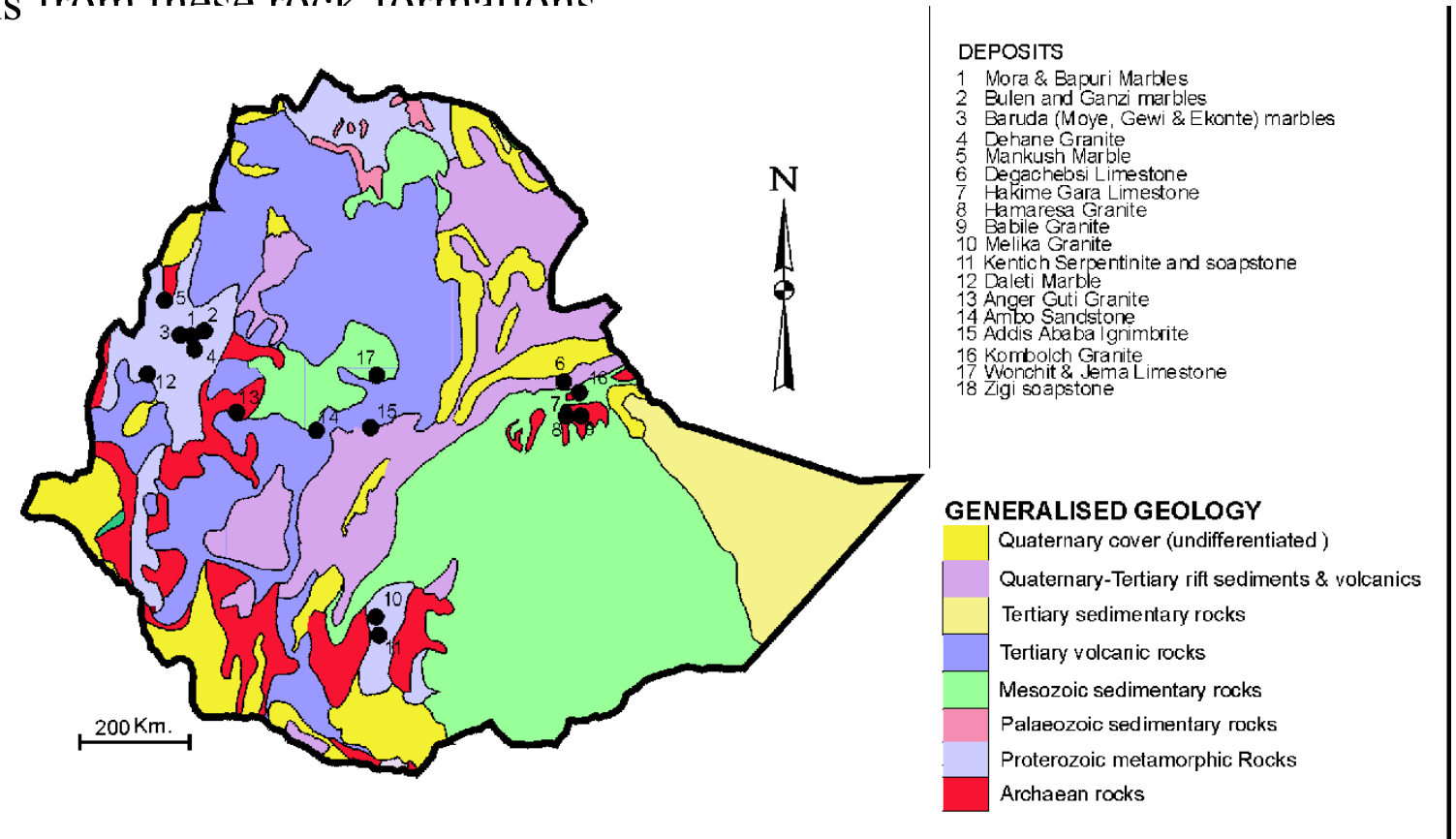
- **Crystalline limestone** is widespread in the basement rocks of Ethiopia.
- Marble has been quarried in such localities as west of **Mekelle and south of Adwa in Tigray.**
- In the east in **Galetti, Soka, Ramis, Rochelle, Kumi and other** valleys of **Chercher Mountain in West Hararghe.**
- In the northwestern also in areas built of Precambrian **schist in Gonder, and the Dabus River and other neighboring river basins in Benishangul-Gumuz and Gojjam.**





Construction stones

- **Basalt, granite, limestone and sandstone** are important building stones.
- For the **surfacing of roads and compaction, basalt, scoria and other volcanic rocks** are extensively used.
- **Mesozoic limestone** is an important raw material for cement and chalk production.
- The earlier cement works at **Dire Dawa** and the recent ones at **Muger Valley, Abay gorge (Dejen), Tigray (Messebo)** are using similar raw materials from these rock formations



Mineral Fuels

- **Petroleum and natural gas**
  - These are **fossil fuels that appear to be organic remains of ancient life accumulated on the sea floor along with sediments**.
  - It is estimated that about 33% of Ethiopia is covered by sedimentary rocks that have economic value for the extraction of petroleum and natural gas.
  - The main areas for such occurrence are **Ogaden, western Ethiopian lowlands, the Abay and Omo valleys.**
- **Coal/lignite**
  - It is said to be the result of decomposed buried forests during the Carboniferous period.
  - Although there is no well developed coal deposit in Ethiopia for a commercial scale mining, Lignite coal is found in **Nedjo (Wellega), Wuchalle(Wollo), Chilga(Gonder), and Abay gorge (Shewa) , Yayu (Illubabaor)**





- Petroleum and Natural gas extraction



The hot steam/Geothermal Energy

- found in association with the hot rocks beneath the surface in some areas of the **rift valley**, which is also a geothermal energy potential region.
- Ethiopia has geologically confirmed large reserves of **geothermal** energy.
- The first geothermal plant is being established North of **Lake Langano** where a promising geothermal source has been found.
- **Thermal springs and fumaroles** are extensively found in the rift valley for therapeutic (curing physical sickness) and recreational value.
- More than **130 thermal springs** are known in Ethiopia of which 30 are situated on the plateau areas, mostly in lake Tana basin and within the rift valley region.





- Geothermal Power

2.5.2. Mineral Potential Sites of Ethiopia

- According to the Ethiopian geological survey, the geologic formations that host **most mineral potentials of Ethiopia** includes **three major greenstone** belts and other formations. These are:
  - **The Western and South-western-greenstone belt**
  - **The Southern greenstone belt:**
  - **The Northern greenstone belt(Tigray):**

1. The Western and South-western-greenstone belt

- They contain various minerals: primary **gold** occurrences (Dul,Tulu-Kape, Oda-Godere, Akobo, Baruda,Bekuji-Motish and Kalaj); **Yubdo Platinum**, **Base metals** of AzaliAkendeyu, Abetselo and Kata;Fakushu Molybdenite and the **iron deposits** of Bikilal, Chago, Gordana and Korre, Benshagul-Gumuz- Marble, Akobo and Asosa placer gold deposits and etc.

2. The Southern greenstone belt:

- It is known as the **Adola belt**, which comprises the primary **gold deposits** and occurrences of Lega-dembi, Sakaro, Wellena, Kumudu, Megado-Serdo, Dawa Digati, Moyale and Ababa River; the columbo-**tantalite** of kenticha and Meleka, and the Adola **nickel** deposit and other industrial minerals.

3. The Northern greenstone belt(Tigray):

- This belt comprises of the primary **gold occurrences** of Terakemti, Adi-Zeresenay, and Nirague.
- The base metals of Terer, Tsehafiemba and other parts of Tigray, Placer gold occurrences of Tigray.

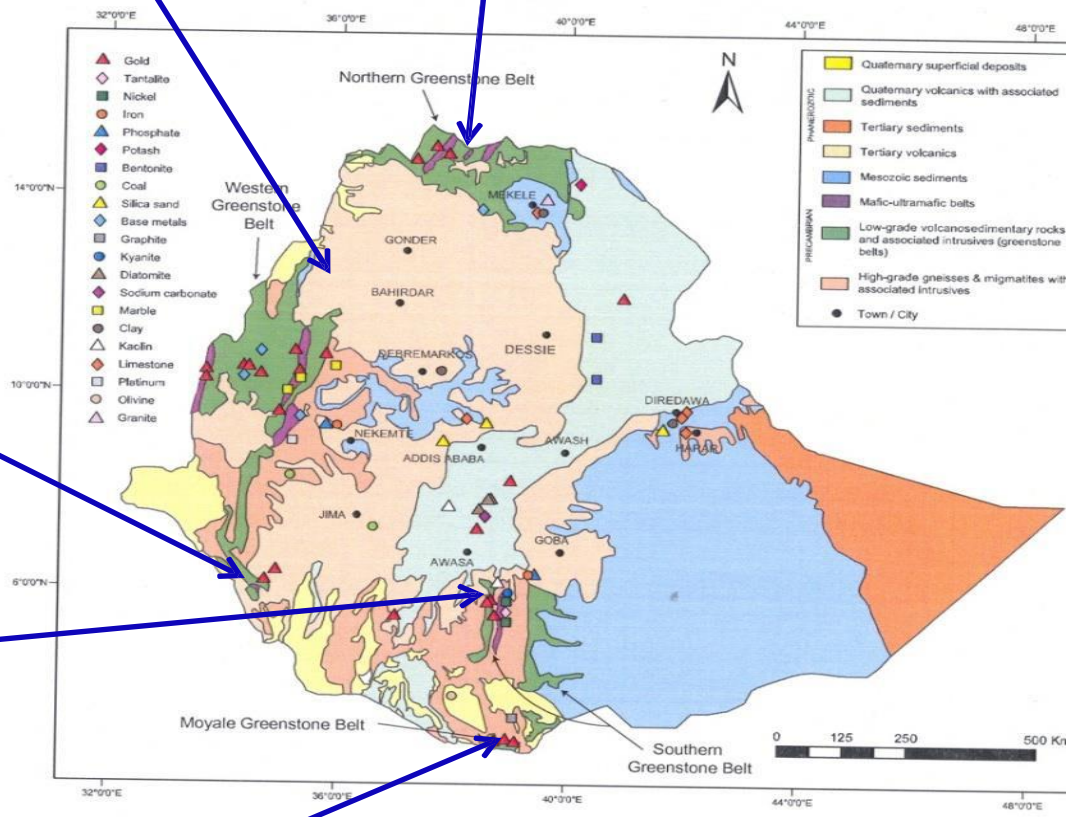
Mineral Potential

Western & SWBelt:-

- Tulu Kape,Dul, Metekel, Chamo, Oda- Godere Pri. **Au** deposits &
- Yubdo Platnium deposit
- Bikilal, Chago ,Gordana, Sirba & Kore iron deposits& mineralization
- Keta & **Abetselo Base metals** Mineraliz.
- Fakusho **Molybdenum** mineraliz.
- **Placer Gold** deposits

Northern Belt:-

- Terkemti, Adi Zersenay, zager
- Primary gold mineralization
- Base Metals mineralization
- Placer gold occurrences & deposits



Southern Belt:- Adola & Moyale

- Lege –Dembi, Sakaro, Moyale, Dawa, OkotePrimary **Gold** deposits & occur.
- Adola **Nickel**
- Kenticha **Tanatalum**
- **Placer gold** deposits
- **Chromite** mineralization

Review Questions

- **Answer the following questions briefly.**

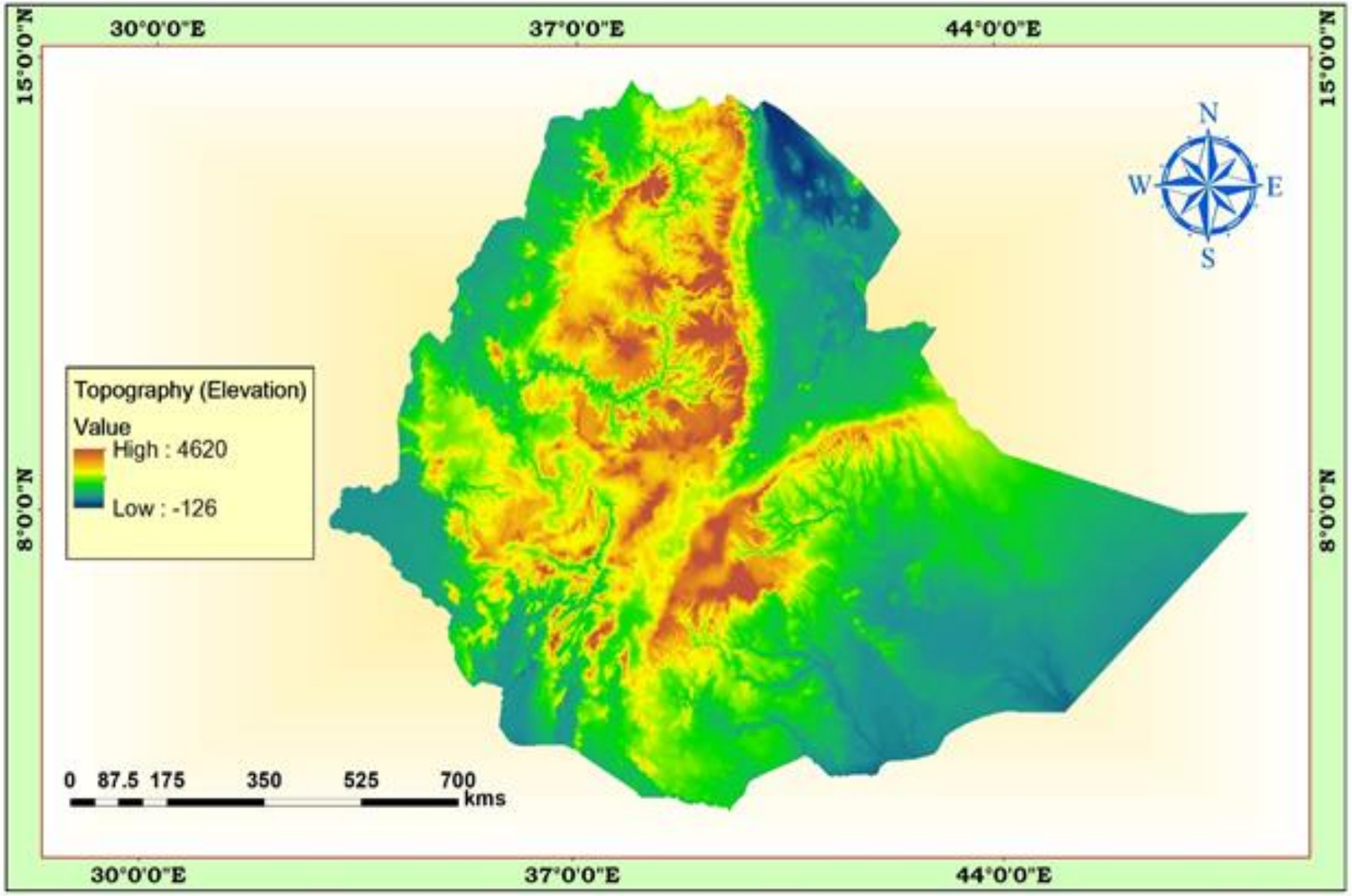
1. What were the major geologic processes of the Precambrian and Paleozoic Eras in the Horn of Africa?
2. Explain the formation of the oldest sedimentary rocks in Ethiopia and the Horn.
3. Discuss the major geologic events of Ethiopia that took place during the Cenozoic Era.
4. Where are the major mineral potential sites of Ethiopia? Discuss each of them with the help of examples.

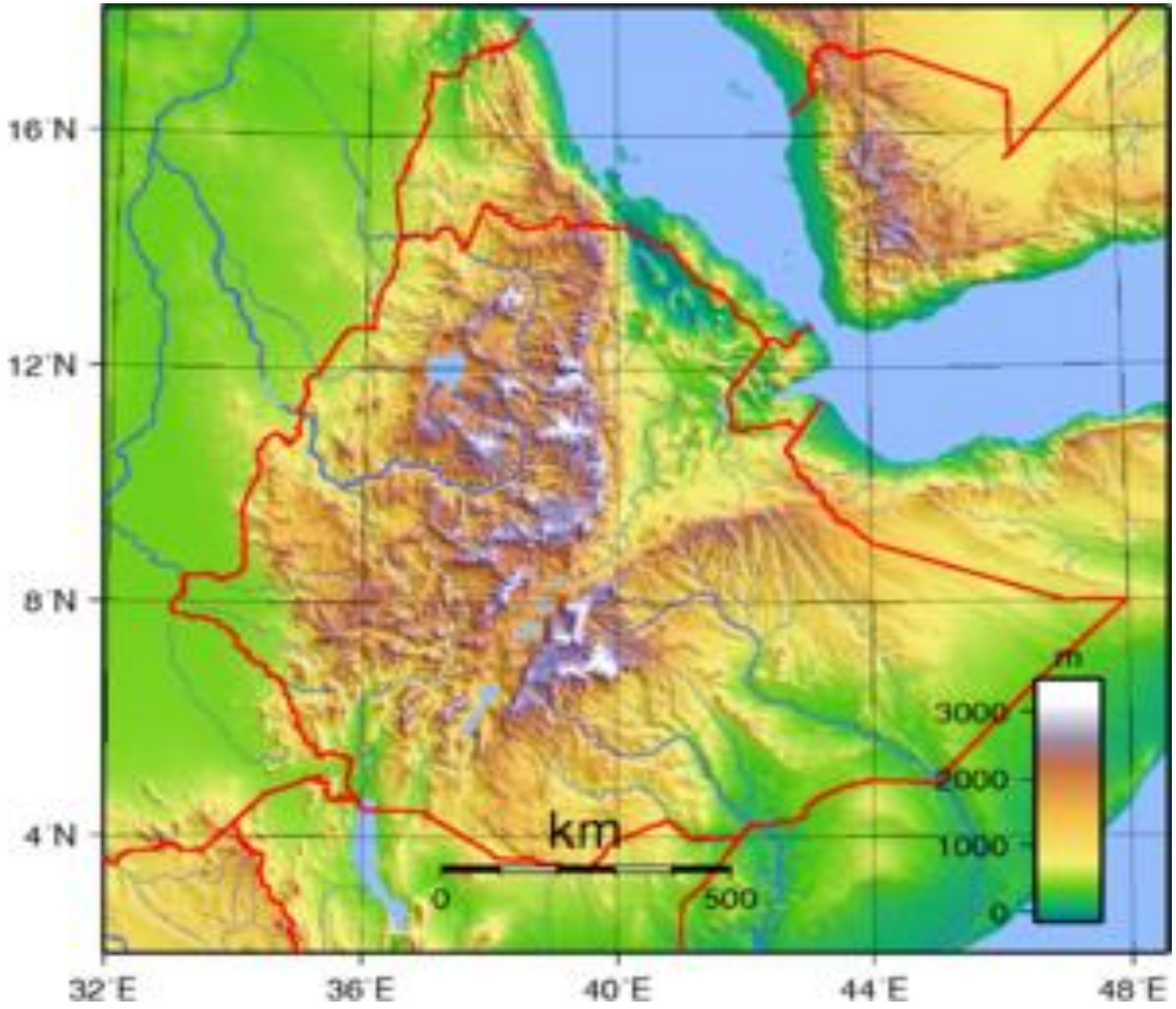
CHAPTER 3:THE TOPOGRAPHY OF ETHIOPIA AND THE HORN

3.1 General Characteristics of the Ethiopian Physiography

- The Ethiopian landform is characterized by great diversity such as **flat-topped plateaus, high and rugged mountains, deep river gorges and vast plains**.
- Altitude ranges from **125 meters below sea level** (**Kobar Sink**) to the highest mountain in Ethiopia, **Mount Ras Dashen (4,620 m.a.s.l),** which is the fourth highest mountain in Africa.
- **Ethiopia has the largest proportion of elevated landmass in the African continent** and it is sometimes described as the **Roof of East Africa** because **more than 50% of the Ethiopian landmass is above 1,000 meters of elevation**; and above 1,500 meters makes 44% of the country.
- For most parts of the country are **sources of many rivers and streams**, and thus the country is described as the "**Water Tower of East Africa**".

Topographic map of Ethiopia




16°N
12°N
8°N
4°N
32°E
36°E
40°E
44°E
48°E
km
0
500
m
3000
2000
1000
0

- Taking the **1,000 meters contour line for the highland-lowland demarcation,** one observes the following contrasting features between the Ethiopian highlands and lowlands:

A/ Characteristics of Ethiopian highlands:

- Moderate and high amount of rainfall (>600 mm per year);
- Lower mean annual temperature (<20^{0}C);
- The climate is favorable for biotic life;
- Rain-fed agriculture is possible;
- Free from tropical diseases;
- Attractive for human habitation and densely settled.

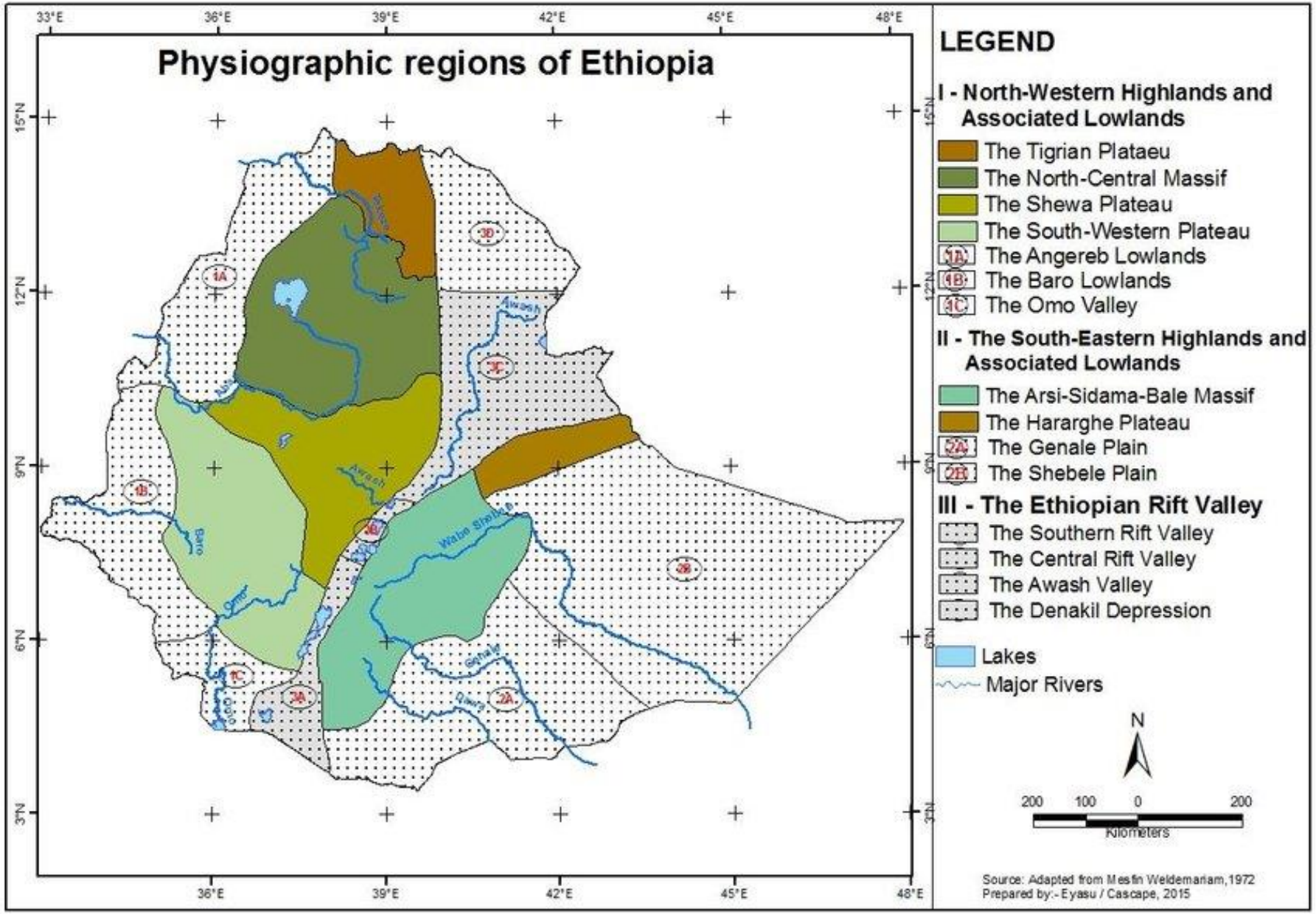
- **Highlands make up nearly 56%** of the area of the Ethiopia.
- The cumulative effect of all this is that, **the highlands have been significant throughout Ethiopian history in the economic, cultural and political life of the people.**
- Highlands further subdivided into lower highland (1,000 - 2,000 m.a.s.l), which make up 35% and higher highland (>2,000 m.a.s.l) constituting nearly 22%.

B/ Characteristics of Lowlands

- Constitute **44%** of the Ethiopian lowlands and are characterized by:
  - Little amounts of rainfall and higher temperature;
  - High prevalence of tropical diseases;
  - Lower population densities;
  - Nomadic and semi-nomadic economic life;
  - Vast plain lands favorable for irrigation agriculture along the lower river basins.

3.2. The Physiographic Divisions of Ethiopia

- Following the structural divisions brought about the geological processes **three major** physiographic units can be identified as:
  1. **The Western highlands and lowlands**
  2. **The South-Eastern highlands and lowlands**
  3. **The Rift Valley**


Physiographic regions of Ethiopia
33°E
36°E
39°E
42°E
45°E
48°E
15°N
12°N
9°N
6°N
3°N
Awash
Baro
Omo
Wabe Shebele
Genale
Dawa
LEGEND
I - North-Western Highlands and Associated Lowlands
The Tigrian Plataeu
The North-Central Massif
The Shewa Plateau
The South-Western Plateau
1A The Angereb Lowlands
1B The Baro Lowlands
1C The Omo Valley
II - The South-Eastern Highlands and Associated Lowlands
The Arsi-Sidama-Bale Massif
The Hararghe Plateau
2A The Genale Plain
2B The Shebele Plain
III - The Ethiopian Rift Valley
The Southern Rift Valley
The Central Rift Valley
The Awash Valley
The Denakil Depression
Lakes
Major Rivers
N
200 100 0 200
Kilometers
Source: Adapted from Mesfin Weldemariam, 1972
Prepared by:- Eyasu / Cascape, 2015

3.2.1. The Western Highlands and Lowlands

- This physiographic region
  - Includes all the area **west of the Rift Valley**.
  - Encompass nearly the whole western half of Ethiopia.
  - Makes up about **44%** of the area of the country.
  - Further subdivided into **four groups of highlands** (76.3%) and **four groups of lowlands** (23.7%).

i) The Western Highlands can be subdivided into the following sub-regions

a. The Tigray Plateau

b. North Central Massifs

c. The Shewa Plateau/central highlands

d. The Southwestern Highlands

a. The Tigray Plateau

- Lies to the southeast of the upper course of **the Mereb/Gash River and to the northeast of Tekeze River Gorge.**
- Constitutes about **13%** of the area of the region.
- Most of the land being in between **1,000 and 2,000 meters** above sea level.
- **Denudation** has created residual features of **granite hills**, **rugged topography, and Ambas**.
- High mountains in this plateau include **Mount Tsibet** (3988 m.a.s.l), **Mount Ambalage** (3291 m.a.s.l), and **Mount Assimba** (3248 m.a.s.l).
- The **famous monastery** at **Debre-Damo** is also located in this plateau region.

b. North Central Massifs

- Much of its northern and southern limit follows **the Abay and Tekeze gorges.**
- In its central part, **it accommodates the Lake Tana basin** surrounded by plains of Fogera and Dembia in the north & an upland plain in its south.
- **58%** of the region is at an altitude of **> 2,000 meters** making it, next to the Shewan Plateau.
- Consists of the **Gonder, Wello and Gojjam Massifs**.





- Among the known mountain peaks, the most popular ones include
  - Mount **Ras Dashen** (4,620 m.a.s.l), Mount **Weynobar/Ancua** (4462 m.a.s.l), Mount **Kidis Yared** (4453 m.a.s.l) and Mount **Bwahit** (4437 m.a.s.l) in the Simen Mountain System
  - Mount **Guna** (4,231m.a.s.l) in the **Debre Tabour** Mountain System
  - **Abune Yoseph** (4,260 m.a.s.l) in the **Lasta highlands of Wello** and
  - Mount **Birhan** (4,154 m.a.s.l) in the **Choke Mountain System** in Gojjam are also part of Simen Mountain System.
- **Yeju-Wadla Delanta land bridge (ridge)** has been significant in history. It served as a route of penetration by the Turks, Portuguese, and Italians etc.





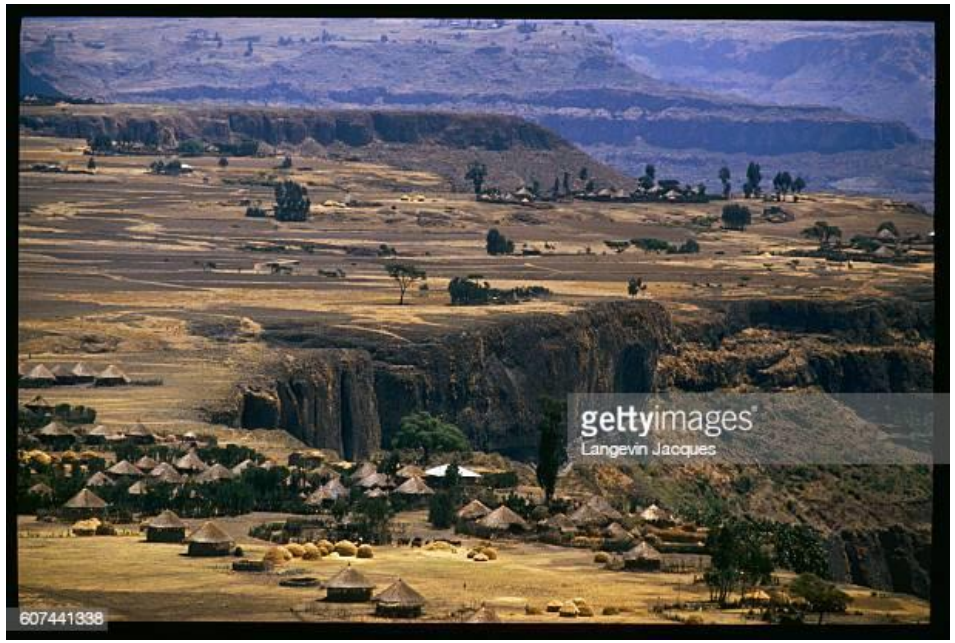
C. The Shewa Plateau/central highlands

- Is **bounded by the Rift Valley** in the east & southeast, by **the Abay gorge** in its northern & western limit, & **the Omo gorge** in the south and west.
- Occupies a central geographical position in Ethiopia.
- Is the smallest of the Western highlands (constituting **11 %**).
- Has the **largest proportion of elevated ground** (**3/4th** of its area-above 2000 m).
- Is drained, outward in all directions by the tributaries of **Abay, Omo, and Awash.**
- The tributaries of **Abay -Guder, Muger, Jema** etc. have cut deep gorges and steep sided river valleys.



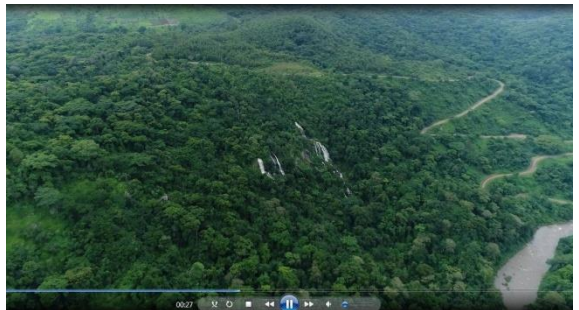
- They have created several **tablelands and isolated plateau units** in the north.
- The tributaries of Omo and Awash have dissected the other sides of the plateau.
- Has **relatively extensive flat-topped uplands**, giving it the **appearance of a true plateau**.
- Highest mountain in the Shewan plateau include **Mount Abuye-Meda** (4,000 m.a.s.l) in Northern Shewa, **Mount Guraghe** (3,731 m.a.s.l) in the south.





d. The Southwestern Highlands

- Consists of the **highlands of Wellega, Illuababora, Jimma, Kaffa, Gamo and Gofa.**
- Is separated from the adjacent highlands by the **Abay and Omo river** valleys.
- Extends from the Abay gorge in the north to the Kenya border and Chew Bahir in the south.
- Accounts for 22.7% of the area of the region & is the second largest in the Western highlands.
- About 70% of its area is lies within **1,000-2,000 meters** altitude.
- **Is the wettest in Ethiopia** & is drained by Dabus, Deddessa (tributaries of Abay), Baro, Akobo and the Ghibe/Omo rivers.
- Accommodates the most **numerous and diverse ethnic linguistic groups** in Ethiopia.
- **Guge** Mountain (4,200 masl) is the highest peak in this region.





ii) The Western Lowlands

- Are the **western foothills & border plains** that extend from **Western Tigray** in the north to **southern Gamo-Gofa** in the South.
- Make 23.7% of the area of the physiographic region.
- The general elevation ranges between **500 and 1000 masl**.
- Is further subdivided into four by the protruding ridges.
- These are **Tekeze lowland**, **Abay-Dinder lowland**, **Baro lowland**, and **Ghibe lowland** from north to south.
- With the **exception of the Baro lowland**, the region is generally characterized by **arid or semi-arid** conditions.
- **Pastoral or semi-pastoral** economic activities dominate the area.
- The **Ghibe/Omo** lowland structurally also belongs to the **Rift Valley**.
- In the Western lowlands, **there are small but important towns** (in terms of agriculture, history, or are simply border towns and frontier ports);These are **Humera, Metema, Omedla, Kurmuk, Gambella** etc.

3.2.2. The Southeastern Highlands and Lowlands

- Is the **second largest** in terms of area & accounts for **37%** of the area of Ethiopia.
- The **highlands** make up **46%** of the physiographic division while the rest **(54%) is lowland**.

i) **The Southeastern Highlands constitutes**

A. The Arsi-Bale-Sidama Highlands

B. The Hararghe Plateau

The Arsi-Bale-Sidama Highlands

- Make up 28.5% of the area of the region and 62% of the south - Eastern Highlands.
- The well-known mountains in **Arsi Highlands** are Mount **Kaka** (4,180 m.a.s.l), Mount **Bada** (4,139 m.a.s.l) and Mount **Chilalo** (4,036 m.a.s.l).
- The **Bale highlands** are separated from the **Arsi highlands** by the head and main stream of **Wabishebelle**.

- The **Afro-Alpine** summit of Senetti plateau is found in Bale high lands.
- The highest mountain peaks in Bale Mts. are **Tulu-Demtu** (**4,377** m.a.s.l) and **Mount Batu** (4,307 m.a.s.l).
- The Arsi-Bale Highlands are important **grains producing areas** with still high potential.
- The **Sidama Highlands** are separated from the Bale Highlands by the **Ghenale river** valley.
- The prominent feature in Sidama highland is the **Jemjem plateau**, an important **coffee growing area.**











- Rivers **Wabishebelle and Ghenale** along with their tributaries have dissected this physiographic region.
- Specially, **Weyb River**, tributary of Ghenale, has cut an underground passage (**Sof Omar cave, i.e. a spectacular cavern**) through the Mesozoic Limestone rocks.
- The cave is found near Bale Mountains.





B. The Hararghe Plateau

- Is a north-easterly extension of the south-eastern highlands & extends from the **Chercher** highland in the south-west to **Jigjiga** in the east.
- Makes up 38% of the South Eastern highland and 17.4% of the whole physiographic region.
- Has the smallest proportion of upper highland (>2,000 meters) & is a **low lying** and **elongated region**.
- The left-bank tributaries of Wabishebelle drain it.
- The highest mountain here is Mount **Gara-Muleta** (3,381 m.a.s.l).





ii) The Southeastern Lowlands

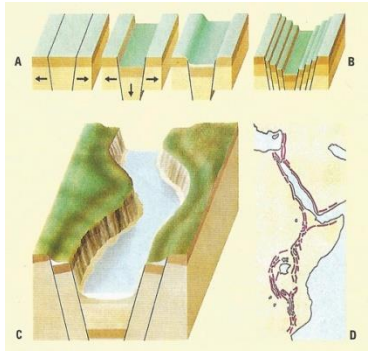
- Are located in the southeastern part of the country and are the most **extensive lowlands** in Ethiopia.
- Make up 54% of the area of the physiographic region & around **one-fifth** of the country.
- Is divided into **Wabishebelle plain** (60%) and the **Ghenale** Plain (40%).
- Include the plains of **Ogaden, Elkere**, and **Borena**.
- Because of the **harsh climatic conditions**, these lowlands are **little used & support very small population.**
- Are **sparsely inhabited by pastoral and semi-pastoral communities**.
- Its economic potential includes **animal husbandry**, **irrigation**, agriculture and perhaps **exploitation of petroleum & natural gas**.

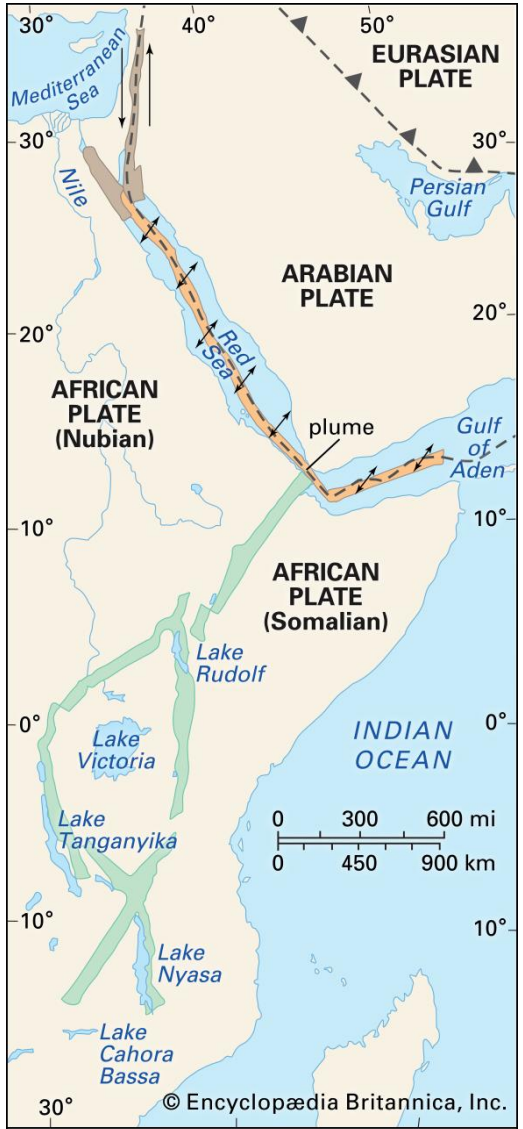


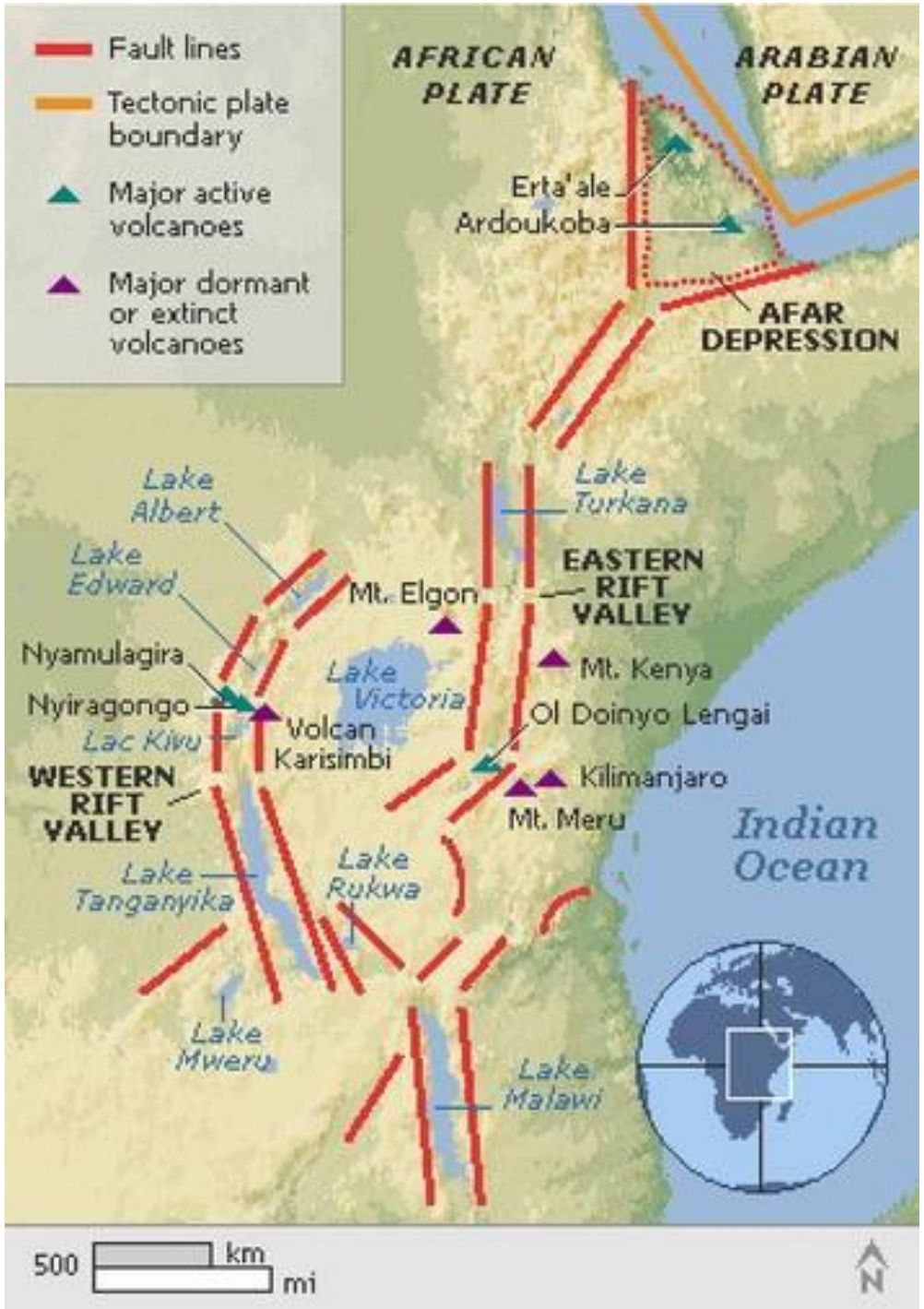


3.2.3. The Rift Valley

- Has separated the Ethiopian Highlands and Lowlands into two.
- It extends from the **Afar triangle** in the north to **Chew Bahir** for about **1,700 km**.
- It covers 18% of the area of Ethiopia.
- It is **elongated and funnel shaped**, with a NE-SW orientation.
- It opens out in the **Afar Triangle**, where it is the **widest**, and **narrows down to the south.**
- The floor of the Rift Valley is made up of interconnected **troughs, grabens and depressions.**
- **Volcanic rocks**, **fluvial** and **lacustrine** deposits cover the floor.




30°
40°
50°
EURASIAN PLATE
Mediterranean Sea
30°
Nile
Persian Gulf
ARABIAN PLATE
20°
Red Sea
AFRICAN PLATE (Nubian)
plume
Gulf of Aden
10°
AFRICAN PLATE (Somalian)
Lake Rudolf
0°
INDIAN OCEAN
Lake Victoria
Lake Tanganyika
0 300 600 mi
0 450 900 km
10°
Lake Nyasa
Lake Cahora Bassa
30°
© Encyclopædia Britannica, Inc.
Aulacogen (East African Rift System)
Spreading ridges
Transform faults
Direction of opening of the Red Sea and the Gulf of Aden
Direction of movement along transform fault


Fault lines
Tectonic plate boundary
Major active volcanoes
Major dormant or extinct volcanoes
AFRICAN PLATE
ARABIAN PLATE
Erta'ale
Ardoukoba
AFAR DEPRESSION
Lake Albert
Lake Turkana
Lake Edward
EASTERN RIFT VALLEY
Mt. Elgon
Nyamulagira
Lake Victoria
Mt. Kenya
Nyiragongo
Volcan Karisimbi
Ol Doinyo Lengai
Lac Kivu
WESTERN RIFT VALLEY
Kilimanjaro
Mt. Meru
Indian Ocean
Lake Tanganyika
Lake Rukwa
Lake Mweru
Lake Malawi
500 km
mi
N

- In many places, numerous **volcanic domes, hills and cinder cones** rise from the floor.
- Altitude in the floor ranges from **125 m.b.s.l** at **Dallol Depression**, to as high as **2,000 m.a.s.l** in **the Lakes region**.
- From the floor to the edge of the escarpment, the heights vary from **200 to 1500 meters**.
- Because of its altitudinal variation and positional differences, the climate also varies from warm, hot and dry to cool and moderately moist conditions.
- Similarly, the social and economic life reflects this pattern.
- Sparsely inhabited by **pastoralists** where as in others parts people practice some rain-fed agriculture.
- Further subdivided into **three physiographic sub-regions**:
  - **the Afar Triangle,**
  - **the Main Ethiopian Rift, and**
  - **the Chew Bahir Rift.**

i. The Afar Triangle

- Is the **largest and widest** part of the Rift Valley & makes up **54%** of the Rift Valley area.
- Is bounded by the high western and eastern escarpments in the west and east respectively, and by the Afar and Aisha Horst in the northeast.
- The area is generally of **low altitude** (300-700 meters).
- Is **triangular-shape** lowland.
- Its elevation drops from **1,000 meters** in the SW to below sea level in the North (Danakil depression) and in the east, where the shores of Lake Asal, fluctuating at around **125 meters** below sea level
- The lowest point in this region represents the lowest subaerial point of the African continent.
- The depression, which hosts one of the most **hostile environments** on Earth (Max. temp. **> 50 Degree Celsius**, at Dallol, i.e. Northern tip of Afar).

- The area is characterized by **faulted depressions** (grabens), **volcanic hills, active volcanoes, volcanic ridges, lava fields and low lava platforms.**
- Lakes (**Abe, Asale, and Afrera**) occupy some of these basins.
- A prominent feature in this region is the **Denakil Depression (Kobar Sink).**
- Separated from the Red Sea by a 200 meters high land barrier, much of it lies below sea level.
- A larger part of this is covered by **thick and extensive salt plain**.
- Lake Asale and Lake Afrera occupy the lowest parts of this sunken depression.
- The Afar Triangle is generally hot and dry.
- The only respite one gets in the Southern part is from the waters of the **Awash River**.
- The **economic importance** of this region includes **salt extraction**, **irrigation** along the Awash River & **electric potential** from **geothermal energy**.








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ii. The Main Ethiopian Rift/Central Rift

- Refers to the **narrow belt of the Rift Valley** that extends from **Awash River** in the north to **Lake Chamo** in the south.
- Is the **narrowest & the highest** with an average width of **50-80 kms** and general elevation of **1,000-2,000** m.a.s.l.
- The floor in many places is **dotted by cinder cones & volcanic mountains.**
- The big ones include Mount **Fentale, Boseti-guda** (near Adama), **Aletu** (north of Lake Ziway) & **Chebi** (north of Lake Hawasa).
- Contains **numerous lakes** formed on **tectonic sags and fault depressions.**
- Other resource bases include the **recreational value of the lakes**, the **agricultural importance** of some streams and lakes, and the **geothermal energy potential**.


Getty Images







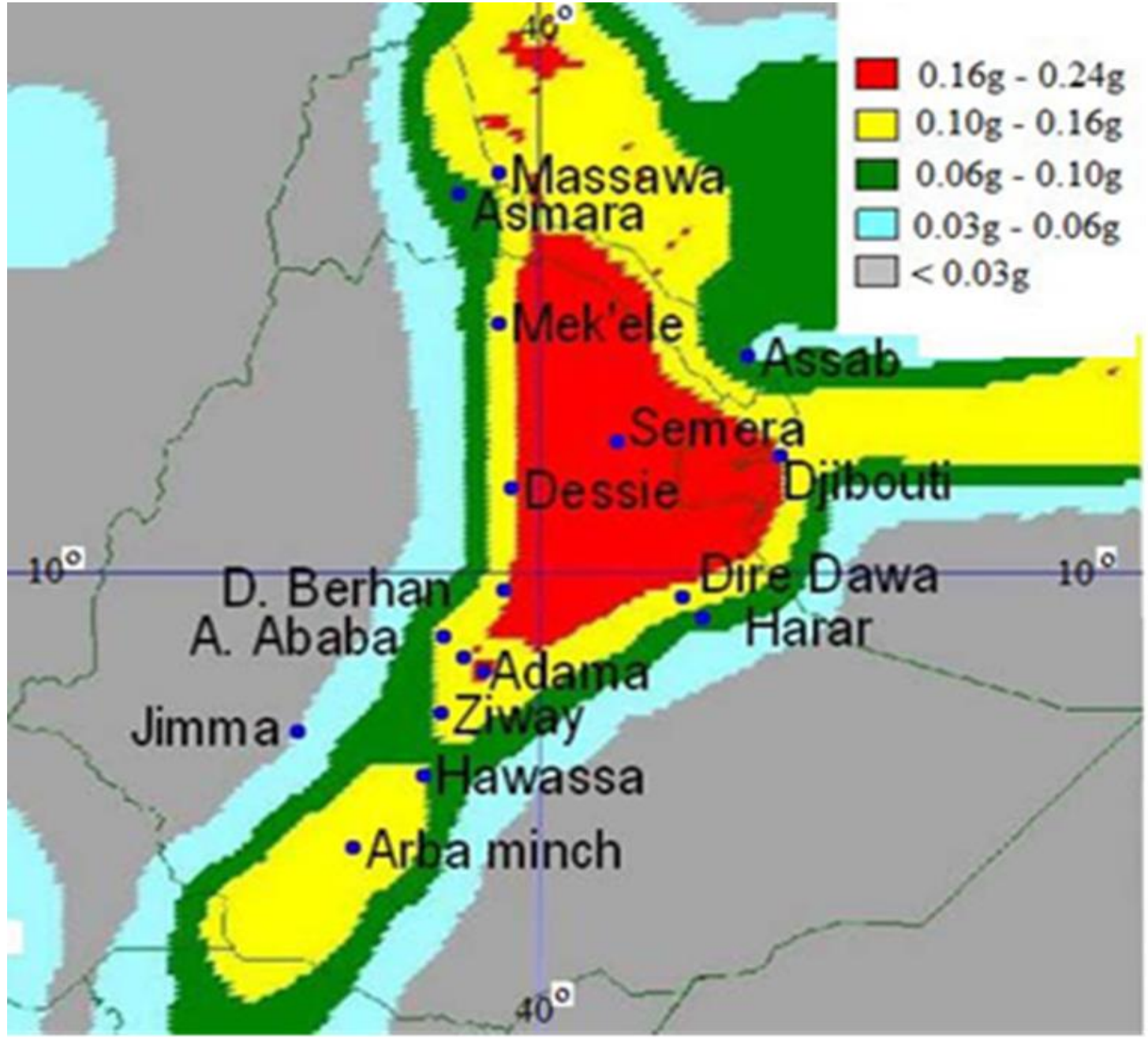
iii. The Chew Bahir Rift

- Is the **smallest** and the **southern-most part** of the Rift Valley.
- Gneissic highlands of **Konso** and the surrounding highlands separate it from the Main Ethiopian Rift to the north.
- The characteristic feature of this region is the **broad and shallow depression**, which is a **marshy area** covered by tall grass, into which the Segen and Woito streams empty.








0.16g - 0.24g
0.10g - 0.16g
0.06g - 0.10g
0.03g - 0.06g
< 0.03g
Massawa
Asmara
Mek'ele
Assab
Semera
Djibouti
Dessie
D. Berhan
Dire Dawa
Harar
A. Ababa
Adama
Ziway
Jimma
Hawassa
Arba minch
10°
40°

3.3. The Impacts of Relief on Biophysical and Socioeconomic Conditions

- The highly dissected character of the landscape over much of the country influence the various socioeconomic aspects of Ethiopia as presented hereunder.

1. **Agricultural practices**
   - Relief influences **farm size and shape** in that in an area of rugged terrain the farmlands are **small in size and fragmented** and tend to be **irregular in shape**.

- Choice of **farming techniques** and **farm implements** are highly influenced by relief.

- Relief influences **crop production**.

- The practice of **animal husbandry** is also influenced by relief.

2. Settlement pattern

- Highlands of Ethiopia are **densely settled**.
- Rugged & difficult terrain hinders the **development of settlement and its expansion.**
- The **highlands** of Ethiopia are characterized by **sedentary life** & **permanent settlements** while **lowlands** that are inhabited by **pastoralists** have **temporary settlements**.

3. Transportation and communication

- The highly dissected nature of the landscape is a **barrier** to the development of transportation.
- The difficult terrain makes infrastructure development & maintenance **costly**.
- **TV and radio** communications are also highly influenced by relief.
- The rugged topography rendered rivers **less navigable** due to the **waterfalls, deep gorges and steep cliffs**.

4. Hydroelectric power potential

- The great **difference in altitude** coupled with **high rainfall** created **suitable conditions** for a very high potential for the production of HEP in Ethiopia.

5. Socio-cultural feeling

- The rugged terrain as a result of excessive surface dissection resulted in **isolation of communities** that led to the occurrence of **cultural diversity.**
- People who live in the highlands - ***degegnas*** **(mountaineers**) and those who live in the lowlands-***kollegnas*** **(lowlanders**).

6. Impacts on climate

- The **climate of Ethiopia** is a result of the **tropical position** & the **great altitudinal variation**.
- **Highlands** with **higher amount of RF & lower rate of Evaporation** tend to be **moisture surplus** compared to the moisture deficit lowlands.

7. Impacts on soil

- **Steep mountain slopes** have **shallow and little developed soils**.

8. Impacts on natural vegetation

- Relief through its effect on climate and hydrology affect the type of **natural vegetation** grown in an area.

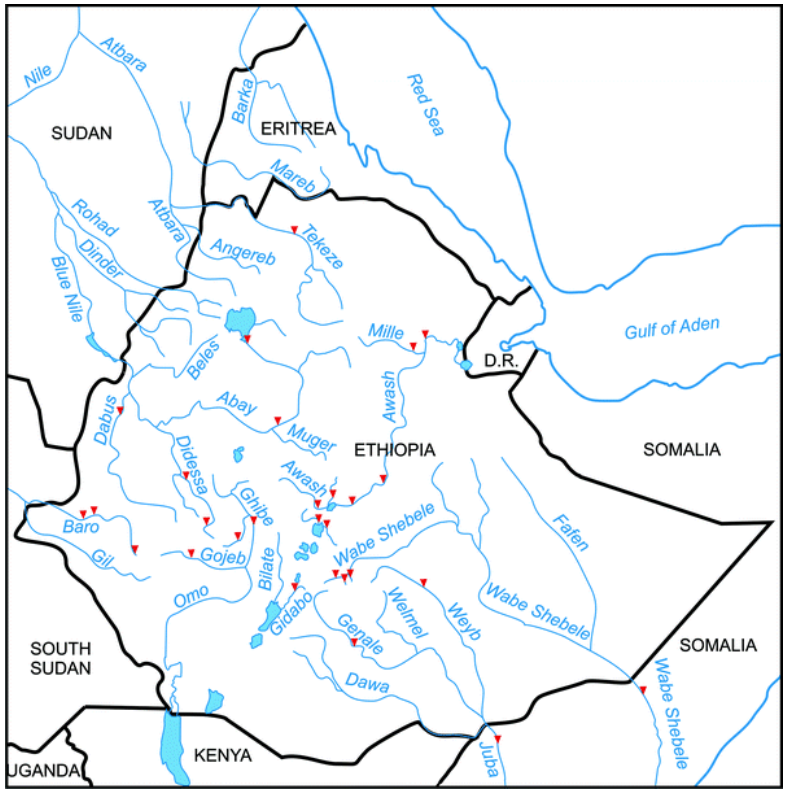
Chapter Four
DRAINAGE SYSTEMS AND WATER RESOURCE OF ETHIOPIA AND THE HORN

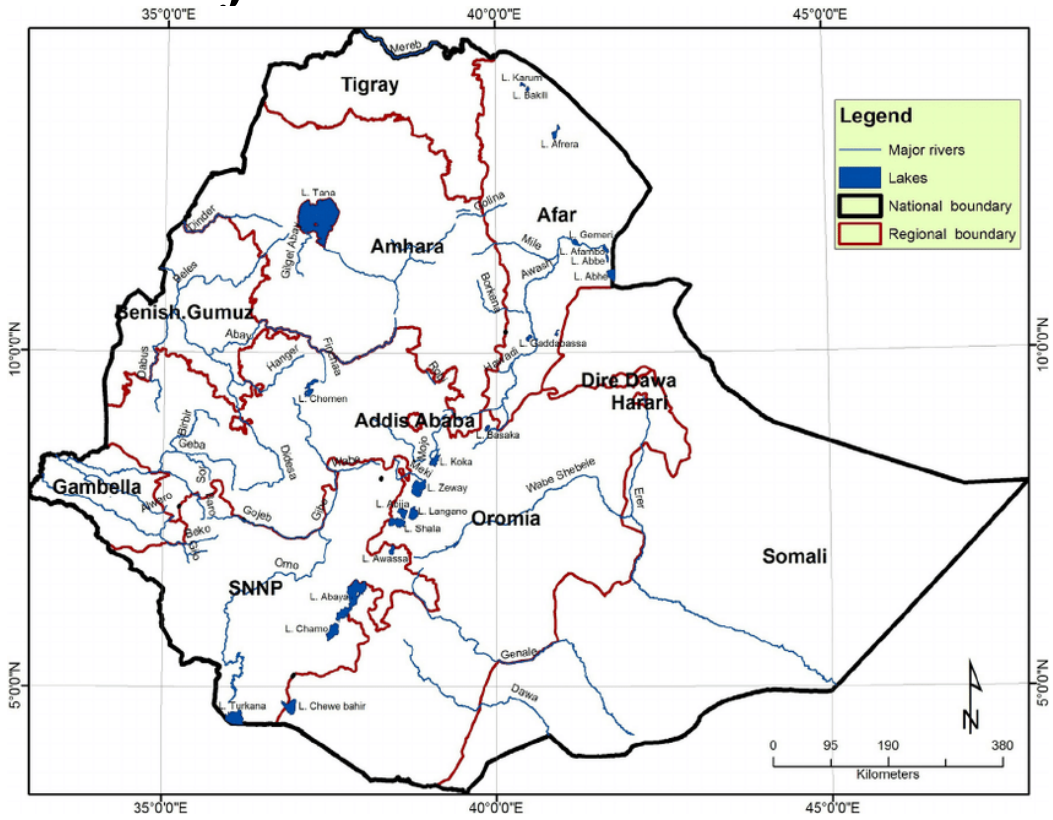


4.1 Introduction

- About **71%** of the earth's total surface is covered by water bodies majorly occupied by **seas and oceans**.
- Of the earth's total water surface, nearly **97.5% is alkaline** accumulated in seas and oceans.
- **Alkaline water** is water that is **slightly basic**. It contains basic minerals such as **calcium, magnesium, or bicarbonate**. These compounds bind to hydrogen ions in solution, making the water more basic.
- The remaining **2.5% is fresh water**, of which nearly **68.7% is deposited in glaciers**, **30.1% in ground water**, **0.8% in permafrost and 0.4% in surface waters.**
- Water in lakes, rivers, atmosphere, soils and wetlands are considered as **surface waters**.
- Surface and ground waters are by far **the most abundant and easily available** fresh waters.
- However, fresh water is distributed **unevenly** throughout the world following varied **latitudinal locations, climatic and topographic setups**.

- The **diverse topographical setup**, relatively **higher rainfall** and its **nearness to equator** made the country to have **larger volume of ground and surface water.**
- Around **0.7 %** of the total land mass of Ethiopia is covered by water bodies.
- The country's **surface water potential is estimated to be 124.4 billion cubic meters** (BCM).
- Many call Ethiopia, **the water tower of *"Eastern Africa"*.**





4.2. Major Drainage System of Ethiopia

- The flow of water through well-defined channel is known as **drainage**.
- **A drainage system** is made up of a **principal river and its tributaries** (the rivers that flow into it).
- **a drainage system is branched network of stream channels** together with the adjacent land slopes they drain.
- The drainage pattern of an area is the **outcome of the geological processes, nature and structure of rocks, topography, slope, amount and the periodicity of the flow.**
- A drainage basin is the topographic region from which **a river and its tributaries collect both the surface runoff and subsurface flow**.
- It is bounded by and separated from other river basins by **a water divide** or ***topographic divide***.

- The general patterns of major river basins in Ethiopia are determined by **topographical structures** which can be classified as:

  a. The topography of the outward sloping of the **Western and South eastern plateaus**

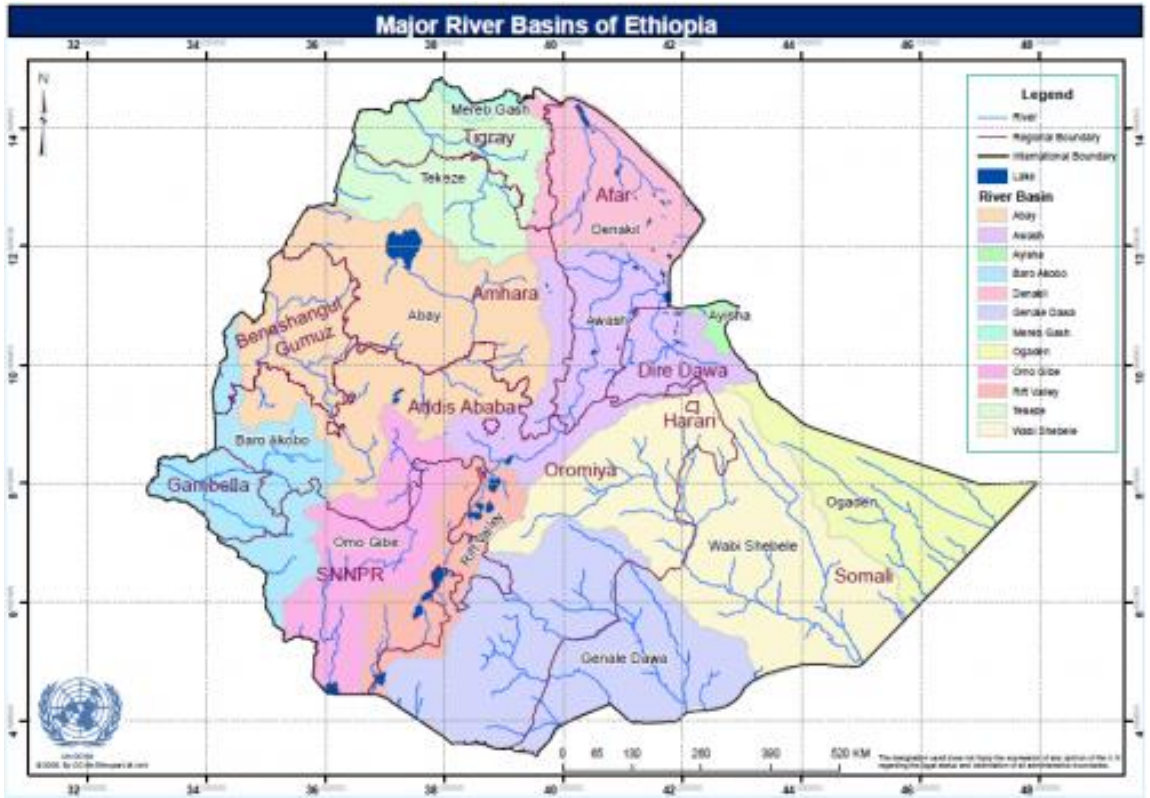
  b. The structural formation **of the Rift Valley with its in-ward-sloping escarpments resulting mainly in an inland drainage system**.

  c. **Faults and joints** that structurally influence part of the courses of many rivers.

The Major Drainage Systems

- Following the complex physiographic setup and geological makeup, Ethiopia possesses **three** broadly classified drainage systems namely **Western, Southeastern and Rift Valley Drainage Systems with a total of 12 river basins.**
- Western and the Southeastern drainage systems **are separated by the Rift Valley system.**

1. ***The Western Drainage Systems***
2. ***The Southeastern Drainage Systems***
3. **The Rift Valley Drainage System**



4.2.1. The Western Drainage Systems

- The Western Drainage Systems are **the largest** of all drainage systems **draining 40 percent** of the total area of the country and **carry 60 percent of the annual water flow**.
- This drainage system comprises **four major river basins** namely the **Tekeze, Abay, Baro-Akobo, Ghibe (Omo).**
- Unlike other river basins in the system, the **Ghibe (Omo) flows southward**.
- The Abay, Tekeze and Baro **flow westward** ultimately joining the **Nile** which finally ends at Mediterranean Sea.

i) Abay River

- The **largest river** both in **volumetric discharge and coverage** in the western drainage systems is the **Abay.**
- Abay river basin **covers an area of 199,812 km^2**, covering parts of Amhara, Oromia and Benishangul-Gumuz regional states.
- Together with its tributaries, most of which are left-bank tributaries; it carries **65 percent** of the annual water flow of the region.
- Abay which **rises from Lake Tana** (some sources indicate its origin from **Sekela, Choke mountain**) **flows about 1,450 kilometers** and **joins the White Nile in Khartoum, Sudan to form the Nile River.**
- More than **60 streams** drain the Abay within elevation ranging between 500 - 4261 meters above sea level. The largest of these is **Ghilgel Abay** (Little Abay).
- **Abay flows eastward, turns 180 degree** to make a large bend and after cutting an impressive and deep gorge emerges out in the west.








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C1HPGH
www.alamy.com

ii) Tekeze River

- Similarly, the **Tekeze and its tributaries**, carrying **12 percent** of the annual water flow of the region drains **82,350 Km2** of land surface within elevation ranging between 536-4517 meters above sea level.
- Erosion in the basin resulted in large **tablelands, plateau blocks and isolated mountain** groups.
- The basin has two main tributaries (**Angereb and Goang**) which rises in the central highlands of Ethiopia.
- Tekeze River is termed **Atbara in Sudan**, which is a tributary of the Nile.
- The total mean annual flow from the basin is estimated to be **8.2 billion** metric cubes (BMC).





iii) Barro-Akobo

- **The Baro-Akobo and Ghibe / Omo rivers** drain the **wettest** highlands in the south and southwestern Ethiopia.
- They carry **17 percent** and **6 percent** of the annual water flow respectively.
- The Ghibe/ Omo river basin drains an area of **79,000 km2** with an estimated mean annual flow of **16.6 BMC**.
- In the lower course, the Baro River flows across an extensive marshy land.
- Baro/Akobo river basin has an area of **75,912 km2**, covering parts of the Benishangul-Gumuz, Gambella, Oromia, and SNNPR.
- The total mean annual flow from the river basin is estimated to be **23.6 BMC.**
- The Baro together with Akobo forms the **Sobat River in South Sudan**.





iv) Ghibe River

- The Ghibe / Omo River finally empties in to the **Chew-Bahir** at the mouth of **Lake Turkana** (an elongated Rift Valley lake) thereby forming an inland drainage.





4.2.2. The Southeastern Drainage Systems

- Nearly the entire physiographic region of southeastern part of Ethiopia is drained by the southeastern drainage systems. The basin which is mainly drained by **Wabishebelle** and **Ghenale,** slopes south-eastwards across large water deficient plains.
- Major highlands of this basin include plateaus of Arsi, Bale, Sidama and Harerghe.
- **Wabshebelle and Ghenale rivers cross the border into Somalia**, carrying **25 percent** of the annual water flow of Ethiopia.

ii) Ghenale River

- **Ghenale River basin** has an area of **171,042 km2**, covering parts of Oromia, SNNPR, and Somali regions.
- Ghenale, which has **fewer tributaries** but **carries more water than** Wabishebelle, reaches **the Indian Ocean**.
- The Ghenale basin flows estimated to be **5.8 BMC** within elevation ranging between 171-4385 meters above sea level.
- Ghenale river is named as ***Juba River i***n Somalia.

ii) Wabishebelle River

- **Wabishebelle** with a total catchment area of **202,697 km2**, is the **largest river in terms catchment area**.
- It is the **longest river in Ethiopia** and drains parts of Oromia, Harari and the Somali regions.
- Its tributaries are mainly **left bank and, most of them, are intermittent**.
- Despite its size, the Wabishebelle **fails to reach the Indian Ocean** where at the end of its journey it flows parallel to the coast before its water disappears in the sands, **just near the *Juba River***.





4.2.3. The Rift Valley Drainage System

- The Rift Valley drainage system is **an area of small amount of rainfall**, **high evaporation** and **small catchment area**.
- The Rift Valley drainage system is therefore left with the **slopes of the escarpment** and the **Rift Valley floor** itself as the catchment area.
- *The rift valley drainage system is drained by few rivers mainly b Awash, Ghibe and several* **intermittent** rivers descending from the escarpment of the rift valley.

i) Awash River

- The only major river basin in the rift valley drainage system is the **Awash River**.
- Awash river basin has a catchment area of **114,123 km^2** and has an average annual discharge of **4.9 billion cubic meters**.

- The Awash River originates from ***Shewan plateau*** *in Central highlands of Ethiopia, and flows 1250 kms.*
- *It covers parts of the Amhara, Oromia, Afar, Somali, Dire Dawa, and Addis Ababa City Administration.*
- *Awash is the* ***most utilized river*** *in the country.*
- Following the rift valley orientation the Awash flows in a **NE direction.**
- It finally ends in a maze of small lakes and marshy area; the largest of which is **Lake Abe** on the Ethio-Djibouti border.







ii) Afar Drainage

- The **Afar drainage sub-basin** has practically **no stream flow**.
- It is an area of **little rain**, **very high temperature** and **very high evaporation.**
- Lake **Afrera** and **Asale** are the only main surface waters in the basin which are not the result of any meaningful surface flow.
- Their formation is related to tectonic activities.

iii) The Rift vally Lakes Region Drainage

- The **Southern part** of the Rift Valley sub-basin is characterized by a **number of lakes** and **small streams**.
- It is also described as **lakes region**.
- The lakes occupy fault depression

- There are small streams that drain down from the nearby mountain slopes which supply water to the lakes.
- For example,
  - ✓ ***Meki and Katar Rivers*** *flow into Ziway*
  - ✓ ***Bilate*** *into Abaya and*
  - ✓ ***Segen*** *into Chew Bahir.*
- *Likewise, some of these lakes are interconnected.*
- *Lakes Ziway and Langano drain into Lake Abijiata through the small streams of Bulbula and Horocolo respectively.*



4.3. Water Resources: Rivers, Lakes and Sub-Surface Water

4.3.1. The Ethiopian Rivers

- Unlike many other African countries, Ethiopia is endowed with **many rivers.**
- Majority of the rivers originate from **highland areas** and **cross the Ethiopian boundary.**
- Altogether, Ethiopian rivers form **12 major watersheds** separating the Mediterranean Sea from the Indian Ocean drainage systems.

Table 4.1: Data on major Ethiopian rivers

River	Catchment Area(km^2)	Annual Volume BMC	Terminus/ Mouth	Major tributaries
Abay	199,812	**54.5**	Mediterranean	Dabus, Dedessa, Fincha, Guder, Muger, Jema, Beshilo
Wabishebelle	**202,697**	**3.4**	Coast of Indian Ocean	Ramis Erer, Daketa Fafan
Genale Dawa	171,042	6	Indian-Ocean	Dawa, Weyb, Welmel, Mena
Awash	114,123	4.9	Inland (within Ethiopia)	Akaki, Kesem, Borkena, Mile
Tekeze	87,733	8.2	Mediterranean	Goang, Angereb
Gibe (Omo)	79,000	16.6	Lake Turkana	Gojeb
Baro Akobo	**75,912**	23.23	Mediterranean	Akobo

General Characteristics of Ethiopian Rivers

- Owing to the **highland nature** of the Ethiopian landmass, **surface ruggedness**, the **outward inclination of the highlands**, and the **climatic conditions**, Ethiopian rivers have the following characteristics.
  - ❖ Almost all major rivers **originate from the highlands** elevating more than 1500 meters above sea level,
  - ❖ Majority of Ethiopian rivers are **trans-boundary**,
  - ❖ Due to the marked seasonality of rainfall, Ethiopian rivers are characterized by **extreme seasonal fluctuation**.
  - ❖ In the **wet season**, **runoff is higher** while during the **dry seasons** they became mere **trickles of water/even dry up**,
  - ❖ Due to surface ruggedness they have **rapids and waterfalls** along their course,
  - ❖ They have **cuts, steep-sided river valleys** and **deep gorges** along their courses,
  - ❖ Rivers in Ethiopia flow on **steep slopes** having steep profiles.
  - ❖ Some of the rivers *serve as **boundaries**, both international and domestic administrative units.*

4.3.2. The Ethiopian Lakes

- Relatively Ethiopia is **rich in lakes** and almost all of them are the result of **tectonic process** that took place during ***Quaternary period of Cenozoic era***.
- The lakes in the drainage are mainly formed on faulted depressions and are clustered along the system forming linear pattern.

i) Highland Lakes

- **Lake *Tana***, *the largest lake in Ethiopia* ***occupies a shallow depression in the highlands.***
- The Tana depression is believed to be formed following **slower sinking** and **reservoir by lava** flow between Gojjam and Gonder massifs.
- Lake Ashenge (Tigray) is formed on a tectonic basin.

Lake Tana



ii) Crater Lakes

- Ethiopia is also gifted with **crater lakes**. These include the lakes at and around **Bishoftu, Wonchi (near Ambo), Hayk (near Dessie) and the Crater Lake on top of Mount Zikwala.,**







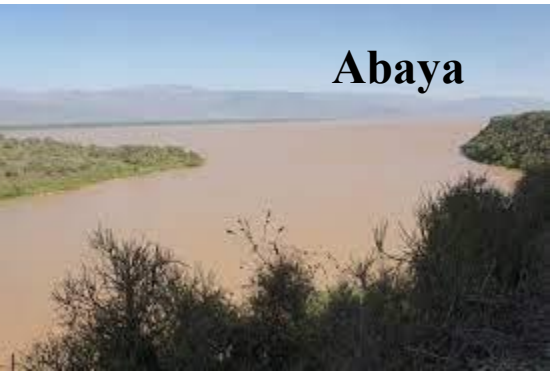


iii) Rift valley Lakes

- Except few Ethiopian lakes, majority of lakes are located within the Rift Valley System including Lake **Abaya , Chamo, Hawassa, Langano, Shala, Ziway, Turkana, Chew Bahir, etc.**. Cluster of lakes are lined up within main Ethiopian rift.
- Lake Abaya is the largest of all the lakes in the system while **Shala and Ziway** are the deepest and the shallowest lakes in the central Ethiopian Rift
- The southern tip of the Rift Valley forms the marshy land called the Chew Bahir which is drained by Segan and Woito.







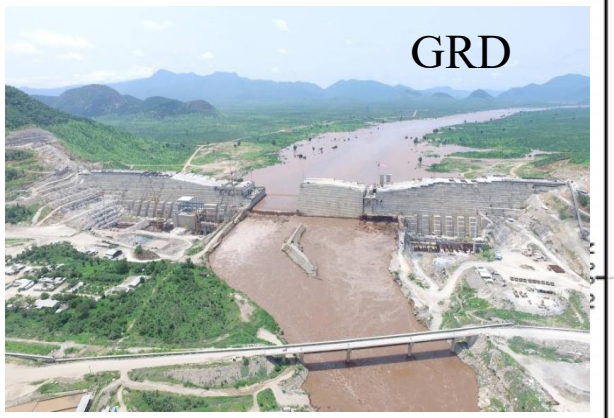


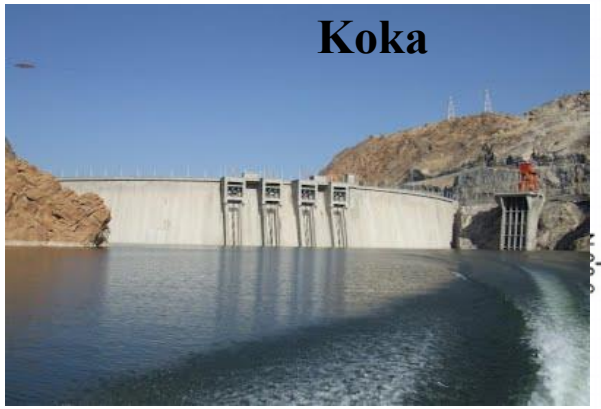




iv) Man-made Lakes

Other types of lakes in Ethiopia are **man-made** such as **Lakes Koka, Fincha and Melka Wakena**, and many other lakes dammed following hydroelectric power generation projects.





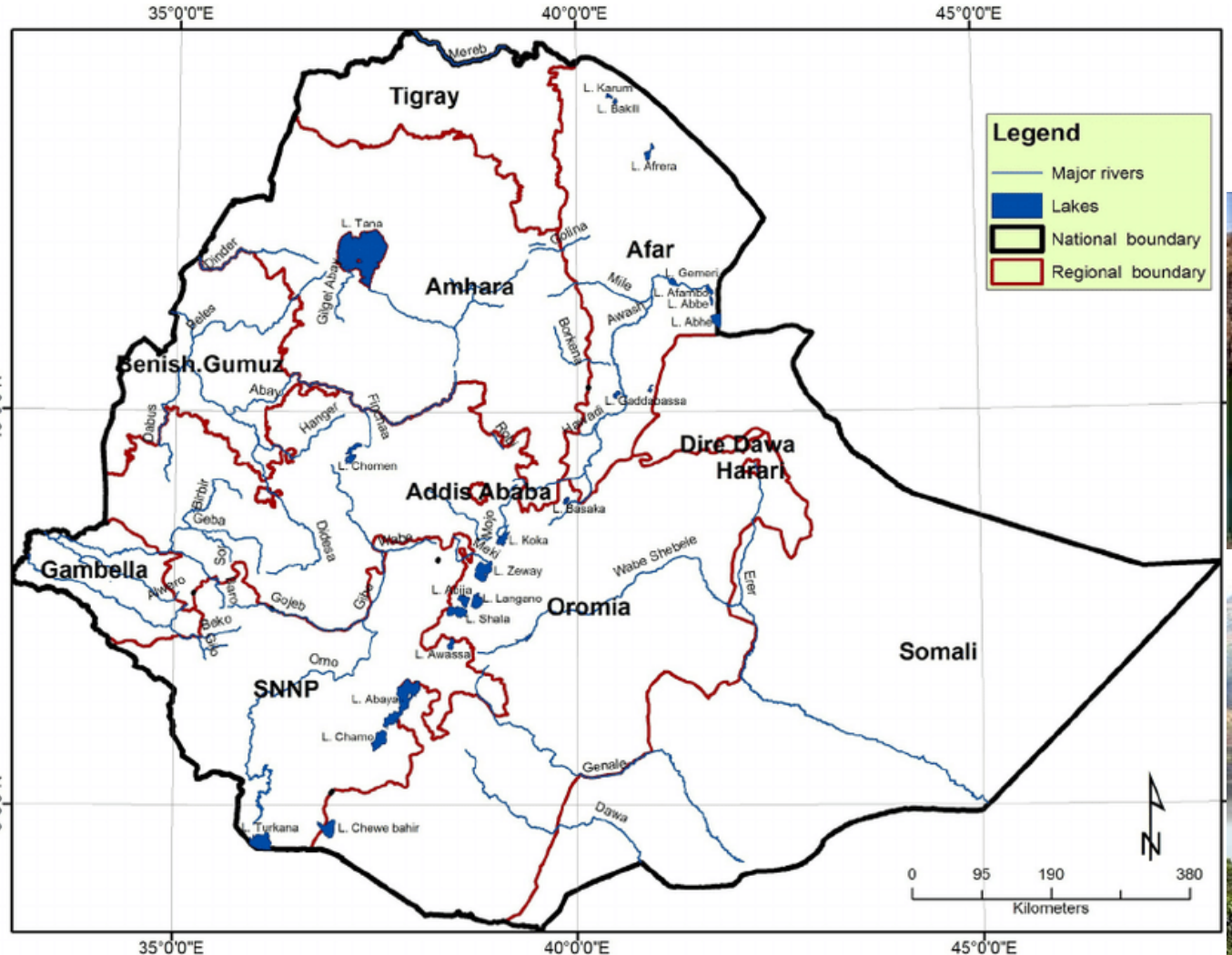






Table 4.2: Area and depth of some of Ethiopian Lakes

Lake	Area (km^2)	Max. Depth(m)	Lake	Area (km^2)	Max. Depth(m)
Tana	**3600**	9	Abijata	205	14
Abaya	1162	13.1	Awassa	129	10
Chamo	551	13	Ashenge	20	25
Ziway	442	**8.95**	Hayk	**5**	23
Shala	409	**266**	Beseka	48.5	11
Koka	205	9			

Table Ethiopian Lakes

Name	Area (km²)	Elevation (MAMSL)	Type	Region	Remarks
Lake Abaya	1,162	1,285	Freshwater lake	SNNPR	
Lake Abbe	320	243	Salt lake	Afar Region	
Lake Abijatta	205	1,573	Salt lake	Oromia Region	
Lake Afambo	35	339	Freshwater lake	Afar Region	
Lake Afrera	100	-102	Salt lake	Afar Region	
Lake Ashenge	20	2,409	Salt lake	Tigray Region	
Lake Hawasa	129	1,686	Freshwater lake	SNNPR	
Lake Bario	22.16	339	-	Afar Region	
Lake Basaka	42.6	950	Salt lake	Oromia Region	
Lake Chamo	317	1,110	Freshwater lake	SNNPR	
Lake Chew Bahir	343	570	Salt lake	SNNPR	
Lake Chomen	180	2,224	Reservoir	Oromia Region	
Grand Ethiopian Renaissance Dam	1,874	155	Reservoir	Benishangul-Gumuz Region	Under construction on the Blue Nile (2018)
Lake Gummare	60	-	Freshwater lake	Afar Region	
Lake Hayq	23	2,030	Freshwater lake	Amhara Region	
Lake Haramaya	-	2,010	Freshwater lake	Oromia Region	Dried out from 2011
Lake Kadabassa	-	562	-	Afar Region	
Lake Karum	50	-120	Salt lake	Afar Region	
Koka Reservoir	180	1,595	Reservoir	Oromia Region	
Lake Langano	230	1,585	Freshwater lake	Oromia Region	
Lake Shala	329	1,558	Salt lake	Oromia Region	
Lake Tana	3,060	1,788	Freshwater lake	Amhara Region	
Lake Turkana	6,405	361	Salt lake	SNNPR	Most of the lake is in Kenya
Lake Zway	440	1,636	Freshwater lake	Oromia	
Lake Zengena	1	2,500	Freshwater lake	Amhara Region	Crater lake

4.3.3. Subsurface (Ground) Water Resource of Ethiopia

- As compared to surface water resources**, Ethiopia has lower ground water potential.**
- However, there exists higher total exploitable groundwater potential.
- **Climatic** and **geophysical conditions** determine the availability of groundwater resource.
- Based on existing scanty knowledge, the groundwater potential of Ethiopia is estimated to be ***2.6 - 6.5 BMC***.
- *However, this estimate is now considered* ***underestimated***.
- *Considering various separate studies, Ethiopian potential of groundwater is believed to range* ***between 12-30 BMC***.

4.4. Water Resources Potentials and Development in Ethiopia

- The enormous water resource potential of Ethiopia is **underutilized** due to so many factors.
- However, there are plenteous of opportunities that can transform the resource into our collective social and economic needs.
- The followings are **some of potential development uses** of water resource of Ethiopia.

a) Hydro-electric Potential

b) Irrigation and Transportation

c) Fishing

d) Recreation

a) Hydro-electric Potential

- Ethiopian rivers have a very high potential for generating electricity.
- The exploitable potential of hydroelectric power is estimated at about **45000 megawatts**.
- The **first hydroelectric power** generation plant was installed on **Akaki River (Aba Samuel) in 1932.**
- Currently many hydroelectric power dams are operating and many others are under construction to realize Ethiopia's ambitious energy goals.
- **Grand Ethiopian Renaissance Dam** (GERD) is the country's largest dam under construction aiming to generate **6400 megawatts**.
- **Gilgel Gibe III** hydropower project has gone operational generating **1870 megawatts**.



Gilgel Gibe

Table 2.5. Irrigation and hydropower potential of river basins.

River Basin	Area (Km^2)	Runoff (Bm^3)	Potential Irrigable Land (ha)	Gross Hydro-electric potential Gwh/year	Estimated ground water potential (Bm^3)
Tekeze	82,350 ***	8.2	83,368	5,980	0.20
Abbay	199,812	54.8	815,581	78,820	1.80
Baro-Akobo	75,912	23.6	1,019,523	13,765	0.28 0.13Recharge/year
Omo-Ghibe	79,000	16.6	67,928	36,560	0.42 (.10)Rech /yr
Rift Valley	52,739	5.6	139,300 *	800	0.10
Mereb	5,900 ***	0.65	67,560	-	0.05
Afar /Denakil	74,002 ****	0.86	158,776	-	-
Awash	112,696	4.9	134,121	4,470	0.14
Aysha	2,223 ****	-		-	-
Ogaden	77,121 ****	-		-	-
Wabi-Shebelle*	202,697 **	3.16	237,905	5,440	0.07
Genale-Dawa	171,042 **	5.88	1,074,720	9,270	0.14
Total	1,135,494	124.25	3,798,782	155,102	2.86

Source: Integrated River Basin Master Plan Studies, carried out during 1997-2007 (MoWR 1996, 1997, 1998a, 1998b) Irrigable land

Table 2.3: List of dams and hydropower plants

Name	Installed capacity		Basin	Financing	Remarks
Fincha	134 MW	1973	Fincha (Blue Nile)		also for irrigation
Gilgel Gibe I	180 MW	2004	Gilgel Gibe River	World Bank	
Tekezé	300 MW	2009	Tekeze (Atbara)	Chinese	
Beles	460 MW	2010	Lake Tana (Blue Nile)	Ethiopian government	Irrigation of 140,000 ha
Gilgel Gibe II	420 MW	2010	Omo River (no dam, fed by GG I)	Italy and EIB	
Gilgel Gibe III	1,870 MW	2012-13	Omo River	Italy	
Halele Worabese	440 MW	2014	Omo River	FairFund?	
Gilgel Gibe IV	2,000 MW	2014	tributary of the Omo River	Chinese	
Chemoga Yeda	278 MW	2013	tributary of the Blue Nile, near Debre Markos	Chinese	Will consist of 5 interconnected dams
Genale Dawa III	256 MW	awarded in 2009	b/n Oromo and Somali state	Chinese	
Grand Ethiopian Renaissance Dam	6000 MW	2014	Blue Nile River	Government	

- Currently Ethiopia is administering **14 hydroelectric power plants** constructed on Lake Aba Samuel, Koka, Tis Abay, Awash, Melka Wakena, Sor, Fincha, Gibe/Omo, Tana Beles and Tekeze, generating close to **4000 megawatts of energy**.
- Besides the domestic use of generated electricity, the country is **exporting electricity** to the neighboring countries.
- The **major problem** related to the use of Ethiopian rivers for the generation of hydroelectric power is the **seasonal flow fluctuations and impact of climate change and variability.**
- The **severe erosion from the highlands and sedimentation** in the reservoirs is also a critical problem for hydroelectric power generation.

b) Irrigation and Transportation

- The terrain in Ethiopia is so **rugged** that **it limits the uses** of Ethiopian rivers both for irrigation and transportation.
- In the **highlands**, steep slopes, rapids, waterfalls, narrow and deep valleys and gorges are important obstacles.
- But on the **lowlands**, their demand for irrigation is high.
- Regardless of existing physiographic setups, Ethiopia's **potential of irrigation** is estimated to be **5.3 million hectares**.
- The ***Baro-Akobo*** *and* ***Genale Dawa river*** *systems have* ***large irrigation*** *potential compared to other basins.*





Table 2.7. Irrigation Potential in the River Basins of Ethiopia

Basin	Catchment Area (Km²)	Irrigation potentials (Ha) (Respective recent master plan studies)				Irrigation Potential (WAPCOS 1995)		
		Small-scale	Medium-scale	Large-scale	Total	Total Drainage Area (km²)	Irrigable Area (Ha)	percent Irrigable Area of the Country
Abbay	198,890.7	45,856	130,395	639,330	815,581	201,346	1,001,000	27
Tekeze	83,475.94	N/A	N/A	83,368	83,368	90,001	3,17,000	8.5
Baro-Akobo	76,203.12	N/A	N/A	1,019,523	1,019,523	74,102	9,85,000	26.5
Omo-Ghibe	79,000	N/A	10,028	57,900	67,928	78,213	4,45,000	12
Rift Valley	52,739	N/A	4000	45,700	139,300	52,739	1,39,000	3.7
Awash	110,439.3	30,556	24,500	79,065	134,121	112,697	2,05,000	5.5
Genale Dawa	172,133	1,805	28,415	1,044,500	1,074,720	117,042	4,23,000	11.4
Wabi Shebele	202,219.5	10,755	55,950	171,200	237,905	102,697	200,000	5.4
Denakil	63,852.97	2,309	45,656	110,811	158,776	74,102	-	-
Ogaden	77,121				-	77,121	-	-
Ayisha (Gulf of Aden)	2,000				-	2,000	-	-
Total	1,118,074.53				3,731,222	982,060	3,715,000	100

Note: The national water resources master plan (WAPCOS 1995) was a desk study without significant field investigation.

Source: Seleshi Bekele and et.al, 2007.

- *Despite the untapped irrigation practice, more than **60%** of the area under irrigation so far is located in **Rift Valley Drainage System**.*
- *Except few, **majority of hydro-electric reservoirs** are **multi-purpose** and are expected to contribute for irrigation.*
- Majority of Ethiopian rivers **are not suitable for transportation**.
- The ***Baro*** *at its lower course is the only navigable river.*
- *Comparatively, **Ethiopian lakes** are much suitable for transportation than rivers.*
- ***Lake Tana*** *and **Abaya** are relatively the most used for transportation*

c) Fishing

- The majority of Ethiopian lakes are **rich in fish**.
- Currently the annual production of fish is estimated to be **31.5 thousand tons.**
- The exploitable **potential** is however, by far greater than the **current production.**
- **Exploitable fish potential in lakes varies**.
- Currently **Lake Tana leads the potential** by estimated 8,000-10,000 tons per year.
- Fish production from **Lake Chamo** is estimated at 4,500 tons per year.
- More than **60% of fish supplies** are coming from Ethiopian main Rift Valley lakes.
- However, some of the lakes are currently threatened by **sedimentation**, **invasive species**, **over exploitation** and **expansion of investments around lakes.**





D) Recreation

- There are a **variety of fish, birds** and **other aquatic life** forms in the lakes, some of which are only **endemic to Ethiopia**, with immense scientific purposes.

Kuriftu

- This and the scenic beauty of the lakes, the hot springs, the spectacular river gorges & the most impressive waterfalls make Ethiopian rivers & lakes **important recreational & tourist attractions**.
- The existence of fish, birds and other wild life in the lakes and rivers such as **flamingoes in Abijiatta**, **hypppotamus and crocodiles in Ziway and Abaya**; the beautiful scenery of the lakes and the hot springs around them; the spectacular river gorges, and the impressive waterfalls like the Tis- Isat falls on the Abay river make Ethiopian rivers and lakes important recreational and tourist attractions.





- **Lake Langano** is the most utilized of the lakes as it is a weekend resort. Lake Tana is one of the most popular tourist destinations in Ethiopia known for island monasteries with many treasures of the medieval arts.
- **Lake Tana and Awassa** are important tourist attractions and resort sites as they are located at the vibrant and beautiful cities of Bahir Dar and Hawassa.





- 1. Discuss the major drainage systems of Ethiopia.
- 2. Explain the hydroelectric and fishing potential of Ethiopian water resource.
- 3. State the major characteristics of Ethiopian rivers?
- 4. How do you justify the origin and flow direction of Ethiopian rivers?

CHAPTER FIVE

THE CLIMATE OF ETHIOPIA AND THE HORN

5.1 Introduction

•**Weather:** is the daily conditions of the atmosphere elem ents like temperature, atmospheric pressure, humidity, wi nd speed and direction, cloudiness and precipitation in th e troposphere.

•**Climate:** the average conditions of the atmosphere over l ong time periods.

5.2. Elements and Controls of Weather and Climate

5.2.1. Controls of Weather and Climate

•The determining factors for the variations of weather an d climate between places are called controls of weather a nd climate

a. Latitude/ angel of the sun

•Equatorial area has low angle & polar areas has slanting angle deter mine the strength of sun shines

•Ethiopia found around the equator that has:

- high average temperatures and sun shines
- high daily and small annual ranges of temperature
- no significant variation in length of day and night between su mmer and winter

b. Inclination of the Earth's Axis

1)solar distance

•The earth inclination 23 ½ ° from their axis and also 66 ½ ° from th eir Orbit. This bring the difference in solar distance between polar & equatorial areas.

i. **aphalion**: polar distance estimated 152,106,000 kms
ii. **perhelion**: Equatorial distance from the sun estimated 147,10 3,000 kms.

2) Position of the sun

- The earth rotate on its axis determines day and night and also revolves on its orbit around the sun determine the seasons.

a) Equinox

- overhead sun appears around the equator when the position of the sun seen across equator
  - equal length of day and night
  - high sun light reach around equatorial areas.
- Equinox appears twice a year.

i. The Vernal (spring) equinox: March 21st

- Appear when the Sun's rays crosses the equator and directly reach to the Northern Hemisphere
- marks the offset of the vernal equinox.

ii. The Autumn equinox: 23rd

- appears when the sun crosses the equator
- on 23rd of September marks the offset of the Autumn equinox

b) Solstice

- overhead sun appears either northern or southern hemisphere

i. The summer Solstice: June 21st

- sun is directly over head on Tropic of Cancer 23 ½ °N
- The northern hemisphere has maximum tilt towards the sun experiencing longest daylight of the year.

ii. The winter solstice: December 22nd

- overhead sun directly above 23 ½ ° S on the Tropic of Capricorn.
- maximum insolation reach to the Southern Hemisphere.

c. Altitude

- The height of location above the sea level.
- When increasing elevation temperature will be decrease.
- lapse rate: is the average rate of temperature changes per unit of altitudinal change.
- The normal lapse rate is 6.5°C/km per rise in altitude.

Types of lapse rate

- Three types of lapse rates are identified;

i. Dry adiabatic laps rate/rise dry air (unsaturated)

- temperature changes occur not only by the result of heat additi on or withdrawal from outside sources, but rather are the cons equence of internal processes of expansion and contraction of air parcel.

- adiabatic lapse rate is the rate at which the temperature of an air parcel changes in response to the expansion or compression proc ess associated with a change in altitude.
- the major cause of adiabatic is vertical displacements of unsatura ted air
- When unsaturated air rises, it expands(cover large volume) becau se it has less weight & pressure in air parcel.
- unsaturated air has the rate of 10°c/1000m to heating or cooling i n every of change in elevation.

ii. Wet Adiabatic laps rate/rise wet air (saturated)

- saturated or precipitating air rising or sinking that chang es its temperature. This process is called wet adiabatic t emperature change.
- The rate of cooling of wet air is approximately 5°c/1000 m.

iii. Environmental lapse rate or Atmospheric lapse late

- the actual change of temperature with altitude increase or decrease.
- lower layer of atmospheric is more warmer tha n upper layers. Because:
  - atmospheric heat is received directly from the ear th's surface and only indirectly from the sun.
  - high density of the air
  - more water vapor and dust particle are exist
- The rate of change is 6.5°c/1000.

5.3. Spatiotemporal Patterns and Distribution of Temperature and Rainfall in Ethiopia

5.3.1. Spatiotemporal Distribution of Temperature

- Altitude and Latitude are the two most important element in determining temperature in Ethiopia. humidity and winds also have significant impacts on temperature conditions in Ethiopia.

Altitude (meter)	*Mean annual Temp °C*	*Description*	*Local Equivalent*
3,300 and above	*10* or	Cool	*Wurch*
2,300 - 3,300	*10*	Cool Temperate	*Dega*
1,500 - 2,300	*15*	Temperate	*Woina Dega*
500 - 1,500	*20*	Warm Temperate	*Kola*
below 500	*25 and above*	Hot	*Bereha*

5.3.2. Spatiotemporal Distribution of Rainfall

- Rainfall system in Ethiopia is needs an understanding of the position of Inter Tropical Convergence Zone (ITCZ), pressure cells and Wasterlies &Trade Winds. Thus, the rainfall system in Ethiopia is characterized by spatial and temporal variabilities.

Inter Tropical Convergence Zone (ITCZ)

- convergence of Northeast Trade winds and the Equatorial Westerlies
- It is a low-pressure zone
- oscillation of ITCZ position is a causes for variation in the Wind flow patterns over Ethiopia and the Horn

Seasonal or Temporal Variabilities

i. Summer (June, July, August)

- sun overheads north of the equator(Tropic of Cancer 23 ½ °N)
- High pressure cells develop on the Atlantic and Indian Oceans arou nd the tropic of Capricorn
- Equatorial Westerlies (Guinea monsoon) winds from southwest Atl antic Ocean and South easterly from Indian Ocean blow to Ethiopia except lowlands in Afar and Southeast most parts of Ethiopia receive rainfall.

ii. Autumn (September, October and November)

- ITCZ shifts towards the equator weakening the equatorial westerlies
- Weak south easterlies from Indian Ocean showers the low lands in southeastern part of Ethiopia.

iii. Winter (December, January and February)

- ITCZ shifts towards farther to Capricorn 23 ½°S.
- High pressure cells develop on Arabian Land
- northeasterly winds originating from the landmass of Asia crossing the Red Sea carry very little moisture and supplies rain only to the Afar lowlands and the Red Sea coastal but most parts of Ethiopia is stay drying.

iv. Spring (March, April and May)

- ITCZ again shift to equator twice a year
- South-easterlies from the Indian Ocean provide rain to the highlands of Somalia and to the central and southeastern l owlands and highlands of Ethiopia.

Rainfall Regions of Ethiopia

i. Summer rainfall region

•except lowlands in Afar and Southeast parts of Ethiopia others par ts receive rainfall.

•having rainfall of 1,000 mm or more/year

ii. All year-round rainfall region

•southwestern Highlands and windward side Mountain

•average rainfall varies from 1,400 to over 2,200 mm/year

iii. Autumn and Spring rainfall regions

•South eastern lowlands of Ethiopia receive rain during autumn an d spring seasons

•varies from less than 500 to 1,000 mm/year.

iv. Winter rainfall region

•Red sea and some parts of the Afar region receive rain.

•Very low amount of rain receive

5.4 Agro-ecological Zones of Ethiopia

Zones	Altitude (m)	Mean annual rainfall (mm)	Length of growing periods (days)	Mean annual temperature (^{0}C)	Area share (%)
Wurch (cold to moist)	>3,200	900-2,200	211–365	Below 10	0.98
Dega (cool to humid)	2,300 - 3,200	900-1,200	121–210	≥11.5–17.5	9.94
Weyna Dega (cool sub humid)	1,500 - 2300	800-1,200	91–120	>17.5 – 20.0	26.75
Kola (Warm semiarid)	500 - 1,500	200-800	46–90	>20.0 – 27.5	52.94
Berha (Hot arid)	<500	Below 200	0–45	>27.5	9.39

sms

✓Climate change refers to a change in the state of the climate. (mea n, variability) over time due to either natural variability or human acti vities.

5.5.1. Current Trends of Climate in Ethiopia

- Ethiopia ranked 5th out of 184 countries in terms of its risk of drought.

- 12 extreme drought events were recorded between 1900 and 2010. seven of them are occurred since 1980.
- resulted in famines, harvest failures, livestock deaths

Trends in Temperature Variability

- rise mean annual temperature from 0.2°C to 0.28°C/decade over the last 40-50 years.
- 1.3°C has been observed between 1960 and 2006.
- Consequently, the country's minimum temperature

Trends in Rainfall Variability

- Spring and Summer rainfall have shown decline by 15-20% between 1975 and 2010.
- The average Precipitation has remained fairly stable over the last 50 years but Rainfall variability is increasing that has strong implications for crop production & husbandry

5.5.2. Causes of Climate Change

A. Natural Causes

- **Earth orbital changes:** The earth is tilted at an angle of 23.5°c lead to small but climatically important changes in the strength of the seasons.

- **Energy Budget:** Sun's energy output small changes over an extended period of time can lead to climate changes.
  - Sun spot varied from 80 km^2(dark)-80,000km^2(light) each spot has varies life cycle in every 50 years 1°c change in sun energy.
- **Volcanic eruptions:** releases large volumes of sulphur dioxide, carbon dioxide, water vapor, dust, and ash into the atmosphere. can increase planetary reflectivity causing atmospheric cooling.

B. Anthropogenic Causes

- Industrial activities, construction, transportation, b uildings, decomposition of wastes in landfills, agri culture, ruminant digestion and manure managem ent, synthetic compounds manufacturing, clearing of land for agriculture, industrial activities, and ot her human activities have increased concentration s of greenhouse gases

Green house gases and its effects:

-Gases that an ability to absorb heat are called greenhouse gases
-Such gases: Methane, Water vapor, CO_2, CO, NO_2, CFC etc.

-GHs bring global warming

5.5.3. Consequences of Climate Change

- Impacts on human health
- Impact on water resources
- Impact on Agriculture
- Impact on Ecosystem
- Global warming, increase in sea level, desertification, drought, shift of tropics, tropical diseases, loss of bio-diversity etc

5.5.4. Climate Response Mechanisms

•There are three major response mechanisms to climate change namely:

1)Mitigation

2)adaptation

3)resilience.

i. Mitigation and its Strategies

- Taken an actions to reduce and control greenhouse gas emissions or avoid significant human interference with the climate system.
- some mitigation measures are:
  - Practice Energy efficiency
  - Increase the use of renewable energy such as solar
  - Efficient means of transport implementation (electric public transport, bicycle, shared cars etc).

ii. Adaptation and its Strategies

- adjusted to actual or expected future climate and coped with changes in climate and extremes with varying degrees of success

- **Some of the major adaptation strategies include:**
  - building flood defenses,
  - plan for heat waves and higher temperatures,
  - installing water-permeable pavements to better deal with floods and storm water
  - improve water storage and use are some of measures taken by cities and towns.
  - landscape restoration and reforestation,
  - flexible and diverse cultivation to be prepared for nat ural catastrophes
  - preventive and precautionary measures (evacuation plans, hea lth issues, etc.)

iii.resilience and its strategies: recycle & carbon clean met hods

CHAPTER SIX

SOILS, NATURAL VEGETATION AND WILDLIFE RESOURCES OF ETHIOPIA AND THE HORN

6.1. soil formation

- Ethiopia to have varied soil, geological process, varied cli matic and biological diversity.
- the distribution of wildlife and natural vegetation in Ethio pia and the Horn is controlled by many factors among whic h are climate, soil types, drainage, etc.
- Natural vegetation are vital that provide shelter, food, sou rce of fuel, pasture and grazing, raw material for industries
- Ethiopia also have money Unique endemic fauna and flor a resources.

6.2. Ethiopian Soils: Types, Degradation and Conservation

6.2.1. Soil Composition

-Composed from; minerals (45%), air (25%) water (25%) o rganic matter (5%) & parent materials (rocks)

soil formation

•The formation and spatial variability of soils in Ethiopia is largely rela ted the factors of topographic, climatic, parent material (rocks), land use(human activities) and time.

•**Weathering:** the processes of soil formation

•Three types of weathering

i.Mechanical (physical) weathering

- Physical disintegration of rock without changing composition.
- due to the action of rainwater(frost), extremes of temperature(exfoliation), and biological activity(plant & animals)

ii. Biological weathering

- weathering involves the weakening and subsequent disintegration of rock by plants, animals and microb es.
- Bring both chemical and physical compositional cha nge

iii. Chemical weathering

- involves the modification of the chemical and mineralogica l composition of the weathered material.
- different processes can result in chemical weathering.
- most common chemical weathering processes are hydr olysis, oxidation, reduction, hydration, carbonation and so lution

Soil Properties

- Soils have two basic properties:

1)Physical properties

- Physical composition and major soil components. such as texture, structure, porosity etc.
- These properties affect air and water movement i n the soil, and thus the soil‟s ability to function.

2)Chemical Properties

- various chemical constituents.like, availability of m inerals, electrical conductivity, soil pH, etc. Soil che mical properties affect soil biological activity and i ndirectly the nutrient dynamics.

6.2.2. Major Soil Types in Ethiopia

- Soils of Ethiopia are basically derived from cryst alline, volcanic and Mesozoic sedimentary rocks.
- FAO has identified 18 soil associations in Ethiop ia
- major soils (11 soils) cover about 87.4% of the l and area of Ethiopia.

Soil	Environmental condition	Characteristic	Agricultural suitability	Occurrence
Nitosols	associated with highlands with high rainfall, formed on forest	gently land, strongly weathered, reddish-brown, rich in non-soluble:Fe, Al & non soluble like Ka, Ca	productive than other tropical soils, leaching	Trap series volcanics, volcanic ash, and even metamorphic rocks, found in Wellega Kaffa, Illuababora Most highlands.
Acrisols	mostly at high rainfall highlands areas	lost most nutrients low productive capacity	Infertile because inherently& organically degraded chemically	found along with nitosols in southwestern
Vertisols	highlands and moderate climates	swelling when wet and cracks when dry. heavy clay soils	has very high natural chemical fertility. But easily degraded	develop from volcanic, basalt, pyroclastic, sedimentary rocks, colluvial and alluvial plains. Gojjam, Shewa, Arsi, Bale and central Hararghe).
Lithosols, Cambisols & Regosol		young, shallow and coarse textured, low water holding capacity		found in rugged topography and steep slopes, Central Highlands, Rift Valley, Harargh, Danakil and eastern Ogaden

Soil	Environmental condition	Characteristic	Agricultural suitability	Occurrence
Xerosols	subjected to wind erosion, deserts soil, available in arid and semiarid areas	high salt content and low organic content	very fertile if irrigated poor humus & nitrogen, rich in phosphorus and potash	Ogaden and northeastern escarpments
Yermosols and Solanchaks	available in arid and semiarid areas		drier and more problematic than Xerosols	located in salty plains of Afar, especially Solonchaks in Afar salty plains
Fluvisols		recent alluvial deposits in depressions	flat or nearly flat ground, fertile for large-scale irrigation	Omo, Awash, Abay Akobo, Baro, Rift Valley
Luvisols	Develop by alternative of wet and dry seasons	not leaching	good chemical nutrients, best agricultural, intensively cultivated	steep slopes & flat top Lake Tana area, Parts of Northern, Central, Eastern Highlands & Southern lowlands.

6.2.2. Soil Degradation

- change in any or all of soil status or diminished the capacity or deterioration of any physical, chemical and biological properties of soil.
- There are three major types of soil degradation:

i. Physical Degradation:

- deterioration of the physical properties of soil.

Compaction:

- elimination or reduction of structural pores of soil that accelerated runoff and erosion.

Soil erosion:

- The process of wearing away and moving down of soil components due to erosion, leaching and others agents of erosion.
- Ti has three-phase process. **detachment** of individual soil particles, **transportation** and **deposition**.

ii. Biological Degradation

•Reduction in soil organic content, decline in bio mass carbon and decrease the activity and diver sity of soil fauna,

iii. Chemical Degradation

•Nutrient depletion, excessive leaching, decline i n soil pH and a reduction in base saturation.

causes of soil erosion in Ethiopia.

1.Physical Causes:

-Steepness slope,, intensity, duration and seasonal of rain, texture, types and composition of soil, plant co ver etc.

2. Human Causes:

•Deforestation, overgrazing, poor farming, firing cultivation, construction, building, mining, f arm land expanding etc.

6.2.3. Soil Erosion Control Measures

- We have two major soil erosion control mechanisms.

A. Biological Control measures

- include vegetative strips, plantation, and reforestation and afforestation.
- prevent splash erosion, reduces the velocity of surface runoff, increases su rface roughness which reduces runoff and increases infiltration

B. Physical control measures

- Includes terracing, check dams, gabion, trenches, contour ploughing, soil b unds etc.

6.3. Natural Vegetation of Ethiopia

6.3.1. Introduction

- Natural Vegetation mean Plants grow with little or no human interference
- factors controlled the distribution of natural Vegetati on are climate, soil types, drainage, elevation(slope), temperature and precipitation etc.
- Plants can provide shelter, food, source of fuel, pastu re and grazing, raw material for industries, source of timber and non-timber products, moderating local cli mate, home of wild life, medicinal values, minimizing soil erosion etc.

- **xeromorphic plants:** plants which are adapted to dro ught and high temperatures
- **Afro-alpine plants:** plants which are adapted very col d and frost climate.

6.3.2. Major Natural Vegetation Types of Ethiopia

- Taking altitude into consideration natural vegetatio n can be classify into five groups:
  1. Afro-alpine and sub-afro alpine Region
  2. Forest Region
  3. Woodland Savannah Region
  4. Steppe Region
  5. Semi-desert Region

1. Afro-alpine and Sub-afro alpine Region

i. Afro-alpine

- Ethiopia has the largest extent of Afro-alpine and sub afro-alpine habitats in Africa
- Found between 4000 and 4,620 meters above sea lev el
- covers nearly 1.3% of the total landmass of Ethiopia.
- annul precipitation which ranges between 800 and 1, 500 mm with sleet or snow and temperature records 0°c and negative.
- mostly shallow and eroded soils exist.
- Afro-alpine region consists dominantly tussock grassl ands, scrub, scattered mosses and lichens etc.

ii. Sub-afro-alpine

- less extreme environment than the Afro-alpine
- Found between 3,300 and 4,000 meters
- Sub-afro alpine region is dominated by woodland, often degraded to scrub and wet grasslands with p lants like Asta, giberra etc.

2. Forest Region

- predominantly trees are well known.

i. **Highland forests: include** Kosso, tid, kerkha, zigba, Weyra, keraro etc

ii. **lowland forests: like acacia trees**

iii. **Gallery (Riverine) Forests:** stretch along the banks of the lower courses of rivers e.g sholla

3. Woodland Savannah Region

- woodland savannah are known by their xerom orphic characteristics.
- broadly classified into three divisions:

i.Junipers Woodlands (tid):

ii.Acacia woodlands: grass grow with shrubs and A cacia trees (grar) and Konter.

iii.Mixed deciduous woodlands: grass mixed with deciduous trees (large leave drop during winter)

4. Steppe and Semi Desert Regions

- regions in the arid and semiarid parts of the country.
- where the temperature is very high and the rainfall is very low (100 to 550 mm for steppe and 50 to 300 mm for the semiarid)
- The Soils is alkaline and saline
- Xerophytic plants such as short shrubs, scattered tufts of grass species and a variety of acacias and palm trees are very common.

6.3.3. Degradation of Natural vegetation

•Over the past century, Ethiopia has dense forests coverag e (40%) but it disappearing at an alarming rate for differen t reasons:

- Clearing of forests for cultivation
- Timber exploitation practices
- Charcoal burning and cutting for fuel
- Extensions of coffee and tea production areas
- Overgrazing
- Expansion of settlements both rural and urban and cl earing for construction

6.3.2. Natural Vegetation Conservation

•three main approaches used for biodiversity conservat ion:

i.Protection:

i. designation and management of the Protected areas i nclude sanctuaries, national parks, and community conse rvation areas

ii.Sustainable forest management:

i. involving sustainable harvesting of forest products to pro vide a source of financial income

iii.Restoration or rehabilitation:

– The process of recovery of a forest that has been degra ded, damaged or destroyed.

6.4. Wild Life/wild animals in Ethiopia

6.4.1. Introduction

- Ethiopia has a unique fauna, flora, mammals, birds, reptiles, amphib ians and a few groups of arthropods diversity with a high level of en demicity.

•Ethiopia has about:

- 860 avian species (16 endemic species and two endemic genera),
- 279 species of mammals (31 endemic species and six endemic gene ra),
- 201 species of reptiles (14 endemic species),
- 23 species of amphibians (23 endemic species)
- 150 freshwater fish (6 endemic species).

- The wild animals in Ethiopia can be classified into five major groups:
  1) **Common wild animals:** (those animals that are found in many parts of the country (e.g. hyenas, jackals)
  2) **Game (lowland) animal:** include many herbivores like giraffes, wild asses, zebras etc and carnivores like lions, leopards, and cheetahs.
  3) **Tree animals or arboreals:** (which include monkeys, baboons)
  4) **A variety of birds:** in the Rift Valley lakes
  5) **Rare animals:** gelada baboon, Semien fox, walia- ibex, Nyala.

6.4.2. Wildlife Conservation

•To prevent the destruction of wildlife, a total area of nearl y 100,000 square kilometers found in Ethiopia.

- ✓ 21 major national parks
- ✓ 2 major wildlife sanctuaries,
- ✓ 3 wildlife reserves,
- ✓ 6 community conservation areas,
- ✓ 2 wildlife rescue centres,
- ✓ 22 controlled hunting areas,
- ✓ 2 botanical gardens, and 3 biosphere reserves

The Importance of wildlife conservation are:

- for ecological importance, economic importance, inve stigatory importance, conservation of biological divers ities etc.

  - scientific and educational researches (valuable information for medical purposes and environ mental studies)
  - physical and mental recreation (aesthetic value)
  - promotion of tourism (economic value)
  - its potential for domestication
  - maintaining ecological balance
- Ethiopia have different animals include buffaloes, zebras , lions, elephants, ostriches, giraffes, oryx, African wild a sses, flamingos, pelicans, hippopotamus, crocodiles, Se mien fox, gelada baboon, Nyala and Walia ibex.

6.4.3. Challenges of wildlife conservation in Ethiopia

- some of the major challenges are:
- Limited awareness on the importance of wild life
- Expansion of human settlement in protected areas.
- Conflict over resource
- Overgrazing (fodder and wood)
- Illegal wildlife trade
- Excessive hunting
- Tourism and recreational pressure
- Mining and construction material extraction
- Forest fire

CHAPTER SEVEN

POPULATION OF ETHIOPIA AND THE HORN

7.1. Introduction

•Human beings are producers and consumers of wealth from natural r esources.

•Many disciplines like Geography, Demography, Economics, Epidemiol ogy, Sociology and many more study about human population. Their

differences lie in the methodologies and the aspects they emphasize

•Geography explaining regional/spatial differences on population distri bution and densities, population numbers, human-environment intera ctions, population dynamics (fertility, mortality and migration),

as well as population characteristics and qualities (age, sex, education and health composition etc.).

1. what is the importance of knowing population size of a certain city, region or country?
2. what are the determinant factors for the size of world population
3. what are the determinant factors for the population size of a certain country or region
4. what is the total size of world's population? how do you know?

7.2. Population Data: Uses and Sources

- Regular and reliable population data are vital for socio economic development, planning and administration.
- data needed to plan such as schools, hospitals, roads, water, sewerage facilities, housing, establishing voting, district boundaries, estimating future tax revenue and d esigning public programs.
- demographic data are crucial to administrators, busine ssmen, researchers, academicians and planners.
- demography influences production, distribution, consu mption, defense and administrative services at any adm inistrative or natural unit considered.

- There are three conventional sources of population data

A. Census

- The total process of collecting, compiling and publishing demographic, economic and social data pertaining at a specified time to all persons i n a defined territory.

Characteristics of census are:

- **Universality:** inclusion of all persons in a given area during the count.
- **Periodicity:** census undertaking at regular time intervals
- **Simultaneity:** undertaking census in a very limited time duration calle d the census day/night,
- **Government sponsorship:** an expensive endeavor and publication

procedures for collecting census data

•There are two procedures/ approaches

i.Dejure approach

i. it involves counting people according to their usual place of r esidence (where he/she lives most of the time).

ii. permanent population of an area making it suitable for plann ing and administrative purposes.

ii.Defacto approach:

– each individual is recorded at the place where he/she was fo und at the time of the census.

B. Sample Survey

- a method selected with the view that information acquired would represent the entire population.
- have the inherent weaknesses related to sampling errors and inadequate sources.
- This method is advantageous over census as costs, administer and taken much faster.

C. Vital Registration

- a system of continuous, permanent, compulsory and legal recording of the occurrence and the characteristics of vital events like births, deaths, marriages, divorces, and adoptions.
- more precise than census/sample survey because of time series data

7.3. Population Dynamics: Fertility, Mortality and Migration

Introduction

- Based on projected data from CSA, Ethiopia has an estimated 2019 population of 112 million
- a large and fast-growing population ranking 2nd in Africa after Nigeria and ranks 12th in the world
- the current growth rate is about 2.6%.
- 80% is rural based population
- 65 % of the population is below 24 years of age

7.3.1. Demographic Measurements

- In Ethiopia, fertility and mortality are the two princ ipal determinants of population growth as internati onal migration is insignificant.
- Some of the basic demographic measurements incl ude:

A. Crude Birth Rate (CBR)

- The number of living birth from one 1000 people in a given years and a restricted region
- Example P= 70,686,000 B= 270,300 CBR?

CBR= $\frac{B}{P}$XK
- B → Birth
- P → Total population
- K → Constant (1000)

$$CBR=\frac{B}{P}k=\frac{270{,}300}{70{,}686{,}000}=3.8/1000$$

B. General Fertility (GFR)

- The Number of living births in a year per 1000 women at a reproductive age
- **birth from specific age of women to number of women in the specific age group**
- GFR= k P1= population of women between 15-49 years ago.

. Example:

- P= 50,000,000 P1= 1/5 of total population
- B= 500,000
- Year 2000
- P1= 1/5 x 50,000,000 = 10,000,000
- = 50/1000

$$GFR=\frac{Total\ annual\ live\ birth}{Total\ women 15-49}x1000$$

- q

C. Total Fertility Rate

- The average number of children that a wom an would have at the end of her reproductive period
- 5 years female age specific birth from 15-49 is call ed cohort

TFR=(∑ASBR)5 or $TFR = 5x \sum_{n=1}^{7} \frac{Bi}{Wi}$ or $ASBR = \frac{Bx}{px} x1000$

Where:-

- Bx a number of birth to a mother of x age group (1 5-19, 20-24, 25-29, 30-34, 35-39, 40-45)
- Px- women at age groups of from 15-49

D. Crude Death Rate (CDR)

- the number of deaths per 1000 population in a year

$$CDR = \frac{D}{P} \times 100\%$$

E. Infant Mortality Rate (IMR)

- The number of infant death under age one per 1000 live births in a year. $IMR = \frac{D1}{B} \times 100$

F. Maternal Mortality Rate: death of mothers in connection fr om pregnancy and birth complications per hundred thousand liv e birth.

$$MMR = \frac{\text{Total annual maternal death due to birth complications}}{\text{Total live birth}} x\,100{,}0000$$

G. Life Expectancy at birth: the average number of years t hat a newly born baby is expected to live.

- **Natural Rate of Increase:** the difference between crude birth rat e and crude death rate **NRI= CBR - CDR**

7.3.2. Levels and trends in Fertility and Mortality rates in E thiopia

region	fertility	mortality	IMR	Life expectancy	SR
Country	4.6	4.4	54	62.6 years	99.96m/100f
Tigray	4.7	43	3rd highest		females dominants
Afar	5.5	36			male
Amhara	3.7	43			male
Oromia	5.4	45			male
Somalia	7.2	36	2nd least		Male
Benshangule		45	1st highest	47 years	male
Gambella	3.5	49	3rd least		male
SNNP	5.6	36	2nd highest	49 years	females dominants
Harari	4,3	36			male
Addis Ababa	1.8	18	least		females dominants
Dire Dawa	3.1	37			male
females	-	lowest		65.4 years	-
Males	-	highest		61.2 years	-
urban	2.3	lowest	48		females dominants
rural	5.2	highest	62		male

- Now in most developing countries, death rate declining but remain high birth rate due to:
  - —Little family planning practices and lack of populat ion education
  - —Lower status of women
  - —Early marriage, particularly of females
  - —Parents consideration of children as assets
  - —The relatively high infant and child mortality rates that trigger couples to have more births to compe nsate for the loses
  - —Perhaps religious influences.

- Countries of the Horn of Africa have higher population growth rate that exceeds 2.6 %. Because of:
  - low per capita GNP
  - increased unemployment and under -employment
  - mounting social ills such as destitution, begging, theft, prostitution
  - continuous inflation that erodes purchasing power of the currency
  - shortage of cultivated land and food shortages
  - overcrowding of infrastructural and social facilities; housing problems and increase in urban slums and squatter settlements
  - Environmental problems such as deforestation, soil erosion, loss of biodiversity and pollution.

7.3.3. Migration in Ethiopia and the Horn

- mobility or movement of peoples permanent or se mi-permanent change of residence between clearly defined geographic units

The implications of migration are:

- Migration yields an increased level of urbanization;
- It enhances rural-urban linkages in creating an integ rated economy
- It influences spatial population distribution
- negatively influences human fertility and mortality patterns and levels and affects age and sex composi tion of the population.

- It is a means of achieving economic efficiency.
- cause and consequence of inequality and unequal development
- regarded as a cause and consequence of diversity and a mechanism of spreading cultures
- create the condition of strengthening, sense of nat ionhood and national unity
- Motivating to creativity and open to new ideas tha n a homogenous group of people.

A. Internal Migration in Ethiopia

rural out migration during the Derg Regime

- was very low due to:
  - land to the tiller granted land to the rural landless farmers
  - Establishment of urban dwellers association and rural peasant associations
  - urban land nationalization discouraged migration
  - high level of urban unemployment and underemployment
  - Derg was also taking away the youth whoever in the city as soldiers to the warfront

- Now a day, the country has a relatively high level of internal migration 16.6% are labeled as migrant out of the total population.

B. International migration

- accelerated after the 1974 revolution
- Today, Ethiopia Ethiopian migrants are found in the Middle East, USA, Canada, Europe and African countries such as Sudan, Kenya, South Africa and Botswana.
- Diaspora estimated to be about 4,000,000. one of the countries that has a large number of emigrants from all African countries

- The causes of cross-border migration include:
  - Lack of employment and livelihood opportunities and negativ e attitudes attached with low paying and informal job and po or work ethics amongst the youth.
  - Rural underemployment and lack of resources
  - Unfavorable political context and insecurity,
  - Ethiopia's location and its long boundary that extends over 5 ,328 km which makes border management difficult
  - Existence of large number of local brokers with networks exte nding to destination countries
  - Misinformation and false promises by brokers/traffickers, fami ly and peer pressure
  - Demand-side factors of migration (shortage of labour in low-p aying, informal and perilous jobs, such as domestic work, cons truction, agriculture in destination countries)

- ILO identified migration source areas of Ethiopia
- **Dessie area:** Kemise, Bati, Kalu,Tehuledere, Haiq, Wore babo, Mersa, Dessie and Woldia
- **Shashemene area:** Shashemene-Zuria, Kofele, Kore and Assassa.
- **Jimma area:** Kaffa, Wolega and Iluababora, Dedo, Agaro , Setema, Sigmo, and Gomma
- **Mekelle/Tigray area:** Alamata, Kobo, Raya, Erob, E dagahamus, Gulomehadi, Etsebi, and Weneberta.
- **Other source areas:** Assela-Zuria, Adama-Zuria, Ambo, Fitche, Chancho, Hirna, Gelemso, Shewa Robit, Debre Bi rhan, and Debre Tabor

7.4. Age and Sex Structure of Ethiopian Population

1.Age structure (age ratio)

- Age Structure mean the distribution of population by age groups.
- age groups categorizing In every five-year (0-4, 5-9, 10 -14, ..., 60-64, 65 and above). but, broadly used Age g roups are three groups (0-14, 15-64, 65 and above).th ey known as young age, working age and old age resp ectively.

Measure of Age Structure

i.young to old ratio

$$\text{YDR} = \frac{youth(15-64)}{old(>65)} \times 100$$

- Ethiopia has heavy youth dependency ratio that b ring serious problems and implication on socioec onomic development of the country. like;
  - Imposition of heavy burden on the working age
  - allocation of budget to food and other needs affect th e investments and saving habits of the population.
  - diversion of limited resources on social services (scho ols, hospitals, medicines etc).
  - Burden to create expansion of employment opportun ities for booming babies that require a job.
  - further promotion of high-level fertility by increased n umber of women entering to reproductive age (ages 1 5-49) annually.

ii. Age dependency ratio

– the young and old ages are dependent on the workin g age population.

– negatively affecting capital formation, investment and development in both household and national level.

$$\text{ADR} = \frac{children(0-14) + older(>65)}{adult(15-64)} \times 100$$

iii. old age index

– old age population as the percentage of the working age population

2. Sex Structure (Sex Ratio)

- the ratio of male population to female population at di fferent age groups.

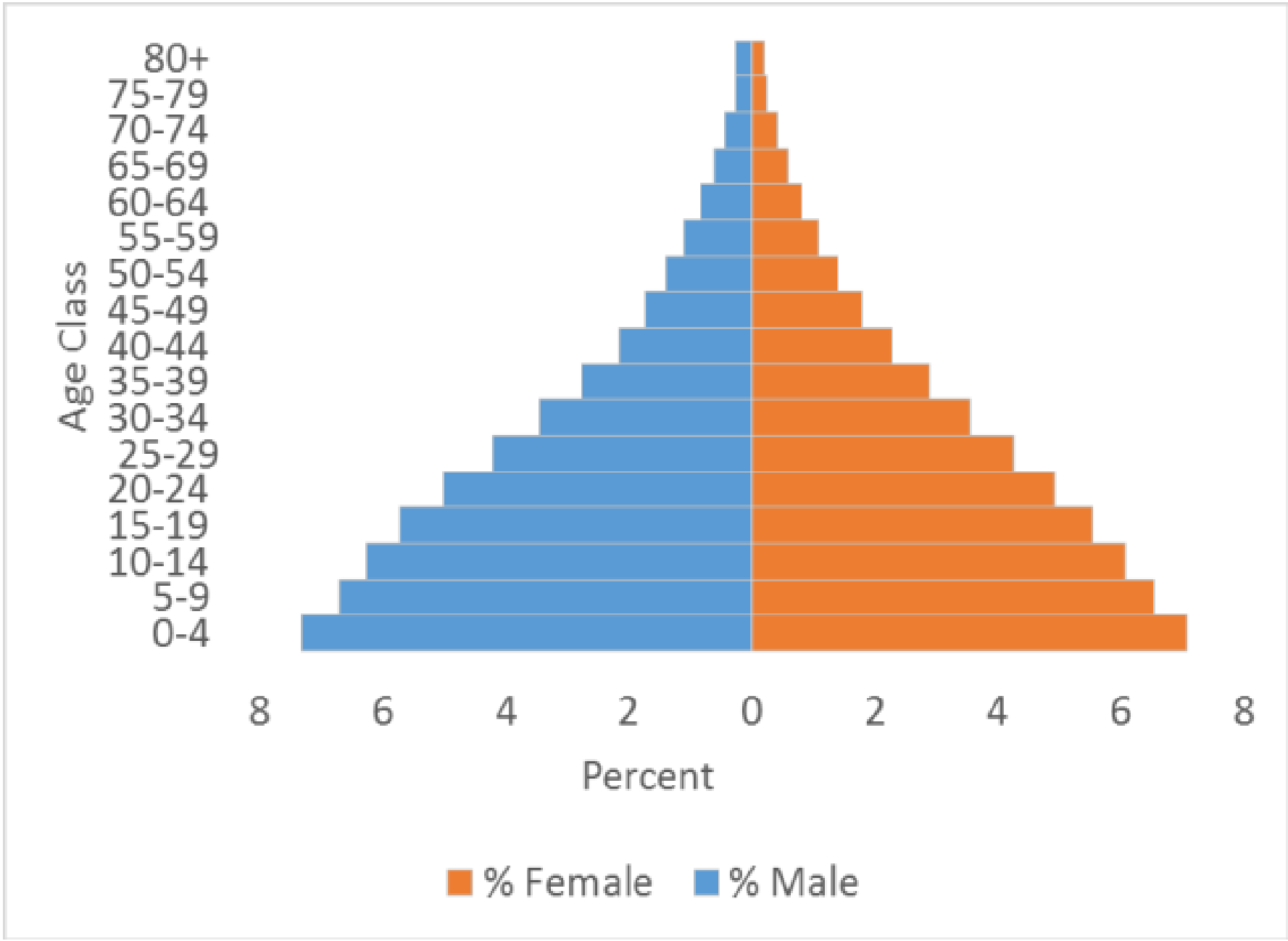
$$\text{SR} = \frac{\text{male}}{\text{female}} \text{ x } 100$$

- Sex ratios also vary with age
- At birth and young ages males tend to be greater i n number but in reality greater number male child ren mortality in Ethiopia.

Population Pyramid (Dynamics) of Ethiopia

- A Graphical Illustration of Population Cases.
- It shows sex cohorts, age, number population dep endency ratio, youngest, oldest and children popul ation, developing or developed, fertility, mortality
- negatively affecting capital formation, investment and development
- seriously
- both household and national

The graphic Illustration of Ethiopian Population Pyramid



7.5. Population Distribution in Ethiopia

- the arrangement of people over space that is pr ovided for them to settle and make a living thro ugh exploiting resources.
- physical factors that affect population distributi on include climate, soil, vegetation, drainage an d slope. the historical pattern of population mov ement, the type of economic activity, urbanizati on, industrialized and the demographic variab les are same of them.

7.5.1. Measures of Population Distribution

Population Density

– the number of people per unit area.

a.Crude density (Arithmetic density)

$$CD = \frac{\text{total population}}{\text{total area}} = \frac{TP}{TA}$$

- it does not show variations of population distribution i n the uses of parcel of land.
- low density regions are Gambella, Somali, Afar and Be nishangul-Gumuz.
- the largest population density regions are SNNP, Am hara, Harari.

- Wanago, Damot Gale, Aleta Wendo, Yirga Chefe, Dara, Kacha Bira, Angacha, Sodo Zuria, Shebe dino and Kedida Gamela have crude densities ex ceeding 500 people/km2
- low densities areas are Guba,Gog, Vaso, Dolo Bay , and Liben less than 10 people/km2.

b. Physiological Density

- a ratio between total population and arable pa rt of a country (estimated 969,680 km2)
- Ethiopia has very low physiological density.

$$PD = \frac{total\ population\ (TP)}{total\ arable\ land\ area (TAL)}$$

c. Agricultural rural density

- takes only agricultural population as a numerator and cultivated land as a denominator.
- in most developing countries there is not a signific ant difference between rural and agricultural popu lation.
- It measures a better indication of the pressure of population on land resources.

$$Ad = \frac{\text{rural population}}{\text{cultivated land}} = \frac{RP}{CL}$$

- cultivated land and urban population are the two main factors makes variations in agricultural densi ty

7.5.2. Factors Affecting Population Distribution in Ethiopia

- Both physical and human factors makes uneven populat ion distribution in Ethiopia.

•Regional Population Distribution

- Oromia, Amhara, SNNP are a major populated regions b ut Gambella and Harrar are a least populated regions.

i.Physical Factors

- Climate factors: mainly rainfall and temperature, and ve getation.
- other physical factors include drainage and slope, soil, a ttractiveness of a region etc.

ii. Human Factors

A. The historical pattern of population movement

– decline of the Axumite , establishiment of central highlands kingdom, northward movement of the Oromos population, a significant explanation of A mhara, Agew and Guraghe populations are some of examples.

B. Types of economic activities.

– pastoral herding, cultivated lands, the developme nt of commercial farms, expansion of urban areas, industrial growths, transportation routes are some of the important elements for population re-distri bution over time.

7.6. Socio-cultural Aspects of Ethiopian Population: Education, Health and Languages

7.6.1. Education

- It have gender parity, regional differences, urban-rural differences, age parity, infrastructural limitation & access differences, structure of educational system(primary, secondary, high school, TVT, college and university) etc.

7.6.2. Health

- health in Ethiopia is related to potentially preventable, communicable diseases and nutritional disorders
- Ethiopia has poor health status

- **Some root causes for poor health status are:**
  1. Lack of access to clean water: rivers and lakes rem ain the most important sources of water
  2. Disease related to beliefs, behaviors and traditiona l practices which have a negative effect on health s tatus include circumcision, early marriage, and low value of girls and children
  3. Lack of adequate nutrition:
  4. Lack of health services:

7.6.3. Languages Families and Languages of Ethiopia.

- Ethiopia is a country where about 80 language s are spoken most Languages belong to two Supp er families.

A. Afro-Asiatic

1) **Cushitic:** largest number of speakers and the widest coverage. Oromigna, Somaligna, Sidamigna, Afarigna, Ke mbatigna, Hadiyigna, Alabigna, Gedeogna, and others.
2) **Semetic:** Amarigna, Tigrigna, Guragigna, Siltigna, Aderign a, and Argobigna.
3) **Omotic:** Wolaitigna, Gamogna, Kullogna, Kefigna, and Ko ntigna

B. Nilo Saharan

- It include Kunamigna, Bejigna, Gumuzigna, Maogna, Kew amigna, Nuerigna, Annukigna, and others.
- spoken in the western lowlands of Ethiopia along the bor der with Sudan.

7.7. Settlement Types and Patterns

7.7.1. Types of Settlement

Rural Settlement

- can be temporary or permanent depending on whether
- majority of the Ethiopian population still lives in ru ral settlements either hamlets or villages.

Temporary / Mobile Settlements

- most parts of the Rift Valley, peripheral areas, hot and dry mostly pastoral herding and mobile settle ments. Because searching for new sites for water and grass.

Permanent Settlements

- where crop cultivation is practiced are permanent
- Permanent settlements are of two types
  1. scattered (diffused or dispersed)
  2. Clustered or Nucleated

7.7.2. Urban Settlements and Urbanization in Ethiopia

1.Urbanization is the process of society change or tr ansformed from being rural economy, culture and lif estyle to being of urban.

2.Today, the urban population is about 20 %of the c ountry's population.

- The major criteria used to classify settlements as urban in Ethiopia are:

  i. Minimum of 2,000 people;

  ii. Two-thirds of the population engaged in non-agric ultural activities;

  iii. Chartered municipality;

  iv. The presence of social services and amenities

An overview of the History of Urbanization in Ethiopia

–Before Addis Ababa, only prominent urban centers were Axu m, Lalibela and Gondar

–**Modern urbanization** associated with the establishment of Addis Ababa

–Ethiopia is regarded as a vast urban desert.

- different factors contributed to the growing and permanency of Addis Ababa
  - Introduction of the fast-growing Australian eucalyptus tree
  - Water supplies
  - Italian occupation and establishment of small-scale industries and institutions, road construction
  - The construction of roads and railway radiate from Addis Ababa
  - Introduction of modern schools, hospitals and health centers, hotels, cafes, bars, bakeries, butcher's shops, cinema, post office, modern cathedrals, telephone and telegraph system, bank, printing press etc.

- low level of urbanization can be attributed to:
  - self-sufficiency of agriculture which reinforced rural peasant life
  - Low level of industrialization, low level structural tra nsformation and economic development
  - morphology of the country hindered transportation and communication
  - continual warfare between kingdoms
  - and the frequent changes of the royal residence
  - Lack of employment and housing shortage in urban areas that discourage in migration
  - Political instability, ethnic conflict and social unrest during the late 1970s, 1980s and even the current.

- Now a day, urbanization process grows at faster rate in Ethiopia. Due to driving factors of urbanization:
  - establishment, expansion and its permanency of Addis Ababa as a capital city
  - Ethio-Djibouti railway line
  - Italian occupation contributed to road building and small-scale industries
  - Integration of the major administrative provincial with capital cities
  - Political decentralization
  - Proximity, agglomeration and metropolisation effects

- High Population density and growth rates
- New and large commercial farms, mining areas and agro -industries
- Sugar factories, mega projects, airports, highways and dr y ports Opening of Universities and job opportunities,
- Tourism assets and attractions
- Development of border towns

Distribution of Urban Centers in Ethiopia

•The distribution of urban centers in Ethiopia shows spatial variation and hierarchy.

hierarchy of urban centers in Ethiopia

•***Addis Ababa Metropolitan cluster :*** Addis Ababa and its surrou nding towns.

•***Secondary city clusters:***

•**Lake Tana Urban Cluster:** Bahir Dar, Gondar Debre Tabour, Debre Markos

•**South Rift Valley Urban Cluster:** Hawassa, Shashemene, Dila, Hos ana, Sodo, Arba Minch

•**Eastern Urban Cluster:** Dire Dawa, Harar, Jigjiga

•**Mekelle Urban Cluster:** Mekelle, Adigrat, Shire, Axum

•**Dessie- Kombolcha Urban Cluster:**

•**Jima Urban Cluster:** Jima, Agaro, Mizan, Tepi, Gambella

Tertiary urban clusters

- **Nekemte Urban Cluster**:Nekemte, Dembidolo, Gimbi, Metu, Assosa
- **Gode - Kebri Dar** city network
- **Semera-Mille, Asaita** city network

Growth Rate of Urban Centers

- Ethiopia's towns are characterized by wide range of growth rates . classified into three broad categories:

1)Declining Towns

- declining in absolute numbers because net out migration is greater than natural increase (EM>RNI)
- include, Axum, Goba and Maichew.

2. **Slow Growing Towns**

- grow at the rate which is less than the rate of natural increase (<RNI)
- such as Holeta, Harar and Gore are grow slow ly in the recent past

3. **Fast Growing Towns**

- growth rates of greater than the natural rate of increase (>RNI)

CHAPTER EIGHT

ECONOMIC ACTIVITIES IN ETHIOPIA

8.1. Introduction

- Geographers classify a nation's economy into primary, secondary, tertiary(service sectors), quaternary and quinary.
- This categorization is seen as distance from the natural environment. It starts with the primary sector of an economy that extracts or harvests products from the earth.

- **primary sector** include agriculture (subsistence a nd commercial), mining, forestry, farming, grazing, hunting and gathering, fishing and quarrying.
- **secondary sector** of the economy manufactures fi nished goods.
- **The tertiary sector** of the economy is the service industry include retail and wholesale sales, trans portation and distribution, entertainment, restaurants , clerical services, media, tourism, insurance, bankin g, healthcare, and law

8.2. Mining Activity in Ethiopia

8.2.1. Introduction

- Search and extraction of minerals from the crust of th e earth
- Minerals are naturally occurring organic and inorganic substances.
- 1.5 % of GDP (USD 32 billion).

8.2.2. Status of the mineral sector investment in Ethiopia

- The Mining laws of Ethiopia have been issued in 1993 and amended recently.
- The mining stakes in Ethiopia are held by a Govern ment organization named as National Mining Corpora tion EMDSC(2000) and a privately owned Ethiopian enterprise Midrock Gold work.

- Allana Potash a Canadian mining company start mining for potash in the Afar Regional and Indian Sainik Potash has been working in the Dallol depression.
- Presently a number of world class mining companies are operating in Ethiopia for different mineral commodities.

8.2.3. The importance of Mining sector in Ethiopia

1)Economic benefits

- Generates revenue from sales, taxes, royalty
  - federal government collecting 4.4 million USD from the large-scale production of gold every year
  - regional administrations also collect small scale production of precious minerals

2) Generates foreign currency earnings:

- 135 million dollars every year for the last couple of years from the sales of export of minerals such as gold, tantalite concentrate platinum, decorative dimension stones and gemstones
- Small employment opportunity in the sector because, mining is not a labour-intensive industry.
- several hundred thousand Artisana involve in mining works.

3) Social Benefits

- Expansion of infrastructures such as roads, electric power, telecommunication
- Expansion of social services such as health facilitie s, schools service, schools, water facilities for the l ocal communities
- Train local communities to engage in other form of business
- Promote small entrepreneur's engagement in the l ocal community services such as hotel, mini- mar kets, stationeries

8.2.4. Environmental issues and management related to mining

- extraction of minerals leads to disturb the environment
- mining legislation is a compulsory criteria related to environmental issues.
- careful and systemic manage the surface and ground water, soil, rock stability, deforestation grass land, farm land, spillage chemicals or metals, air, noise, dust, aesthetic values of the area, cultural and tourist heritages
- licensees ensure based on mobilization of the environmental management and mine closure plan

8.3. Forestry

- related to exploiting forest products, which include gather ing of fuel wood, production of timber and charcoal, and c onstruction of houses.

Economic significance of forest

- contribution to the national economy (GDP)
- Fuel wood
- Timber household furniture
- For building and construction

•commercial exploitation of forest resources in Ethiopia is still in its infancy stage. Because:

- Rapid deforestation
- Low demand for timer
- Lack of modern lumbering technology
- Inaccessibility of natural forest

8.4. Fishery

8.4.1. Introduction

- involves harvesting of fish resources from water bodies.
- Fishing is known to be one of the oldest human activities.
- fishing is the most widely spread economic activity both in private and modern societies.
- Ethiopia has great potential for fishing.

Main Fish Speciesin Ethiopia are:

- Barbus, Clarius, Tilapia, Nile perch, Bagrus, Barbus Clarious, Labeo, Ti lapia, Barbus, Nile perch, Bagrus, Barbus, Clarious,Cyprinus, Tilapia, b arbus Clarious, Cyprinus, Tilapia, barbus Clarious, Tilapia, Barbus Clar ious, etc

8.4.2. Fishing Grounds in Ethiopia

1.Lakes fishery

- Ethiopian lakes are estimated to cover a surface ar ea of about 7000 km2
- include Lakes Tana, Ziway, Langano, Hawassa, Aba ya and Chamo.
- Among the 101species of fish available in the coun try 25 of them are commercial fish species and are found in the lakes.

2.River Fishery

- Most rivers of Ethiopia have high fish resources. F or example, river Baro

8.4.3. Demand and consumption of fish

- low level of local fish consumption
- fish has not been integrated into the diet of fo od.
- religious influences on consumption patterns, and demand for fish during fasting seasonal.
- limited supply of the product and its high price .
- distance of the fish production sites from d ensely populated areas and its perishability

- Socio-economic contribution of the fishery sector
  - Role of fisheries in the national economy: Gross Dom estic Product
  - Trade: exports and imports significant amounts of fi sh
  - Food security: achieve food security for the growing population
  - Employment: considerable workforce is employed bo th directly and indirectly of capture fisheries
  - Fish meal as animal feed source: excellent source of protein and amino acids
  - Sport fishing as a recreational value: port fishing is co mmon in southern part of Lake Tana.

8.4.4. Constraints and opportunities of the fishing sector

- Factories, agriculture and sewage are the sources of major pollutants affecting Ethiopian water bodies an d their fisheries.
- Inadequate legal and policy frameworks
- limited human resource availability with an acute sh ortage of trained personnel.
- serious constraints on fishery management and t echnical and extension support services.
- Public and private investment in fishery and aquacul ture is low and inadequate infrastructure
- Traditional methods of fishing
- Low transportation & facilitating services

8.5. Agriculture in Ethiopia

8.5.1. Introduction

- Cultivating of crop and rearing of animals for local consumption or commercial porpuse.
- The sector is dominated by small-scale farmers that practice rain-fed mixed farming by employi ng traditional technology, adopting a low input an d low output production system.
- majority of the poor reside engage in this sector.

8.5.2. Contributions, potentials and characteristics of agri culture in Ethiopia

- The contributions of agriculture in Ethiopia
  - accounts for GDP
  - foreign exchange earnings
  - derives its livelihood directly from agriculture
  - provides raw materials for the processing industries
- **The Agricultural Resource Base/potentials of Ethiopia**
  - abundant agricultural resource and varied agro-ecolo gical zones
  - has also vast grazing land and livestock population st anding first in Africa and 10th in the world

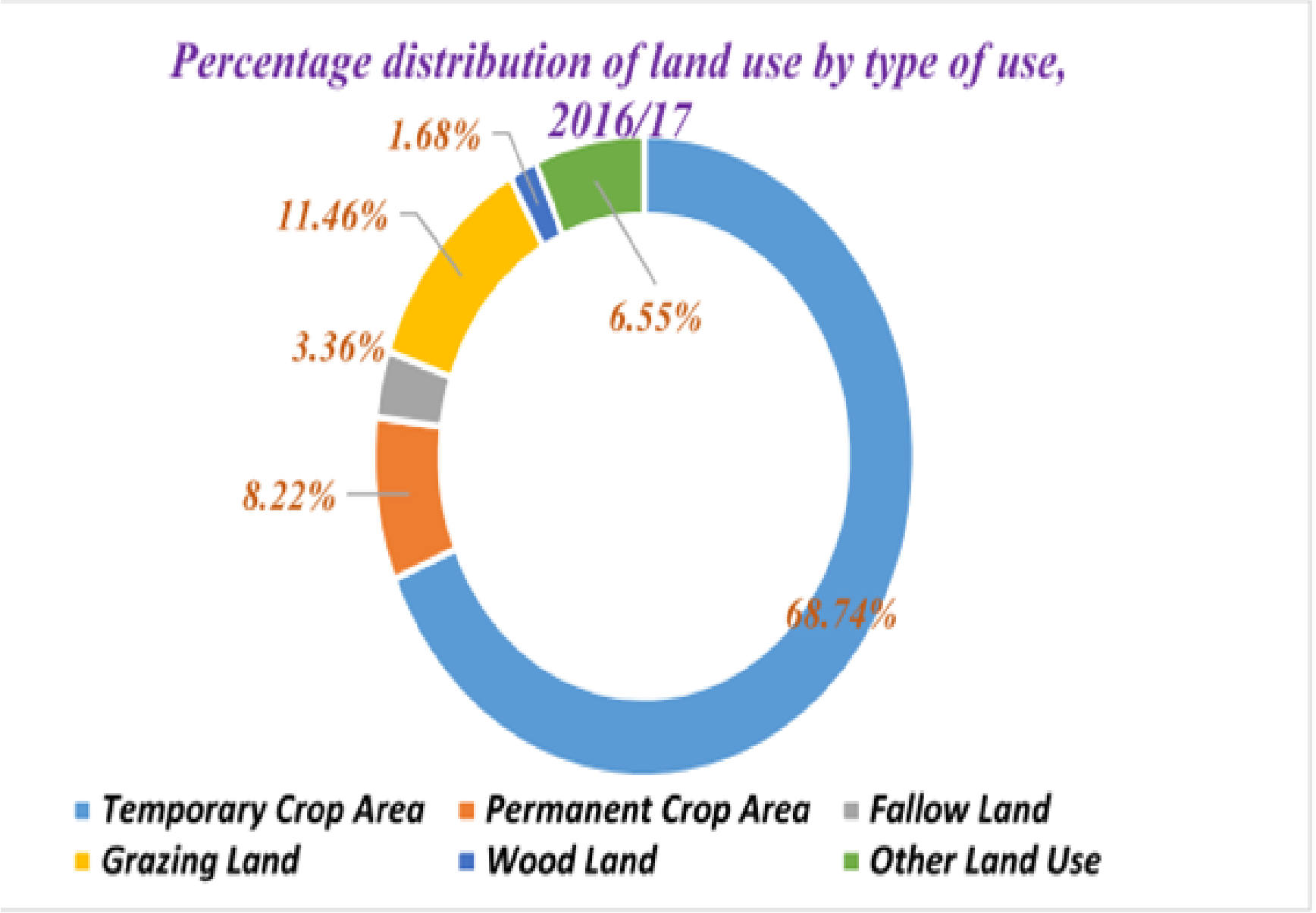
Land Use in Ethiopia

- Ethiopia has a total land area of about 113,000,000 he ctares
- 12.6million hectares moderately cultivated,
- 12.5%) of the total area is intensively cultivated,
- 6.9%, forest and wood land, 30.5% grassland
- lowland areas and river basins are more suitable for ir rigation.

Agricultural Land Use in Ethiopia

- Land resource highly dependent on agriculture. Becau se:
  - Agriculture is a land-based activity
  - Quality of land direct bearing on the productivity of agricult ure

- Agricultural land refers to the share of land ar ea that is arable under **permanent crops** and u nder **permanent pastures**.
- **permanent crops:** includes land under floweri ng shrubs, fruit trees, nut trees, and vines, but excludes land under trees grown for wood or t imber.
- **permanent pastures:** Permanent pasture is la nd used for five or more years for forage, inclu ding natural and cultivated crops.


Percentage distribution of land use by type of use, 2016/17
1.68%
11.46%
6.55%
3.36%
8.22%
68.74%
Temporary Crop Area
Permanent Crop Area
Fallow Land
Grazing Land
Wood Land
Other Land Use

Cropping seasons & Cropping Pattern in Ethiopia

– There are two main crop seasons in Ethiopia.

- **Belg crop season**
  - – is officially crop harvested between March and Au gust.
  - – the main season and produces 90-95%.
- **Meher crop season**
  - – is crop harvested between September and February.
  - – provides the remaining 5-10%

•NB: Cropping Pattern & quantities of production of cereals (teff, w heat, barley, maize, sorghum, etc are vary from place to place and ti me time due to different factors.

Animal Husbandry

- About 75% of the cattle and sheep population are found in the highland zone while
- 70 % of the goat and 100% of the camel population are found in the lowlands. than
- 90% of the livestock population is found in Oromia, Amhara and SNNPR.

Characteristics of Ethiopian Agriculture

1)Subsistence Orientation

2)Fragmentation of farm and Small size of Holdings

3)Low Use of Inputs

4)Susceptibility to Disasters

5)Limited practice of irrigation & dependent on rain-fed

8.5.3. Agriculture Systems in Ethiopia

8.5.3. Agriculture Systems in Ethiopia

- Agricultural system is terms of similar resource basis, enterprise patterns, household livelihood, constraints, could have similar development strategies, interventions and also share more related socio-cultural, economic and livelihood structures and patterns.
- The agricultural systems of Ethiopia can be classified based on:

1. The Agro-ecological patterns
2. Dominant types of crops or animals
3. Farming methods and tools used
4. Level of technological input & disposal of output

- Based on the dominant corps cultivated or animals reared and t he main implements used in cultivation, the major farming syste ms are classified as:
  1) Highland mixed farming system
  2) Lowland mixed farming system
  3) Pastoral system
  4) Shifting cultivation
  5) Commercial agriculture

1. Highland mixed farming system

- areas of over 1500m.a.s.1 in dega woina dega agro- ecological z ones.

A. Grain-plough complex

- Grain-plough is an integration of crop and livestock production with a complex farming activity.
- Farmers mainly depend on cereals crops (Barley and wheat, teff) with pulses and oil crops for subsistence.

B. The Horticulture-Hoe complex

- This agricultural system is primarily found in the hi gh rainfall humid regions of southern and southwe stern parts
- The major crops grown include enset, coffee, chat and various other horticultural corps.
- Maize is the major cereal crop in the region Grown both wood and fruit-trees using multiple cropping methods with coffee, Chat and Enset is known as i nter-mixed Agro-forestry farming system .
- Livestock, mostly cattle are an integral part of hort iculture hoe system

2. Lowland mixed agriculture

- common in the hot and dry mountain foothills, Rift Val ley and northern areas of Awash River
- Grown the varieties of sorghum, maize, wheat, teff, oil corps and lowland pulses mixed with Livestock raised.

3. Pastoral complex

- practiced in the arid and semiarid lowlands of Ethiopia.
- Afar, Somali, Borena, lowlands of Omo and Lake Turk ana area.
- Cattle, camels, sheep and goats are important animals
- Almost all the camels, ¾ of goats, ¼ of sheep, 20% of c attle are raised by pastoral communities.
- About 90% of the live animals exported from areas.

4. Shifting cultivation

- This is the cut-burn cultivation system because, clearing the plo t until its natural fertility is exhausted
- Practice in Benishangul-Gumuz, Gambella and Southern Regio ns but now a time, it is decline due to government intervention and sedentary farming introduced (villagization).
- livestock rearing is limited in this areas due to tsetsefly infestatio n

5. Commercial agriculture

- modern farming practice involving the production of crops or ani mal for market using mechanization and hired laborers.
- Mechanized farms were concentrated in the Awash valley, Arsi, s outhern Shoa and Humera-Metema
- the main problem are lack of infrastructure, access to land, confl icts with local inhabitants, hot and disease ridden areas

8.5.4. Major problems of Ethiopian agriculture

1) Land degradation
2) Variable rainfall
3) Land fragmentation
4) Land tenure insecurity
5) Backward technology
6) Poor rural infrastructure

8.6. Manufacturing Industry in Ethiopia

8.6.1. Introduction

– a higher-level economic activity than the production o f primary materials.

Employment in Industries

- The number of persons engaged in machinery and equipment grew tremendously.
  1. micro enterprises employ
  2. small enterprises employ
  3. medium enterprises employ
  4. and large enterprises employ

The value of industrial production

- The Ethiopian large and medium size manufacturing sector is dominated by food and beverages.
- It accounted the largest proportion of the overall manufacturing value added

8.6.2. Types and characteristics of manufacturing

Types of Manufacturing Industries

a)The Cottage (traditional)industries

- various kinds of weaving, woodcarving, pottery, metal works, basketry etc. are known for a long time.
- most important factor for the development of cottage industries are:
- negative traditional attitudes towards such activities, c onsidered of low social cast activities and limited avail ability (minerals, market, agricultural raw materials an d sources of power)

b) Manufacturing Industries

- In Ethiopia manufacturing industries are at a low l evel of development

i. light industries

- process consumer goods like textiles, food, tobacc o etc. Heavy industries that manufacture capital g oods are very small

ii. Heave industries

- Bulk machinery and construction materials handle in large quantity and produced large quantity. E.g chemical, vehicles, air craft etc industries.

Top 10 towns 2009/10		
Top 10 towns	Share (%)	Employment
Addis Ababa	40.3	45.9
Hawassa	3.6	2.5
Dire Dawa	2.8	1.9
Mekelle	2.8	1.9
Adama*	2.4	3.1
Bahr Dar	2.3	1.4
Bisheftu*	1.7	2.5
Sebeta*	1.7	1.5
Burayu*	1.6	1.3
Hosana	1.1	0.3

Industrial parks/zones in Ethiopia

- To day, Ethiopia gives top priority for manufacturing industrial development that established different industrial parks across t he country. Why becuase:
  - strong linkages with the agricultural sector
  - labour intensive
  - Have export potential and low entry barriers

Three mechanisms place for the establishment of IPs/SEZs :

a) fully developed by the federal or regional governmen t;
b) developed by PPPs with the IPDC and
c) by private developers only.

- Industrial parks in Ethiopia can also be categorized based on their focus sector:

1) Textile and garment;
2) Leather and shoes
3) Agro-processing,
4) Pharmaceutical and IT park.

Contributions of Industrial Parks in Ethiopia

- Stimulating investment and creating employment
- Facilitating export growth and foreign exchange ea rnings
- Developing industrial clusters through forward/ba ckward linkages
- Eliciting knowledge transfer and technology spill o ver
- Establishing connections to global value chain
- Fostering Sustainable Growth and social equality
- Enforcing implementation of national industrializa tion strategy

The Sugar Sector Mega Projects in Ethiopia

- The country has huge labor and resources like water, suitable climate and ideal landscape to expand this ex port-oriented manufacturing industry.
- It also has more than 500 thousand hectares of irriga ble land suitable for the sector.
- Wonji and Metehara) have limited capacity to meet d omestic demand.
- Ethiopia is heavily investing in sugar plantations and f actories like Omo-Kuraz Sugar Factory II, Kessem , Arjo Dediessa, Tendaho, Finchaa, Metehara, Wolqa yet Sugar development project, Tana-belese Sugar Fa ctory II etc

8.6.4. Industrial development in Ethiopia: Challenges and Opportunities

Challenges	Opportunities
High logistics and transportation cost	Relatively cheap electricity charge in comparison to
Limited research/study and action on export incentives	Macroeconomic stability and rapidly growing
Low labor productivity	Relatively cheap labor force & increasing number of
High cost of imported raw materials	Access to wide market (large domestic market,
Limited compliance to the international requirements	Competitive incentive packages which include
Underdeveloped rural infrastructure in the potential areas	Integrated Agro-Industrial Parks (one stop shopping for all the services, economies of scale,
Weak supply chain integration, market institutions and information system	Global attention due to its remarkable economic growth and credit worthiness
Low level of technology	

8.7. The Service Sector in Ethiopia

8.7.1. Introduction

- Tertiary economic activity involves the distribution and p rovision of goods and rendering services.
- include wholesale and retail outlets, banking and other fin ancial services, governmental and educational services, m edical facilities, and much other business and service func tions upon which we depend daily. Security services, tran sportation, information and telecommunication services, t ourism, health services, education institutions and researc h centers all are vital tertiary economic activities.

8.7.2. Transportation and communication in Ethiopia: types, roles and characteristics

8.7.2.1. Transportation

- persons, manufactured goods, and property are physically carried from one location to another
- There are five different types of transport in Ethiopia. These are:

a. Traditional Transport

- pack animals (donkeys, mules, horses, camel) and goods carried by humans.

b. Road Transport

i. Road Network

- The radial patterns of network development with the center being Addis Ababa
- the total road network increased rapidly

ii. Road Density

- The rugged terrain in Ethiopia makes road construction diff icult and expensive
- compare the length of roads to the total area of the countr y

iii. Road accessibility

- annual average distance of all-weather roads or the propo rtion of area of all-weather

c. Railway

- Recently a new railway line connecting Addis Ababa to Djib outi
- it helps transport of bulky products
- The other important railway is Addis Ababa Light Rail Trans it (north-south and east- west)

d. Waterways

- two types: **seaways** and **inland** waterways
- the port of Djibouti is very important for Ethiopia's external trad e.
- Inland waterways are classified in to two **rivers** and **lakes**.
- Topography affect rivers flow
- like Baro and Omo and Tana and Abaya surve small-scale transp ortation

e. Airways

- Air transport is the fastest from the other
- reduced distances by minimizing the travel time.
- very essential for a country like Ethiopia, where the topography i s difficult for communication
- expensive and cannot be used for transporting bulky products.

8.7.2.2. Communications

- the process of conveying messages to others
- include radio, television, internet, satellite, print
- publications, fixed and mobile telephones, and post offices

The Contribution of Transportation & communication to Development

- It creates job opportunity
- It promotes investment opportunities
- plays a big role for both national and international trade.
- It serves as a source of income generation
- It contributes to the maintaining a country,s peace, political wellbeing and stability
- Plays the role of linking rural-rural interconnecting
- contribution to the development of tourism, entertainment, sports and peaceful relationship among people

8.7.3. Trade in Ethiopia

- Ethiopia has two types of trade. These are inter nal trade and external trade.
- Structure of Commodity Export of Ethiopia
- two types of trade. These are internal trade and external trade.
- export is determined by agricultural products.

Particulars	2015/16	2016/17	2017/18
	%share	%share	%share
Coffee	25.2	30.4	29.5
Oilseeds	16.6	12.1	14.9
Leather and Leather products	4	3.9	4.7
Pulses	8.1	9.6	9.5
Meat & Meat Products	3.4	3.4	3.6
Fruits & Vegetables	1.9	1.9	2.2
Live Animals	5.2	2.3	2.2
Chat	9.2	9.4	9.3
Gold	10.1	7.2	3.5
Flower	7.9	7.5	8
Electricity	1.1	2.5	3
Others	7.4	9.7	9.7

The major export destinations for Ethiopian goods

1)Continental distination

- Asia accounted for 39.8 %
- Europe (28.7%)
- Other 20.9 %

2)individual country

- five countries (China, USA, Netherlands, Saudi Arabia and Djibouti) are important

8.7.4. Tourism in Ethiopia: Types, major tourist attraction sites, challenges and prospects

- the activities of persons traveling to and staying in places outside their usual environment for not more than one consecutive year for leisure, business and other purposes
- the contributions of tourism income to GDP as well as export earnings are growing in recent years.

Types of Tourism & attraction sites of Ethiopia

1. Historic Attraction sites
2. The Obelisk of Axum:
3. The Churches of Lalibela
4. The Castles of Gondar
5. The Walls of Harar
6. Natural Attraction sites
7. The Blue Nile Falls/ Tississat
8. Simien Mountains
9. The Rift Valley Lakes
10. The National Parks

The Role of Tourism in the Economy of Ethiopia

- the third source of export revenue after agriculture and industry.
- an important source of employment
- important economic sector in the country.

Challenges of the sector to be addressed

1. Weak institutional framework, skilled human and financing
2. Very limited capacity of international standards for leisure touri sm with investment concentrated in main cities and hotels
3. Lack of basic and IT infrastructure
4. Weak private sector associations
5. Unplanned destination development
6. Poor visitor management in natural and heritage sites
7. Poor tourism statistics